TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

Department of Electrical Engineering

Deep Learning Architectures for Two-Hop Relay Communication: A Comparative Study of Classical and Neural Network Relay Strategies

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
In Tel Aviv University

by

Gil Zukerman

March 2026

TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

Department of Electrical Engineering

Deep Learning Architectures for Two-Hop Relay Communication: A Comparative Study of Classical and Neural Network Relay Strategies

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
In Tel Aviv University

by

Gil Zukerman

This research work was carried out at Tel-Aviv University
in the School of Electrical and Electronics Engineering
under the supervision of Dr. Anatoly Khina and Prof. Raja Giryes

March 2026

Abstract

This thesis compares classical and learned relay strategies for two-hop communication. One configuration is fixed throughout the core study and never departed from: a single-relay SISO link, independent and identically distributed Rayleigh fast fading applied identically on both hops, complex baseband, Gray-coded QPSK, and uncoded transmission, with the relay function as the only variable. Nine relay strategies are implemented, spanning amplify-and-forward (AF), symbol-wise decode-and-forward (DF), a multilayer perceptron (MLP), a hybrid switch, generative models (a variational autoencoder and a conditional generative adversarial network), and sequence models (Transformer, Mamba-S6 and Mamba-2 SSD); eight are evaluated in the final comparison, the conditional generative adversarial network being excluded on cost grounds. All results are Monte Carlo estimates reported with 95% confidence intervals, and the simulator is first validated against closed-form BPSK error probabilities on an additive white Gaussian noise channel, which serves only as that calibration reference; symbol-wise DF on the canonical Rayleigh channel is independently checked the same way, against the closed-form two-hop composition, and matches to within one percent at every measured SNR.

Under the canonical benchmark, the best learned relays outperform AF at low signal-to-noise ratio and attain BER comparable to symbol-wise DF at medium and high SNR, while DF achieves the lowest BER of the relays evaluated at no parameter cost. This should not be read as a general optimality claim for DF. It is an empirical statement about uncoded, per-symbol relaying over the SNR range measured here; the information-theoretic results for block-based DF concern coded transmission and asymptotic blocklengths, and neither they nor the reported measurements establish that DF cannot be beaten by a coded or soft-information relay; a coded, hard-decision relay is measured directly below, though a soft-information one remains open. A complexity study indicates that performance is governed by capacity rather than architecture at high SNR, where an MLP of 169 trainable parameters attains the BER of models two orders of magnitude larger; at low SNR the architectures separate, the feedforward relays tracking DF while the sequence models trail.

Two further studies are reported separately because each requires a configuration the canonical one cannot absorb. Raising the constellation to 16-QAM places four levels on each axis and breaks the per-axis relay formulation, producing a structural error floor that a joint two-dimensional classifier over the full constellation removes, restoring parity with DF rather than surpassing it. The second measures the coding caveat left open above directly: an outer rate-1/2 convolutional code with a genuine block-DF relay beats the uncoded DF baseline above roughly 6–8 dB but is worse below that threshold, which sharpens rather than closes for stronger codes (constraint length swept up to 7). Against that coded classical baseline the learned relays lose clearly at low-to-mid SNR and win at high SNR, and instrumenting the relay output identifies why, which is not that a network out-decodes Viterbi: Viterbi places roughly half as many symbol errors on the air, but block-DF re-encodes after decoding, so its residual errors arrive as valid-but-wrong codewords the destination cannot repair, while a relay that only denoises leaves the code's redundancy intact and has 57–74% of its errors repaired downstream. Forwarding soft rather than hard decisions improves the learned relay at every SNR on identical weights, but does *not* rescue block-DF, which localizes the liability precisely: it is consuming the code's redundancy at the relay, not quantizing there. Held to equal spectral efficiency rather than equal E_s/N_0, moreover, the rate-1/2 code does not pay for itself below 20 dB at all, and where it does the gain is $1.35\times$ while costing $20\times$ the buffering latency and $50\times$ the arithmetic of the learned relay. Letting the code rate itself adapt to the channel, over a grid spanning both constellations and rates $1/2$ through $3/4$ plus uncoded transmission, bounds the relay-decoding question further: maximizing delivered goodput shows rate selection matters an order of magnitude more than which relay strategy decodes the code, and the optimal choice at high SNR is to switch coding off entirely, delivering under a third of the channel's unconstrained Shannon capacity even there, a gap set by the finite constellation and residual frame-error rate rather than by relay strategy. That bound is itself conditional on the link being free to choose any blocklength: because a relay that decodes and re-encodes must buffer a full frame twice against a denoising relay's once, imposing a round-trip latency budget locks block-DF out of coding sooner and reopens the relay-strategy gap, reversing it by up to $1.5\times$ at SNRs where the unconstrained comparison had favored the classical relay.

The principal contribution is organised as a ladder of four layers, each removing exactly one assumption from the layer above while the classical–learned comparison is repeated unchanged. At layer 1, the canonical setup, classical processing wins: symbol-wise DF is not improved upon, which is a property of the problem rather than of the training, since on a memoryless matched channel hard slicing is the optimal per-symbol relay function. Layer 2 adds channel memory in the form of an unknown three-tap intersymbol-interference filter and holds channel knowledge fixed. Here the memoryless classical relays fail outright, their error rate saturating and then rising with transmit power as hard decisions propagate the interference, while a windowed learned relay of 169 parameters restores the link; given the same filter, a Viterbi maximum-likelihood sequence detector remains 1 to 1.5 dB ahead of it, so what this layer separates is the function class, and what the learned relay buys is constant cost per symbol against a trellis cost exponential in channel memory. Layer 3 degrades the channel estimate to a finite pilot budget, and the classical receiver's advantage is found to be real but purchased: it holds down to roughly twenty pilots and is lost by ten. Layer 4 removes the channel family altogether, redrawing the channel every block, and the learned relay leads the best blind classical equaliser while avoiding the instability of decision-directed sequence estimation.

The results delineate operating regimes rather than declaring a winner. Where the assumed channel model matches the true channel, classical processing is not improved upon and should be used. Where the channel is unknown, or its estimate is degraded below roughly ten pilot symbols, or no per-block estimate is available at all, the learned relay is the option that continues to work, at fixed cost per symbol and with no identification stage – generalizing across realizations of the trained-on channel family rather than to channel families never presented during training.

Acknowledgments

I would like to thank my supervisors, Dr. Anatoly Khina and Prof. Raja Giryes at Tel Aviv University, for their guidance and support throughout this research. This work was conducted in the School of Electrical and Electronic Engineering, Tel Aviv University.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

Relay communication is a key technique in contemporary wireless systems that uses intermediate nodes between a source and a destination to extend coverage, enhance reliability, and increase throughput. Classical relay strategies such as amplify-and-forward (AF) and decode-and-forward (DF) have been extensively analyzed and are widely used as baseline protocols; in several canonical relay-channel models, block-based DF is capacity-achieving or capacity-approaching under asymptotically long codewords [1, 2, 3, 4, 5]. As clarified in Remark 4.2, these results concern coded block operation and do not directly characterize the uncoded, symbol-wise setting studied in this thesis.

Recent advances in machine learning have motivated the application of neural networks to physical-layer communication problems, including detection, decoding, and end-to-end system optimization. Neural models can approximate nonlinear mappings that are analytically intractable and have been reported to offer advantages in low-SNR or non-ideal regimes.

This motivates the central question of this thesis: *under what conditions can neural network–based relays improve upon or complement classical relay strategies in two-hop relay communication?*

This thesis answers that question within a unified simulation framework in which the relay function is the only variable. Nine relay methods are evaluated: two classical relay strategies (`AF` and `DF`) and seven neural relays spanning supervised learning (`MLP`, SNR-adaptive `Hybrid` relay), generative modeling (variational autoencoder `VAE` and conditional GAN, `cGAN`, with `WGAN-GP`), and sequence-based architectures (`Transformer`, `Mamba-S6`, and `Mamba-2 SSD` with structured state-space duality).

The objective is not to propose a new relay design but to answer the question above with a boundary rather than a verdict: to say where learning helps, where it does not, and why. Existing studies typically vary several things at once, a set of architectures and a set of channel models together, which makes it difficult to attribute a difference in error rate to model structure, to parameter count, or to the channel.

Method: one assumption at a time. This thesis answers the question by holding the comparison fixed and relaxing the receiver's assumptions in a monotone sequence. The same relays, the same channel, the same metric and the same simulation budget are used at every step; the only thing that changes from one step to the next is a single assumption that is withdrawn. Four such layers are examined, and reading their outcomes in order *is* the answer to the central question.

At **layer 1** the channel is memoryless and perfectly known, which is the canonical setup of Chapter 5. Classical processing is not improved upon here, and the reason is structural rather than a matter of training effort: on a memoryless matched channel the optimal per-symbol relay function is hard slicing, which is precisely what decode-and-forward already does at no parameter cost. **Layer 2** (Chapter 7) adds channel memory and holds channel knowledge fixed. The memoryless classical relays now fail outright rather than degrade, and a windowed learned relay restores the link, but a sequence detector given the true channel is still ahead of it; what this layer isolates is therefore the *function class*, together with the cost of realizing it. **Layer 3** degrades the channel estimate to a finite pilot budget, and the classical receiver's advantage turns out to be real but purchased: it survives while enough pilots remain and is lost when they do not. **Layer 4** removes knowledge of the channel family altogether, redrawing the channel for every block, and here the learned relay leads.

The shape of that sequence is the contribution. What degrades from layer to layer is not the classical algorithms, which remain optimal for whatever channel they are handed, but their *preconditions*. The learned relay never becomes a better detector; it becomes the only one whose requirements are still satisfied. Hypotheses H1–H5 are settled at layer 1 and its modulation branch, and H6 at layers 2–4.

Scope of this thesis. This thesis studies one communication problem, held unchanged throughout, the classical source–relay–destination (two-hop) link, under *one canonical setup*: SISO on both hops, i.i.d. Rayleigh fading, complex baseband, QPSK, uncoded BER. The only axis varied is the *relay function*. BPSK appears solely on the AWGN channel used to calibrate the simulator against closed-form error probabilities. A scoped extension to 16-QAM (Chapter 6) and a dedicated unknown-channel contribution (Chapter 7) are included; all other departures from the canonical setup are identified as future work in Chapter 8:

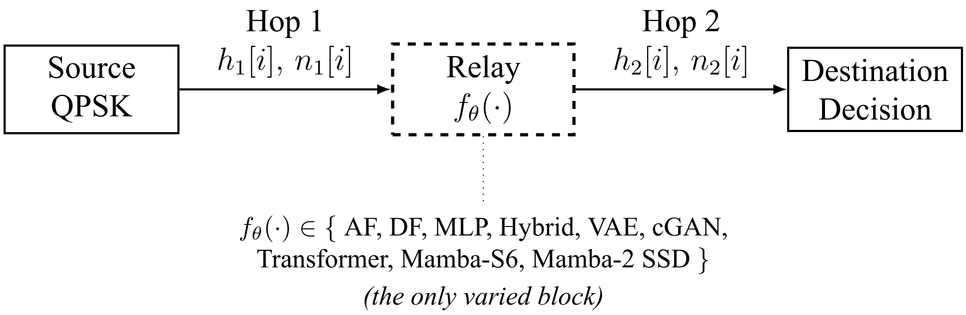

Figure 1.1: Canonical two-hop relay benchmark used throughout the thesis.The source, the channel model on both hops, and the destination are the same in every experiment; only the relay processing function $f_\theta(\cdot)$ (dashed) is varied among the nine candidate strategies. The fading itself is not static: $h_1[i]$ and $h_2[i]$ are redrawn every symbol. Observed BER differences are therefore attributable to the relay strategy alone.

Table 1.1: Scope of the thesis: the canonical setup (QPSK on Rayleigh), the 16-QAM extension, and the unknown-channel contribution. No other topology, relay protocol, or performance metric is evaluated in the main line of this work; all such variations are deferred to future work.

Axis	**Canonical value (future work in parentheses)**
Topology	SISO on both hops *(future work: MIMO second hop)*
Channel	i.i.d. Rayleigh fast fading (canonical), applied identically on both hops; AWGN solely as the analytical baseline the simulator is calibrated against (Section 5.2); unknown, mismatched, and time-varying channels are studied in the dedicated contribution (Chapter 7) *(future work: Rician and other channel families)*
Relay Function *(the studied axis)*	*Classical:* AF, DF. *Learned:* MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD
Modulation	Each constellation is paired with one channel, giving three configurations in total (Table 4.1): BPSK on AWGN (calibration), QPSK on Rayleigh (canonical, Chapter 5), and 16-QAM on Rayleigh (scoped extension, Chapter 6); QPSK is additionally re-checked under the unknown-ISI channel of Chapter 7 *(future work: 16-PSK, 64-QAM and beyond)*
Performance Metric	Uncoded BER vs. SNR, Monte-Carlo averaged

Only the two classical relay strategies evaluated in this thesis, amplify-and-forward and symbol-wise decode-and-forward, are treated as part of the studied system. Material in the literature review that lies outside the configuration above, such as the general relay channel with a direct link and its capacity bounds, is background context and is not evaluated here.

1.1 System Model

System model studied in this thesis. Throughout this work, a single canonical setup underlies the main line of argument: a half-duplex two-hop link with a single relay and *no* direct source-to-destination path, in which the source S transmits during a listening phase to the relay R, and the relay then transmits during a transmission phase to the destination D. Both hops are single-input single-output (SISO) complex-baseband channels with i.i.d. Rayleigh *fast* fading, a fresh realization $h \sim \mathcal{CN}(0, 1)$ per channel use, known perfectly at the corresponding receiver. The modulation is QPSK, and the performance metric is the *uncoded* bit-error rate (BER) as a function of SNR, averaged by Monte Carlo over the fading ensemble. Because the channel is complex, so is the constellation: QPSK places one bit on each axis, and each receiver applies coherent compensation ($y \cdot h^*/|h|$) per axis, so both dimensions the channel acts in are used and nothing is discarded. The learned component of the system is confined to the *relay processing function* $f_\theta(\cdot)$, the block for which AF and DF are the classical alternatives; the source, the channel model, and the destination demodulator are the same in every experiment, so observed BER differences are attributable to the relay strategy alone. The real-valued AWGN channel is used only as the analytical calibration limit that validates the simulator against closed-form BER (Section 5.2); it is paired with BPSK, the one-dimensional constellation that channel's real axis carries, and it is never the channel the thesis draws conclusions from. A second, independent calibration is available on the canonical channel itself: symbol-wise DF's measured BER matches the closed-form two-hop composition to within one percent at every tabulated SNR (Table 5.4), once the QPSK constellation's E_s/N_0-to-E_b/N_0 conversion is applied. Beyond the canonical core, the thesis makes two further contributions of unequal weight. A scoped extension (Chapter 6) takes the further step to 16-QAM, where per-axis I/Q processing breaks down, including a joint 2D N-class classification formulation of the relay. A smaller, supplementary study (Sections 5.5–5.5.5) revisits the uncoded/symbol-wise scope of the canonical setup itself: it adds a genuine information-theoretic block-DF baseline, coded with a rate-1/2 convolutional code and swept over constraint length on both QPSK and 16-QAM, together with soft-decision counterparts of both the classical and the learned relay, to measure – rather than only assert, as Remark 4.2 does – where coding does and does not change the standing of symbol-wise DF against the learned relays. Its conclusion is architectural rather than algorithmic: a relay that decodes the code spends the code's redundancy before it transmits, so its residual errors cannot be repaired at the destination, which is why a relay that merely denoises can end up ahead of one built on an optimal decoder. That advantage is then bounded, not erased, once the code rate is itself allowed to adapt to the channel: maximizing goodput over a modulation-and-coding grid shows the choice of rate dominates the choice of relay strategy by an order of magnitude or more, so the relay-decoding question this study opened matters far less once the link is doing the more basic thing correctly – and the envelope this adaptation reaches stays well short of the channel's own ergodic Shannon capacity throughout, under a third of it even at the top of the swept range, confirming the achieved goodput is limited by the finite constellation and coding grid rather than by anything left on the table for relay strategy to claim. Imposing a hard latency deadline on top of that adaptive-rate picture reopens the question the rate adaptation had closed: because a relay that decodes and re-encodes must buffer a full frame twice (once to decode, once again at the destination) against a denoising relay's once, a tight enough deadline locks the block-DF relay out of coding altogether while the denoising relay still has headroom, reversing the relay-strategy ordering by up to $1.5\times$ at operating points where the unconstrained comparison had favored the classical relay. The principal experimental contribution (Chapter 7) is a systematic study of learned relaying under *unknown and mismatched* channels that deliberately violate the assumptions of the canonical memoryless model, mapping, along every axis relevant to practical deployment, precisely when a minimal learned relay adds value over the strongest classical baselines and when it does not (H6). That chapter also revisits, under an unknown-ISI channel, whether the QPSK generalization established for the canonical benchmark (Chapter 6) still holds: the memoryless-relay failure mode does, but the ordering between the learned relay and genie-CSI Viterbi MLSE does not (Section 7.1.2). All other departures (other channel families, a MIMO second hop, learned constellations) are identified as future work in Chapter 8. Capacity-theoretic results and the cooperative-diversity order are quoted in Chapter 2 as classical *background only*; they are not the operating point evaluated in the experiments.

Figure 1.1 summarizes this canonical benchmark: the source, the channel model, and the destination are identical in every experiment, while only the relay function $f_\theta(\cdot)$ is varied. "Identical" here refers to the model, not to any particular channel realization: the fading is *fast*, so the coefficients $h_1[i]$ and $h_2[i]$ are redrawn independently for every symbol. Nothing in this thesis is a static or quasi-static link.

This chapter states the canonical setup only at the level needed to follow the argument; the complete system model – per-hop channel equations in both their raw complex and coherently-compensated forms, the SNR convention, and two terminology remarks the rest of this thesis relies on throughout (on the causality of the relay's observation window, and on the symbol-wise versus information-theoretic sense of "decode-and-forward") – is given once, precisely, in Section 4.1 rather than repeated here.

Chapter 2

Background and Related Work

Scope relative to the canonical model. Unless stated otherwise, the relay-processing methods reviewed in this chapter are discussed in the context of the canonical memoryless channel model adopted throughout Chapters 5– 6. Channel-memory effects, sequence detection, and channel models with intersymbol interference (ISI) are intentionally deferred to the unknown-channel study of Chapter 7.

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ outperform the classical relay functions (AF scaling, DF regeneration) by exploiting patterns in the received signal that analytical methods do not capture? This question is particularly relevant at low SNR, where both AF (noise amplification) and DF (decoding errors) have well-known limitations, and where the non-linear decision boundary learned by a neural network may provide a potential advantage.

2.1.1 Amplify-and-Forward (AF)

The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \tag{2.1}$$

The end-to-end SNR for AF relay is the standard harmonic-mean expression [3, 6]: (Equation 2.2)

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} \tag{2.2}$$

This form assumes *CSI-assisted variable-gain* AF, in which the relay knows its instantaneous first-hop gain and normalizes its output power per symbol (Equation 2.1); this is the variant implemented throughout this thesis. The additive $+1$ in the denominator originates from the relay's own noise being re-amplified along with the signal, and it is the term that distinguishes this convention from the fixed-gain (blind, long-term-power-normalized) variant, whose end-to-end SNR takes a slightly different form.

This expression reveals a fundamental limitation: even when one hop has high SNR, the effective SNR is bottlenecked by the weaker hop. Moreover, the relay amplifies both signal and noise from the first hop, leading to noise accumulation at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \text{demod}(y_R), \quad x_R = \text{mod}(\hat{b}_R) \tag{2.3}$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\text{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \tag{2.4}$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.40). For BPSK modulation over AWGN (real-valued noise with variance $1/\text{SNR}$), each hop's BER is $P_e = Q\left(\sqrt{\text{SNR}}\right)$. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 4.2, the "DF" baseline used here is the uncoded symbol-wise slicer, not information-theoretic block-DF; its optimality claims are therefore confined to the uncoded setting.

2.1.3 Other Classical Relay Techniques

AF and DF represent two canonical extremes of the relay processing spectrum: AF performs minimal processing (linear scaling) while DF performs maximal classical processing (full regeneration). Additional techniques (e.g. CF, EF, CoF) occupy intermediate points along this spectrum. In this thesis, the AI-based relay strategies are designed to *learn* the relay processing function from data, potentially discovering strategies that subsume or outperform these fixed classical schemes. The focus on AF and DF as classical baselines is therefore deliberate: on the canonical matched-channel benchmark they bracket the classical design space and serve as representative reference baselines against which the learned strategies are evaluated; they should not be read as strict lower and upper performance bounds, since under channel mismatch a learned relay can outperform DF (Chapter 7). These additional relay strategies are reviewed for completeness; they are not simulated in this thesis because the studied system assumes a two-hop link without a direct source-destination path.

2.2 Machine Learning in Wireless Communication

The application of machine learning to physical-layer wireless communication has gained significant momentum in recent years, driven by the ability of neural networks to learn complex non-linear mappings directly from data. Deep learning has been applied to channel estimation [7], signal detection [8], autoencoder-based end-to-end communication [9], and resource allocation [10]. These approaches learn complex mappings from data, potentially outperforming hand-crafted algorithms in scenarios where analytical solutions are intractable or suboptimal.

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, the **universal approximation theorem** [11] guarantees that a feedforward network with a single hidden layer and a non-linear activation function can approximate any continuous function on a compact domain to arbitrary accuracy, given sufficient width. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \to [-1, 1], \quad f^*(y_{i-w}, \ldots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \ldots, y_{i+w}] \tag{2.5}$$

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \tag{2.6}$$

which is a smooth sigmoid-like function that approaches the hard-decision signum function as $\sigma^2 \to 0$ (high SNR). A neural network with a single hidden layer and tanh output can represent this posterior *exactly for a fixed* σ^2, a single tanh unit with the appropriate input scaling suffices. The deployed relay, however, is trained at SNR $\in \{5, 10, 15\}$ dB and evaluated over 0–20 dB, so one set of weights must approximate an entire *family* of posteriors indexed by σ^2 (and, on the fading channel, a mixture over random effective SNRs). That a 169-parameter network approximates this family closely enough to track the classical baselines is an empirical finding of this thesis, not a consequence of the exact single-σ^2 representation.

Second, the **bias-variance decomposition** provides a framework for understanding model complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma^2_{\text{irreducible}} \tag{2.7}$$

The irreducible noise $\sigma^2_{\text{irreducible}}$ represents the minimum achievable MSE, determined by the channel noise. Increasing model complexity (more parameters) reduces bias but increases variance. For the relay denoising task, the target function f^* is simple (essentially a soft threshold), so the bias term is already small for modest networks. Adding parameters beyond this point primarily increases variance (overfitting), explaining the U-shaped relationship between model size and BER observed in this thesis.

2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

Ye et al. [7] demonstrated that deep neural networks can jointly perform channel estimation and signal detection in OFDM systems, achieving near-optimal performance with lower complexity than separate estimation and detection stages. Their key insight was that the DNN implicitly learns the channel's statistical structure, eliminating the need for explicit pilot-based estimation.

Samuel et al. [8] proposed DetNet, an unfolded projected gradient descent network for MIMO detection. By unrolling iterative optimization into a fixed number of neural network layers, DetNet achieves near-maximum-likelihood detection performance with $O(N_t^2)$ complexity instead of the exponential complexity of exhaustive ML search. This demonstrates the power of incorporating domain knowledge (the iterative structure of the detection problem) into the network architecture.

Dorner et al. [9] proposed treating the entire communication system: modulator, channel, and demodulator: as an autoencoder, trained end-to-end to minimize bit error rate. This approach jointly optimizes all components, potentially discovering novel modulation and coding schemes that outperform separately designed modules. The autoencoder paradigm is conceptually related to the relay processing task: both involve learning an encoder-decoder pair that maps through a noisy channel.

Sun et al. [10] applied deep learning to the NP-hard problem of resource allocation in interference networks, demonstrating that a DNN can learn near-optimal power control policies orders of magnitude faster than conventional iterative algorithms.

In the context of relay-assisted communication systems, most existing works employ neural networks primarily for *relay-selection* and *resource-allocation*, particularly in decode-and-forward (DF) relaying scenarios, where learning-based classifiers or predictors are used to identify the optimal relay based on channel state information, signal-to-noise ratios, or energy constraints, while the underlying relay operation remains conventional [12].

A broader perspective on the role of machine learning in wireless communications is provided in [13], which highlights the potential of learning-based methods across multiple protocol layers, including cooperative communications.

Another related line of work exploits the structural similarities between amplify-and-forward (AF) relay networks and neural networks, treating relay nodes as neurons with nonlinear transfer functions and using deep-learning optimization tools to tune relay gains and exploit hardware-induced nonlinearities [14, 15]. While these approaches demonstrate significant gains over classical linear AF schemes, they focus on analog gain optimization rather than symbol-level detection.

Classical cooperative communication frameworks based on AF and DF relaying remain well understood and widely adopted [5], but they rely on fixed linear forwarding or hard-decision rules. Consequently, despite substantial progress, relatively little attention has been devoted to learning the relay's symbol-level input-output mapping itself, motivating neural network-based detection at the relay as a generalized, data-driven alternative to conventional AF and DF processing.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \tag{2.8}$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w + 1$

received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships. The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1\mathbf{w} + \mathbf{b}_1), \quad \hat{x} = \tanh(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2) \tag{2.9}$$

The sliding window formulation ($i \in [i - w : i + w]$) is motivated by the observation that adjacent received symbols share common noise statistics and channel conditions. Under the canonical model (i.i.d. fast fading, white noise), adjacent symbols are statistically independent, so a single-symbol estimator ($w = 0$) is already a sufficient statistic and a window confers no theoretical benefit. The window is therefore not motivated by temporal correlation in the canonical setting; it is retained because it is required by the unknown-channel study of Chapter 7, where a 3-tap ISI channel couples adjacent symbols and the relay must span that memory. Keeping a fixed windowed architecture across all experiments isolates the effect of the channel assumptions from the network structure.

Multi-SNR training is a key design choice: by training on data generated at multiple SNR levels (5, 10, 15 dB in this thesis), the network learns a denoising function that generalizes across operating conditions rather than specializing to a single noise level. This is analogous to training a robust estimator that operates well across a range of noise variances, a concept related to minimax estimation in statistical decision theory.

A critical question in applying neural networks to relay processing is the relationship between model complexity and performance. Prior work has generally assumed that larger models yield better performance, but this assumption has not been rigorously tested in the relay communication context. The bias–variance framework predicts that for a low-complexity target function like relay denoising, performance should plateau or degrade beyond a modest model size: a prediction that this thesis confirms empirically. Understanding this relationship is essential for practical deployment, where relay nodes may have limited computational resources.

2.3 Generative Models for Signal Processing

Generative models offer an alternative paradigm for relay signal processing. Rather than directly learning a denoising function (discriminative approach), generative models learn the distribution of clean signals $p(\mathbf{x})$ and use this knowledge to reconstruct the transmitted signal from noisy observations via Bayes' rule:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \tag{2.10}$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [16] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\text{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \tag{2.11}$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{Latent space regularity}} \tag{2.12}$$

The reconstruction term encourages the decoder to accurately reproduce the input from the latent code, while the KL term regularizes the latent space to be close to the standard Gaussian prior $p(\mathbf{z})$. The β parameter (β-VAE) [17] in Equation 2.12 controls the trade-off between reconstruction quality and latent space smoothness: $\beta < 1$ prioritizes reconstruction (beneficial for the relay task where accurate signal recovery is paramount), while $\beta > 1$ encourages disentangled latent representations.

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \tag{2.13}$$

Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{2.14}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allowing backpropagation through the encoder.

For relay signal processing, the VAE learns a compressed latent representation of the clean signal manifold. During inference, the noisy received signal is encoded into the latent space, and the decoder maps it back to a denoised estimate. The regularized latent space provides implicit denoising: noisy inputs that map to regions far from the learned signal manifold are pulled back toward high-probability regions during decoding. However, the stochastic sampling introduces additional variance into the estimate, which can be harmful for the deterministic BPSK denoising task: a trade-off that manifests as the consistent VAE under-performance observed in this thesis.

2.3.2 Conditional Generative Adversarial Networks

Conditional GANs (cGANs) [18] learn signal denoising through adversarial training, building on the foundational GAN framework [19]. The original GAN formulates generative modeling as a two-player minimax game between a generator G and a discriminator D: (Equation 2.15)

$$\min_G \max_D \; \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))] \tag{2.15}$$

At the Nash equilibrium, G generates samples indistinguishable from real data and D outputs 1/2 everywhere. The conditional variant conditions both G and D on auxiliary information (the noisy signal $\mathbf{y}$), enabling the generator to learn a noise-conditioned mapping.

A fundamental challenge with the original GAN objective is *training instability*: the Jensen-Shannon divergence that the discriminator implicitly estimates can produce vanishing gradients when the generator distribution and data distribution have disjoint supports (which is common early in training). The *Wasserstein GAN (WGAN)* addresses this by replacing the JS divergence with the Earth Mover (Wasserstein-1) distance: (Equation 2.16)

$$W(p_{\text{data}}, p_G) = \inf_{\gamma \in \Pi(p_{\text{data}}, p_G)} \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\gamma}[||\mathbf{x} - \mathbf{y}||] \tag{2.16}$$

The Wasserstein distance provides meaningful gradients even when distributions do not overlap, enabling stable training. By the Kantorovich-Rubinstein duality, the Wasserstein distance can be computed via a supremum over 1-Lipschitz functions, which the critic network approximates. The *gradient penalty (GP)* formulation [20] enforces the Lipschitz constraint softly by penalizing the gradient norm of the critic at interpolated points: (Equation 2.17)

$$\text{GP} = \mathbb{E}_{\hat{\mathbf{x}}\sim p_{\hat{\mathbf{x}}}}\left[(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2\right] \tag{2.17}$$

where

$$\hat{\mathbf{x}} = \epsilon \mathbf{x}_{\text{real}} + (1-\epsilon)\mathbf{x}_{\text{fake}}, \epsilon \sim \text{Uniform}(0,1). \tag{2.18}$$

The full WGAN-GP training objectives used in this thesis are:

$$\mathcal{L}_G = -\mathbb{E}[D(G(\mathbf{y},\mathbf{z}),\mathbf{y})] + \lambda_{L_1}\|\hat{\mathbf{x}} - \mathbf{x}\|_1 \tag{2.19}$$

$$\mathcal{L}_D = \mathbb{E}[D(G(\mathbf{y},\mathbf{z}),\mathbf{y})] - \mathbb{E}[D(\mathbf{x},\mathbf{y})] + \lambda_{\text{GP}} \cdot \text{GP} \tag{2.20}$$

The L_1 reconstruction loss in the generator objective serves a dual purpose: it provides a strong sample-level reconstruction signal (complementing the adversarial signal which provides a distributional match), and it prevents *mode-collapse* (the generator converging to a single output regardless of input). For relay denoising, the L_1 term dominates at $\lambda_{L_1} = 100$, making the cGAN behave primarily as a supervised denoiser with an adversarial regularizer that encourages outputs to lie on the manifold of clean signals.

2.3.3 Generative vs. Discriminative Paradigms for Relay Processing

The application of generative models to relay processing has, to the best of our knowledge, not been extensively studied. The key question is whether the generative inductive bias (learning the data distribution rather than just the input-output mapping) provides any advantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1,+1\}$), suggesting that the generative overhead may not be justified. This thesis compares VAE-based relay processing against classical and supervised learning methods on a common benchmark, and the results show that the generative paradigm provides no significant advantage for this particular task. A conditional GAN relay is implemented and described in Chapter 4 but is excluded from the reported comparison on cost grounds, so no claim is made about it here. On the canonical setup the VAE tracks the supervised MLP and symbol-wise DF rather than trailing them, reaching 0.00972 at 20 dB against DF's 0.00972 and the MLP's 0.00992 (Table 5.4), and it stays inside the feedforward group at every SNR in the equal-parameter comparison as well (Table 5.5). The generative bias is therefore neither an advantage nor a handicap here: it costs parameters and training time to arrive where a 169-parameter regressor already is. One qualification remains, unchanged by this: the relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic variant that substitutes the posterior mean $\boldsymbol{\mu}$ at test time was not evaluated, so nothing here bears on whether that variant would do better.

2.4 Sequence Models: Transformers, State Space Models

Recent advances in sequence modeling have produced two competing paradigms with distinct computational properties and inductive biases. Both have achieved state-of-the-art results in natural language processing, but their relative merits for physical-layer signal processing (where sequences are real-valued temporal signals rather than discrete tokens) have, to the best of our knowledge, not been previously investigated.

2.4.1 Transformers and the Attention Mechanism

Transformers [21] use multi-head self-attention to capture global dependencies in sequences. The core attention mechanism computes a weighted combination of value vectors, where the weights are derived from the compatibility of query and key vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2.21}$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$ are obtained from the input via learned linear projections W^Q, W^K, W^V. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which would push the softmax into saturation regions with vanishing gradients.

Multi-head attention extends this by running h parallel attention heads, each with independent projections, and concatenating their outputs: (Equation 2.22)

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \text{head}_i = \text{Attention}(\mathbf{X}W_i^Q, \mathbf{X}W_i^K, \mathbf{X}W_i^V) \tag{2.22}$$

Each head can attend to different aspects of the input: for signal processing, one head might focus on the immediate neighbors (local denoising) while another captures longer-range patterns. The total parameter count for multi-head attention is:

$$3d_{\text{model}}^2 + d_{\text{model}}^2 = 4d_{\text{model}}^2 \tag{2.23}$$

for Q, K, V projections and the output projection. The attention matrix:

$$\mathbf{A} = softmax(\frac{\mathbf{QK}^T}{\sqrt{d_k}}) \in \mathbb{R}^{n \times n} \tag{2.24}$$

computes pairwise interactions between all n positions, yielding a time and memory complexity of: $O(n^2)$. For the 11-symbol relay window used in this thesis, this is negligible ($11^2 = 121$ entries). However, the quadratic cost becomes prohibitive for longer sequences (e.g., $n = 1000$ symbols), motivating linear-time alternatives.

Positional encoding is necessary because the attention mechanism is permutation-equivariant (it treats input positions symmetrically). This thesis uses sinusoidal positional encoding: (Equation 2.25)

$$PE_{(pos,2k)} = \sin(\frac{pos}{10000^{2k/d}}), \quad PE_{(pos,2k+1)} = \cos(\frac{pos}{10000^{2k/d}}) \tag{2.25}$$

which injects position information into the embeddings, enabling the model to distinguish between symbols at different positions in the window.

2.4.2 Structured State Space Models

Structured state space models (SSMs) [22] are a class of sequence models derived from continuous-time linear dynamical systems. The continuous-time SSM maps an input signal $u(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a latent state $\mathbf{x}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \quad y(t) = \mathbf{C}\mathbf{x}(t) + Du(t) \tag{2.26}$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ governs the state dynamics, $\mathbf{B} \in \mathbb{R}^{N \times 1}$ controls input injection, $\mathbf{C} \in \mathbb{R}^{1 \times N}$ is the output projection, and $D \in \mathbb{R}$ is the feedthrough. The connection to classical signal processing is immediate: this is a linear time-invariant (LTI) filter, and the state space dimension N determines the order of the filter (number of poles/zeros in the transfer function).

To process discrete-time sequences, the continuous SSM is **discretized** using a step size $\Delta > 0$. Applying the zero-order hold (ZOH) discretization: (Equation 2.27)

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \approx \Delta\mathbf{B} \tag{2.27}$$

yields the discrete recurrence: (Equation 2.28)

$$\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k, \quad y_k = \mathbf{C}\mathbf{x}_k + Du_k \tag{2.28}$$

The **HiPPO (High-order Polynomial Projection Operators)** framework provides a principled initialization for $\mathbf{A}$: the HiPPO-LegS matrix is designed so that the state $\mathbf{x}_k$ stores a compressed representation of the input history as coefficients of a Legendre polynomial expansion. This initialization enables long-range memory without the vanishing gradient problems of standard RNNs.

The S4 model [22] introduced the key insight that structured (diagonal or low-rank) parameterizations of $\mathbf{A}$ enable efficient computation. With a diagonal $\mathbf{A} = \text{diag}(a_1, \ldots, a_N)$, the recurrence decouples into N independent scalar recurrences, each computable in $O(n)$ time. The total cost is $O(nN)$ for a length-n sequence with state dimension N: linear in sequence length.

2.4.3 Mamba: Selective State Spaces

Mamba [23] extends the SSM framework by making the parameters **input-dependent** (selective), transforming the LTI system into a linear time-varying (LTV) system:

$$\Delta_k = \text{softplus}(W_\Delta u_k + b_\Delta), \quad \mathbf{B}_k = W_B u_k, \quad \mathbf{C}_k = W_C u_k \tag{2.29}$$

where W_Δ, W_B, W_C are learned projection matrices. Crucially, the state matrix is parameterized as $\mathbf{A} = -\exp(A_{\text{log}})$, so that $\mathbf{A}$ is diagonal with strictly negative entries (Hurwitz, $\mathbf{A} \prec 0$) and the discretized transition $\exp(\Delta_k\mathbf{A})$ is a contraction for every $\Delta_k > 0$. This sign convention is what makes the selectivity mechanism well behaved: it lets the model dynamically control which inputs are stored in the state and which are forgotten.

A **large** Δ_k (triggered by a relevant input) drives the transition toward zero,

$$\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A}) \approx \mathbf{0}, \tag{2.30}$$

which resets the state and lets the new input dominate via

$$\bar{\mathbf{B}}_k \approx \Delta_k \mathbf{B}_k. \tag{2.31}$$

Conversely, a **small** Δ_k (triggered by an irrelevant or noisy input) makes the transition approximately the identity,

$$\bar{\mathbf{A}}_k \approx \mathbf{I}, \tag{2.32}$$

thereby preserving the current state and ignoring the input.

For relay signal processing, this selective mechanism is particularly well-suited: the model can learn to attend to the actual signal component of each received sample while suppressing the noise component, effectively implementing an *adaptive filter* whose coefficients are conditioned on the input. This generalizes the Wiener filter, the optimal linear MMSE denoiser under wide-sense-stationary assumptions, which cannot adapt its coefficients on a per-sample basis.

Remark 2.1. These aforementioned adaptive-filtering connections become directly relevant only in the unknown-channel study of Chapter 7; on the canonical memoryless channel the relay reduces to a per-symbol denoiser.

The Mamba architecture wraps the selective S_6 layer in a gated architecture: the input is projected to $2 \times d_{\text{inner}}$ channels (expand factor 2), and split into two branches. The main branch passes through a causal 1D convolution (*Conv1D*) to mix local temporal context, followed by a *SiLU* activation and the S_6 selective scan. The second branch acts as a *SiLU* gate. The two branches are then multiplicatively merged and contracted back to d_{model}. This gating mechanism provides a multiplicative interaction that helps the model learn sharp non-linear decision boundaries.

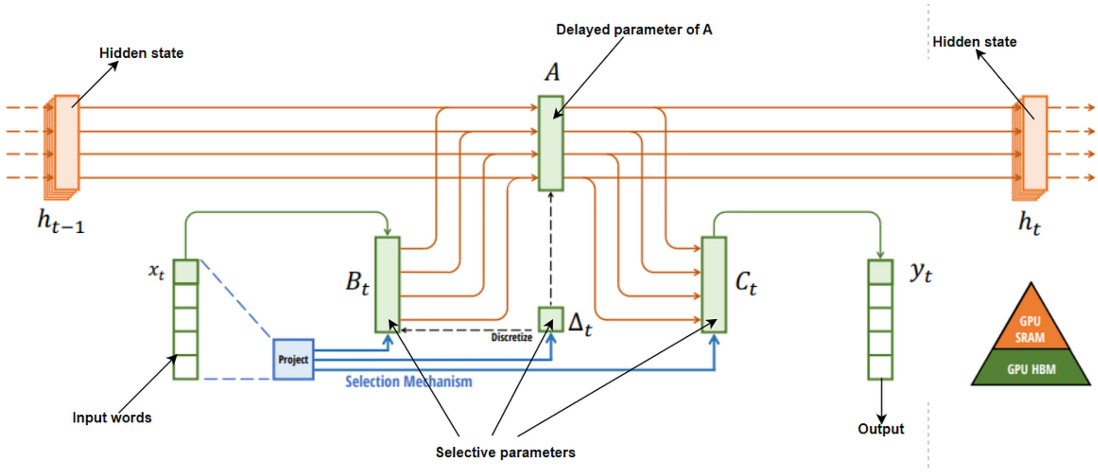

Figure 2.1: Mamba: Linear-Time Sequence Modeling with Selective State Spaces. Structured SSMs independently map each channel (e.g. $D = 5$) of an input x to output y through a higher dimensional latent state h (e.g. state dimension $N = 4$). Prior SSMs avoid materializing this large effective state (DN times batch size B and sequence length L) through clever alternate computation paths requiring time-invariance: the (Δ, A, B, C) parameters are constant across time. The selection mechanism adds back input-dependent dynamics, which also requires a careful hardware-aware algorithm to only materialize the expanded states in more efficient levels of the GPU memory hierarchy.

2.4.4 Mamba-2: Structured State Space Duality

Mamba-2 (SSD) [24] reformulates the selective state space model through an algebraic duality between linear recurrences and structured matrix multiplications. The key theoretical insight is that the SSM output can be expressed as:

$$y_i = \sum_{j \le i} \mathbf{C}_i^\top \left(\prod_{k=j+1}^{i} \bar{\mathbf{A}}_k \right) \mathbf{B}_j u_j \tag{2.33}$$

Defining the **SSM matrix** $M \in \mathbb{R}^{n \times n}$ with entries:

$$M_{ij} = \begin{cases} \mathbf{C}_i^\top \bar{\mathbf{A}}_{j+1:i} \mathbf{B}_j & i \ge j \\ 0 & i < j \end{cases} \tag{2.34}$$

the output is $\mathbf{y} = M \cdot (\mathbf{B} \odot \mathbf{u})$. The matrix M is lower-triangular (causal) and **semi-separable** (each entry factors as an outer product of left and right vectors with a diagonal product in between). This algebraic structure enables efficient chunk-parallel computation:

1. **Divide** the sequence into chunks of length L.
2. **Within each chunk**, build the $L \times L$ local SSM matrix and compute the output via

a single batched matrix multiply.

3. **Between chunks**, propagate the state once per chunk (rather than once per time step).

The per-chunk cost is $O(L^2N)$ for matrix construction and $O(L^2d)$ for the matrix-vector product, where N is the state dimension and d is the model dimension. With n/L chunks, the total cost is $O(n \cdot L \cdot N + n \cdot L \cdot d)$, which for small L (e.g., $L = 8$ or $L = 32$) is effectively $O(n)$ with a favorable constant. The critical advantage over S_6 is that the intra-chunk computation is **fully parallel**: a batched matmul rather than a sequential scan: making it much faster on GPUs where parallelism is cheap and sequential operations incur kernel-launch overhead.

The "*duality*" in the name refers to the equivalence between the recurrent view (state propagation) and the matrix view (structured matmul): the same computation can be expressed either way, and the implementation can choose whichever is more efficient for the hardware and sequence length. This thesis demonstrates this duality empirically: at $n = 11$, the recurrent S_6 view is faster; at $n = 255$, the matrix SSD view dominates.

Mamba-2 can be viewed as a streamlined evolution of Mamba-1: it restructures the block so that the state-space parameters are formed upfront (rather than being generated sequentially along the SSM pathway), and it further stabilizes training by adding an extra normalization stage in the spirit of *NormFormer* [25]. In addition, Mamba-2 emphasizes more parallel, consolidated parameter projection (e.g., producing multiple SSM-related tensors via fewer projections) and reduces per-head redundancy by sharing the B and C projections across heads, conceptually similar to multi-value attention. Figure 2.2 presents a comparison of Mamba-S6 and Mamba-2 SSD [24]:

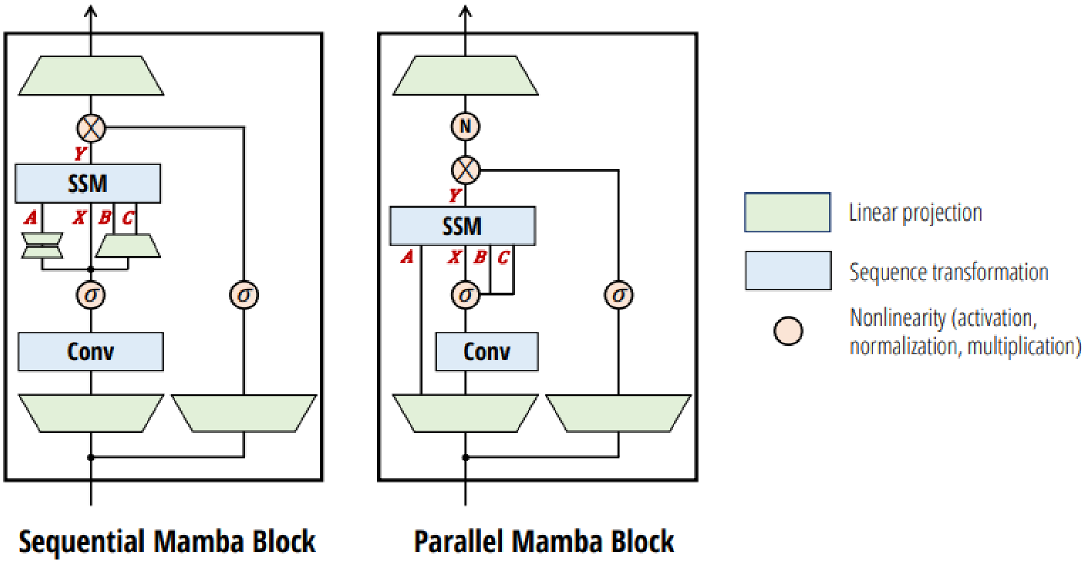

Figure 2.2: The Mamba-2 block simplifies the Mamba block by removing sequential linear projections; the SSM parameters A, B, C are produced at the beginning of the block instead of as a function of the SSM input X. An additional normalization layer is added as in *Norm-Former* [25], improving stability. The B and C projections only have a single head shared across the X heads, analogous to multi-value attention (MVA).

2.4.5 State Space Models for Signal Processing

The application of state space models to physical-layer signal processing is, to the best of our knowledge, largely unexplored. Classical signal processing relies extensively on LTI state space models (Kalman filters, Wiener filters), and the SSM framework provides a natural neural extension of these classical tools. The connection is direct: a trained SSM with fixed (input-independent) parameters implements a learnable linear filter, while the selective Mamba variant implements a learnable *adaptive* filter. For the relay denoising task, this means Mamba can potentially learn the optimal filter structure from data, without requiring manual specification of filter order, cutoff frequencies, or adaptation algorithms.

This thesis presents, to the best of our knowledge, the first comparison of Mamba-S6, Mamba-2 SSD, and Transformers for relay communication, demonstrating that state space models are well suited to this domain and that the SSD formulation of Mamba-2 yields significant speed advantages at longer context lengths.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.35)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{\text{SNR}_{\text{linear}}} \tag{2.35}$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols. The noise variance is inversely proportional to the linear SNR.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/\text{SNR}$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.36)

$$P_e^{\text{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{\text{SNR}}\right) = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{\text{SNR}}{2}}\right) \tag{2.36}$$

where $Q(x) = \frac{1}{2}\text{erfc}(x/\sqrt{2})$ is the Gaussian Q-function and SNR $= P_s/\sigma^2$ is the ratio of signal power to noise variance.

Remark 2.2 (Noise convention)**.** The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes that the one-sided noise PSD is N_0, giving a baseband noise variance of $N_0/2$ per real dimension. In our AWGN implementation the noise is purely real with variance $\sigma^2 = P_s/\text{SNR}$, so $N_0 = 2\sigma^2 = 2P_s/\text{SNR}$ and $E_b/N_0 = \text{SNR}/2$. Substituting yields $Q(\sqrt{2 \cdot \text{SNR}/2}) = Q(\sqrt{\text{SNR}})$, which is the expression used throughout this work.

By contrast, the fading channels add **complex** Gaussian noise with total power $1/\text{SNR}$ (i.e. $\sigma_I^2 = \sigma_Q^2 = 1/(2\,\text{SNR})$) and extract the real part after ZF equalization, halving the effective noise variance per decision dimension. This naturally introduces a factor of 2 inside the Q-function argument for all fading-channel BER expressions (the fading-channel analysis above), making them consistent with the standard textbook forms.

2.5.1.2 Two-hop AF BER

For amplify-and-forward with equal-SNR hops, the effective end-to-end SNR is:

$$\mathrm{SNR}_{\mathrm{eff}}^{\mathrm{AF}} = \frac{\mathrm{SNR}_1 \cdot \mathrm{SNR}_2}{\mathrm{SNR}_1 + \mathrm{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \tag{2.37}$$

where $\gamma = \mathrm{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\mathrm{AF}} = Q\left(\sqrt{\mathrm{SNR}_{\mathrm{eff}}}\right) \tag{2.38}$$

under the same real-noise convention as the single-hop case (Remark 2.2). At high SNR,

$$\mathrm{SNR}_{\mathrm{eff}} \approx \frac{\gamma}{2} \tag{2.39}$$

confirming the well-known 3 dB penalty of AF relaying.

2.5.1.3 Two-hop DF BER

For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination if exactly one hop introduces an error:

$$P_e^{\mathrm{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \tag{2.40}$$

With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\mathrm{AWGN}}$), this becomes

$$P_e^{\mathrm{DF}} = 2P_e(1 - P_e) \tag{2.41}$$

At high SNR, $P_e \ll 1$ and

$$P_e^{\mathrm{DF}} \approx 2P_e \tag{2.42}$$

i.e., the two-hop penalty is approximately a factor of 2 in BER.

2.5.2 Rayleigh Fading Channel: Theoretical Analysis

The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the signal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0, 1), \quad n \sim \mathcal{CN}(0, \sigma^2) \tag{2.43}$$

The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \tag{2.44}$$

with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} = y/h$) becomes $\gamma_h = |h|^2 \cdot \mathrm{SNR}$, which is exponentially distributed.

2.5.2.1 Single-hop BER

Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of γ_h yields the closed-form [26, Eq. 14-4-15] (Equation 2.45):

$$P_e^{\text{Rayleigh}} = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right) \tag{2.45}$$

where $\bar{\gamma} = \mathrm{SNR}$ is the average SNR. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \tag{2.46}$$

This $1/\mathrm{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{\gamma}) \sim e^{-\gamma/2}$), explaining why Rayleigh fading is significantly more challenging. The channel is **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR. Combating this diversity limitation is the primary motivation for the relay architectures studied in this thesis.

2.5.2.2 Two-hop DF BER

The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF,Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \tag{2.47}$$

Chapter 3

Research Objectives

3.1 Main Objective

To systematically evaluate and compare classical and AI-based relay strategies for two-hop communication on a single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER) and to determine the most effective relay strategy as a function of SNR regime and computational constraints.

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, a systematic comparison of relay processing paradigms (spanning classical signal processing, supervised learning, generative modeling, and modern sequence architectures) is, to the best of our knowledge, absent from the literature. The following specific gaps motivate this thesis:

1. **Gap 1: No cross-paradigm relay comparison.** Existing work on AI-based relay processing focuses on individual architectures in isolation. Ye et al. [7] demonstrated DNNs for OFDM detection, Dorner et al. [9] explored autoencoders for end-to-end communication, and Samuel et al. [8] applied unfolded networks to MIMO detection. However, no study systematically compares supervised feedforward networks, variational autoencoders, adversarial networks, attention-based Transformers, and state space models (Mamba) for the same relay denoising task under controlled conditions. Without such a comparison, practitioners cannot make informed architecture selections.

2. **Gap 2: Model complexity-performance relationship is unknown.** It is commonly assumed that larger neural networks yield better performance. However, the relay denoising task (recovering a BPSK symbol from a noisy observation) is a low-dimensional mapping problem. The theoretical minimum description length for this

mapping is small (a soft threshold function), suggesting that large models may overfit. No prior work has rigorously characterized the complexity-performance trade-off for relay processing, nor has a normalized (equal-parameter) comparison been conducted to isolate architectural effects from capacity effects.

3. **Gap 3: State space models for physical-layer signal processing.** The Mamba architecture [23] and its Mamba-2 SSD variant [24] have demonstrated strong performance in NLP and genomics, but their application to relay-level physical-layer signal processing remains, to the best of our knowledge, unexplored. The theoretical connection between SSMs and classical adaptive filters makes this a natural research direction, but empirical validation is needed.

This thesis addresses all three gaps through a unified framework that implements and compares nine relay strategies on the canonical setup, with both original and normalized parameter counts, full statistical analysis, and a dedicated complexity study. The following table summarizes the positioning of this work relative to the literature:

Table 3.1: Research positioning: comparison of this thesis against prior work in deep-learning-based relay communication.

Aspect	Prior Work	This Thesis
Relay strategies compared	1–2 (typically AF vs. DF, or one AI method)	9 (2 classical + 7 AI)
AI paradigms	Single (e.g., supervised only)	4 (supervised, generative, adversarial, sequential)
Evaluation discipline	Single ad-hoc configuration	One canonical setup (SISO, Rayleigh, QPSK), chosen in advance and never varied, with AWGN analytical calibration
Complexity analysis	None	Normalized $\sim$3K-parameter comparison + complexity study (U-shaped BER vs. model size)
State space models	None for relay/physical-layer	Mamba-S6, Mamba-2 SSD, with crossover benchmark
Statistical rigor	Often missing	Wilcoxon test, 95% CI, 10 trials per point

Relationship between the canonical study and the extension study. Hypotheses H1–H5 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H6 studies a deliberately different regime in which the canonical model no longer represents the true channel. It therefore serves as a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup; H5 is an architecture-level hypothesis concerning training compute. H6 is the central hypothesis of the thesis's principal contribution (Chapter 7), posed on channels *outside* the canonical family. The higher-order-modulation extension (Chapter 6) additionally probes the generalizability of H1–H3 to QPSK and 16-QAM and is reported descriptively:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-linear denoising learned by the neural network provides an advantage over AF's noise amplification.

   Rationale: At low SNR, the MMSE-optimal (posterior-mean) denoiser of a BPSK symbol from a single AWGN observation, $f^*(y) = \mathbb{E}[x \mid y] = \tanh(y/\sigma^2)$, is a smooth non-linear mapping that differs substantially from AF's linear scaling; a neural network can approximate f^* closely, whereas AF amplifies noise together with the signal. Note that f^* is optimal in the mean-square sense for the relay's local estimate; it is not claimed to be the end-to-end BER-optimal relay function, which is why H1 is posed as an empirical hypothesis rather than a theorem.

2. **H2 (DF dominance at high SNR).** The symbol-wise DF relay (Remark 4.2) achieves the lowest BER at medium-to-high SNR (≥ 6 dB), outperforming all AI methods.

   Rationale: At high SNR, $f^*(y) \approx \text{sign}(y)$, which is exactly the symbol-wise DF operation. AI relays introduce a small but non-zero approximation error, so DF should be optimal among the per-symbol relay functions considered in this uncoded setting. This dominance is asserted only within the uncoded, symbol-wise family evaluated here and is not a claim of information-theoretic optimality (cf. Remark 4.2).

3. **H3 (U-shaped complexity curve).** There exists an optimal model size for relay denoising beyond which performance degrades due to overfitting; equivalently, BER is U-shaped in model size (U-shaped in accuracy). Specifically, models with $\sim$100–200 parameters achieve performance comparable to models with 10–100$\times$ more

parameters.

Rationale: The bias-variance analysis (Section 2.2.1) predicts that the low-complexity target function f^* requires minimal model capacity. Excess capacity increases variance without reducing bias.

4. **H4 (Architecture convergence at equal scale).** When all AI models are normalized to the same parameter count (∼3,000), the performance differences between architectures narrow significantly, indicating that parameter count is a more important factor than architectural choice.

   Rationale: All architectures are universal approximators and the target function is simple. At equal capacity, architectural inductive biases provide diminishing returns.

5. **H5 (SSM speed advantage at long context).** Mamba-2 SSD trains significantly faster than Mamba-S6 at longer context lengths ($n \gg 11$) due to chunk-parallel computation, while Mamba-S6 is faster at the short relay window ($n = 11$).

   Rationale: The crossover between sequential S6 ($O(n)$ serial steps) and chunk-parallel SSD ($O(n/L)$ parallel matmuls of size L) depends on the ratio of kernel-launch overhead to arithmetic cost.

Unlike H1–H5, which assume the canonical memoryless channel throughout, H6 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

1. **H6 (Principal contribution, unknown-channel mitigation).** On channels with structure unknown to the receiver chain, memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) fail to relay reliably, exhibiting error floors or severe degradation, whereas the minimal 169-parameter MLP relay restores reliable relaying by learning the impairment from data, approaching the performance of the optimal model-aware detector without requiring knowledge of the channel family.

   Rationale: AF forwards the channel impairment untouched, and symbol-wise DF applies a fixed memoryless zero-threshold decision; neither can compensate structure they do not model. Classical theory does provide a remedy *once the channel family is known*, estimate-then-MLSE (pilot-based channel estimation followed by Viterbi equalization) is optimal for a linear FIR channel and is included as a benchmark. The hypothesis is therefore not that learning beats classical theory, but that a single fixed learned architecture can approach model-aware performance with no channel-family knowledge, no explicit identification stage, and complexity independent of channel memory, and that it remains applicable on impairments (e.g., nonlinear bias) for which linear MLSE is mis-specified. How far this holds, and on which impairment structures it breaks down, is determined empirically in Chapter 7. Conversely, on channels that satisfy the classical assumptions, H2 predicts no learned relay can beat DF, so H6 also delineates the precise boundary of the learned relay's value.

3.4 Specific Objectives

1. **Implement and compare nine relay strategies** spanning four learning paradigms: no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE, cGAN), and sequence modeling (Transformer, Mamba-S6, Mamba-2 SSD).
2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with QPSK, validated against closed-form theory (with AWGN/BPSK as the analytical calibration limit), and probe modulation generalizability in a scoped 16-QAM extension.
3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.
4. **Determine whether state space models outperform attention mechanisms** for relay signal processing by comparing Mamba-S6 and Transformer architectures at both original and normalized parameter counts, and by benchmarking at extended context lengths.
5. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of this study, which is confined entirely to the canonical setup (SISO, Rayleigh fast fading, complex baseband, QPSK, uncoded):

- **Modulation:** The core experiments use BPSK. A scoped extension (Chapter 6) evaluates QPSK and 16-QAM, including the joint 2D N-class classification formulation; 16-PSK, 64-QAM and denser constellations are deferred to future work.
- **Channel knowledge:** *In the canonical core* (Chapters 5–6), perfect CSI is assumed at the receiver and channel estimation errors are not modeled. This limitation is deliberately lifted in the principal contribution: Chapter 7 models finite-pilot least-squares channel estimation (a 200-pilot Viterbi baseline and a swept pilot budget) and, in the blind regime, a fully zero-CSI setting with no channel prior at all.

- **Relay topology:** Single relay, two-hop, half-duplex. Multi-relay and full-duplex configurations are excluded.
- **Topology:** SISO on both hops. Multi-antenna (MIMO) configurations are deferred to future work.
- **Training regime:** Offline training on synthetic data. Online adaptation is not implemented.

These delimitations are chosen to enable a clean comparison of relay processing functions in isolation, without confounding factors from protocol-level or system-level complexity.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER. BPSK appears only on the AWGN calibration channel, and 16-QAM only in the extension of Chapter 6; the three pairings are set out in Table 4.1.

Scope of the canonical system model. The system model defined in this chapter serves as the canonical benchmark used throughout the core experiments of the thesis (Chapters 5–6). It assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. Accordingly, all relay comparisons in the canonical study concern per-symbol processing and do not require sequence detection: the current observation $y_R[i]$ is already a sufficient statistic for detecting $x[i]$, and the classical AF and DF baselines operate per symbol. Some neural architectures nevertheless receive a finite window of neighboring observations (Section 4.2); this is an architectural input convention for the evaluated feedforward and sequence models, and under the canonical i.i.d. assumptions it neither introduces channel memory nor supplies additional information about $x[i]$. Channel memory, intersymbol interference (ISI), pilot-based channel estimation, blind equalization, and sequence-detection techniques such as Viterbi MLSE are introduced only later in Chapter 7 as deliberate departures from the canonical assumptions in order to study robustness under model mismatch.

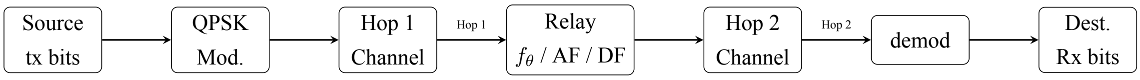

Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between.Each hop applies the channel operator followed by coherent compensation (no equalizer is used in the canonical setup)

Source bits are QPSK-modulated on the canonical setup, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination; no equalization is involved. The same chain carries BPSK on the AWGN calibration channel and 16-QAM in the extension, with only the modulator and demodulator changing. Both hop channels are known perfectly at their respective receivers, per the conventions of Chapter 1.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \tag{4.1}$$

4.1.1 Modulation

Three constellations appear in this thesis. The base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \tag{4.2}$$

with hard-decision demodulation $\hat{b} = \mathbb{1}[\hat{x} < 0]$ at the destination. QPSK and 16-QAM are defined in Section 4.5. All three are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1, 2$ and 4 respectively, giving $E_b = E_s/k$. Further constellations are future work.

4.1.2 Hop Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source → Relay), listening phase:

$$y_R[i] = h_1[i]\,x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0, 1), \quad n_1[i] \sim \mathcal{CN}(0, \sigma^2) \tag{4.3}$$

Slot 2 (Relay → Destination), transmission phase:

$$y_D[i] = h_2[i]\, x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0,1), \quad n_2[i] \sim \mathcal{CN}(0,\sigma^2) \tag{4.4}$$

where $x[i]$ is the complex symbol transmitted by the source at time i (QPSK on this canonical channel; the real BPSK alphabet $\{-1,+1\}$ applies only to the AWGN calibration special case below), $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i}),$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function; for AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$.

Each hop applies a channel function followed by optional equalization (Equation 4.5):

$$y = \mathcal{H}(x, \text{SNR}) + n \tag{4.5}$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case is retained for analytical calibration of the simulator). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel. $\mathcal{H}(\cdot)$ is applied identically on Hop 1 and Hop 2 at the same SNR: a configuration never mixes channel families across hops. A configuration is therefore specified completely by the pair (constellation, channel), and it is that pair which names each experiment below.

Coherent compensation. Because the fading coefficient is known at each receiver, coherent compensation ($y \cdot h^*/|h|$ followed by $\Re\{\cdot\}$) reduces each hop to an equivalent real channel with a random per-symbol effective gain $|h[i]|$,

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \qquad \tilde{y}_D[i] = |h_2[i]|\, x_R[i] + \tilde{n}_2[i], \tag{4.6}$$

with $\tilde{n}[i] \sim \mathcal{N}(0,\sigma^2)$ real per axis. Because the channel is complex, the canonical constellation is two-dimensional: QPSK places one bit on each axis, so both dimensions the channel acts in are used and the relay processes the pair; the real reduction above then applies per axis rather than discarding one. BPSK, being one-dimensional, is used only on the real AWGN calibration channel of Section 5.2, where the closed forms it validates are defined; it is never transmitted over the complex fading channel, which would require discarding the quadrature component. The one constellation that breaks per-axis relay processing, 16-QAM, is treated separately in Chapter 6. The *AWGN calibration limit* used to validate the simulator (Chapter 2) is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

The noise terms are independent and identically distributed with variance

$$\sigma^2 = \frac{P_s}{\gamma} \tag{4.7}$$

where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for every constellation studied, per the normalization of Section 4.1.1) and γ denotes the per-hop linear signal-to-noise ratio (SNR), equal on both hops.

4.1.3 Evaluated Configurations

This thesis evaluates exactly three (constellation, channel) configurations, listed in Table 4.1. They are not a cross product. The governing principle is that **a real channel is paired with a real constellation and a complex channel with a complex constellation**: the signal space the constellation lives in and the signal space the channel acts on are the same space. AWGN with BPSK is the real case, both living on the line; Rayleigh with QPSK and with 16-QAM are the complex cases, both living in the plane, where the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates as well as scales and the constellation has a phase for it to act on.

The pairings excluded by this principle are precisely the ones that would force the simulation to patch over a mismatch. A real constellation on the complex Rayleigh channel must discard the imaginary part of the equalized sample after dividing by h, throwing away half the plane the channel acted in; a complex constellation on AWGN is not wrong, but it exercises none of what makes the complex channel interesting, since there is no phase rotation to undo. Neither is reported here.

Table 4.1: The three evaluated configurations. The channel shown applies to both hops.

Constellation	Channel	Role	Why this pairing
BPSK (real)	AWGN (real)	Calibration	The real case. AWGN is the only channel here with exact closed forms, $Q(\sqrt{2E_b/N_0})$ single-hop and $2P(1-P)$ two-hop DF, so it is where the simulator can be checked against an exact result rather than only against another simulation, and BPSK is the constellation those closed forms describe.
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis of the plane the channel acts in, so the identical I/Q-split relays process it unchanged.
16-QAM (complex)	Rayleigh (complex)	Extension	The same complex channel, driven to four levels per axis, which breaks the per-axis relay formulation. That break is what Chapter 6 studies, and it is studied on the canonical channel.

The same principle also keeps AWGN in its place. It is retained as a measuring stick and needs only the one constellation whose closed form is being checked; evaluating relays on AWGN at higher orders would invite conclusions from a channel the thesis deliberately draws none from.

A consequence worth stating plainly. The SNR argument sets the noise power from the average *symbol* energy, so it denotes E_s/N_0 in every configuration. Since $E_b = E_s/k$, the same numerical SNR corresponds to a different per-bit energy in each row of Table 4.1: E_b/N_0 equals the stated value for BPSK, is 3.01 dB lower for QPSK and 6.02 dB lower for 16-QAM. The three configurations are therefore *not* on a common abscissa, and BER values must not be read across them without that conversion. Where a comparison across constellations is made, this is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure fair comparison:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \tag{4.8}$$

4.1.4 Relay Processing

On Hop 1 the relay's single receive antenna observes, as shown in Equation 4.9:

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \quad \tilde{n}_1[i] \sim \mathcal{N}(0, \sigma^2) \tag{4.9}$$

where the effective observation follows coherent compensation of the Rayleigh fading coefficient (Section 4.1). Note that the per-symbol effective SNR is $|h_1[i]|^2\gamma$ (a random, exponentially distributed quantity) not a constant; the AWGN calibration limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no inter-stream interference at any stage, and the destination demodulates the Hop-2 output directly with no equalizer.

Remark 4.1 (Realizability of the symmetric window)**.** The canonical relay of this chapter is memoryless: on a memoryless channel $y_R[i]$ is already a sufficient statistic for $x[i]$, so $w = 0$, and AF and symbol-wise DF are strictly causal. A relay with a symmetric window $y_R[i - w : i + w]$, $w > 0$, is *not* part of this system model. It is introduced in Chapter 7 together with the channel that motivates it, a three-tap intersymbol-interference filter, and is defined there. Where that window is used it is not causal, since it reads w samples ahead; it is admissible in the half-duplex protocol only because the relay buffers the whole listening-phase block before transmitting, and it costs a fixed w-symbol processing latency. That non-causality is a limitation of the learned relays, recorded as such rather than justified: a strictly causal past-only window $y_R[i - 2w : i]$ is what a streaming deployment would require, and it is not evaluated in this thesis.

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).

3. the model capacity (from 0 to 26K parameters).

This section describes each strategy's architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF), (ii) supervised-learning models (MLP and hybrid MLP–DF), (iii) generative models (VAE and cGAN), and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Gain amplification as shown in Equation 2.1. AF serves as the minimal-processing classical baseline. It performs no learned processing and simply rescales the received signal, preserving both signal and noise. Its end-to-end effective SNR is given in Equation 2.2.

2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean symbols (BPSK on the AWGN calibration channel; QPSK on the canonical channel; per-axis on 16-QAM). DF serves as the maximal-processing classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 4.2). Its theoretical BER follows the error-composition formula of Equation 2.4.

3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

   where $\mathbf{w} \in \mathbb{R}^5$ is a sliding window of received symbols ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5 \times 24 + 24) + (24 \times 1 + 1) = 169$.

   > **Architecture note:** The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \to [-1, +1]$ from a noisy observation window to a denoised symbol estimate: a classical regression task.

   Design rationale: The tanh output activation naturally constrains the output to $[-1, +1]$, matching the BPSK symbol range. The ReLU hidden layer provides the non-linearity needed to approximate the Bayes-optimal soft threshold $\tanh(y/\sigma^2)$.

With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension: sufficient for the low-complexity denoising task while avoiding overfitting. He initialization is used for ReLU layers to maintain proper gradient flow.

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between MLP (low SNR) and DF (high SNR) based on a learned threshold. Combines the AI advantage at low SNR with DF's low-BER, zero-parameter operation at high SNR. Same 169 parameters as MLP.

   Design rationale: The Hybrid relay is motivated by the observation (confirmed empirically) that AI relays outperform DF only at low SNR, while DF achieves the lowest BER among the evaluated symbol-wise relay strategies at high SNR. By introducing a switching threshold, the Hybrid relay automatically selects the better strategy for each operating condition. The threshold is determined empirically from the training data by finding the SNR at which MLP and DF BER curves cross. This approach requires no additional parameters beyond those of the MLP sub-network.

   Training points at the relay node: SNR = **5, 10, 15 dB** (same trained MLP sub-network as the Minimal MLP relay).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder $(7 \to 32 \to 16 \to \mu, \sigma^2(8))$, decoder $(8 \to 16 \to 32 \to 1)$. Total: 1,777 parameters. Trained with β-VAE loss ($\beta = 0.1$) for 100 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP):** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: $(7 + 8 \to 32 \to 32 \to 16 \to 1)$, Critic: $(1 + 7 \to 32 \to 16 \to 1)$. Total: 2,946 parameters. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$), and L1 reconstruction loss ($\lambda_{\text{L1}} = 100$) for 200 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The high $\lambda_{\text{L1}} = 100$ ensures that the generator is primarily supervised by the reconstruction loss, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 128. Total: 17,697 parameters. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. The 11-symbol window provides the same temporal context as the Mamba-S6/SSD models. With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 128 ($4 \times d_{\text{model}}$) follows the standard Transformer expansion ratio.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies: LayerNorm $\rightarrow$ expand ($32 \rightarrow 64$) $\rightarrow$ split to Conv1D/SiLU/S6 main branch and SiLU gate $\rightarrow$ contract ($64 \rightarrow 32$) $\rightarrow$ residual. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The expand factor of 2 ($32 \rightarrow 64$) doubles the internal dimension during the S6 scan, providing richer state dynamics. The state dimension $N = 16$ means each S6 layer implements a 16th-order adaptive filter: substantially more expressive than classical Wiener or matched filters. LayerNorm and residual connections prevent gradient degradation across layers. The SiLU gating provides multiplicative interactions that help the model learn sharp decision boundaries.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that replaces the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with SiLU gating and residual connections. Total: 26,179 parameters. Each block applies: LayerNorm $\rightarrow$ parallel gate/SSD branches $\rightarrow$ SiLU gate $\rightarrow$ contract ($64 \rightarrow 32$) $\rightarrow$ residual. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained

for 100 epochs with Adam optimizer (lr $= 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The chunk size of 8 is chosen to balance parallelism (larger chunks = fewer sequential inter-chunk passes) against memory cost (the $L \times L$ SSM matrix scales quadratically with chunk size). At the 11-symbol window, this yields 2 chunks (8 + 3), with a single inter-chunk state pass. Gradient clipping at norm 1.0 prevents the exponential blowup of gradients through the cumulative matrix products in the SSD computation. The S4D-style initialization $A_{\log} = \log(1, 2, \ldots, N)$ provides geometrically spaced decay rates, enabling the model to simultaneously capture short-range and medium-range dependencies.

Remark 4.2 (Terminology: symbol-wise DF vs. information-theoretic DF)**.** The term "decode-and-forward" is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [1, 2], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless and, over a single link, can operate reliably at any rate below the first-hop capacity.

The system studied in this thesis is *uncoded*: one QPSK symbol carries two information bits, one per axis, and no outer error-correction code is applied. Consequently, the "DF" baseline throughout this work refers to *symbol-wise slicing*: per-symbol demodulation followed by re-modulation, i.e., $f_{\mathrm{DF}}(y_R[i]) = \mathrm{sign}(y_R[i])$ with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used here, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance. This caveat is measured directly, not left as a standing assumption: Section 5.5 adds a real information-theoretic block-DF baseline (a rate-1/2 convolutional code with soft-decision Viterbi decoding, swept over constraint length and both canonical constellations) and reports where it does and does not beat the symbol-wise DF used elsewhere in this thesis.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000

- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(b_i \neq \hat{b}_i) \tag{4.10}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $N = 10{,}000$ is the number of bits per trial. By the central limit theorem, for large N and moderate BER (say $P_e \geq 10^{-3}$), the per-trial BER estimate is approximately normally distributed with variance $P_e(1 - P_e)/N$.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\mathrm{CI}_{95\%} = \bar{P}_e \pm t_{0.025,M-1} \cdot \frac{s}{\sqrt{M}} \tag{4.11}$$

where $\bar{P}_e$ is the sample mean BER across trials, s is the sample standard deviation, and $t_{0.025,9} = 2.262$ is the critical value of the Student's t-distribution with 9 degrees of freedom. At low BER ($\hat{P}_e \lesssim 10^{-4}$), the normal approximation may become unreliable due to the small number of observed errors; however, in this regime the relay methods converge and the relative ranking is stable.

BER resolution. With $N \cdot M = 100{,}000$ total bits per SNR point, the minimum detectable BER is approximately $1/N = 10^{-4}$ per trial, or 10^{-5} for the aggregate. BER values below this threshold are reported as 0.

Per-experiment simulation budgets. The budget above ($M = 10$ trials $\times$ $N = 10{,}000$ bits $= 100{,}000$ bits per SNR point) applies to the canonical relay comparison and the parameter-normalization/complexity study (E2–E3). Other experiments use different budgets, stated in their respective sections: the channel-model validation (E1, Section 5.2) uses $20 \times 50{,}000 = 1{,}000{,}000$ bits per SNR point for tight validation intervals; the main unknown-ISI study, the flat-channel control, and the QPSK generalization check (Chapter 7, Sections 7.1, 7.1.1, 7.1.2) use $10 \times 100{,}000 = 1{,}000{,}000$ bits per SNR point, matching the project-wide $M = 10$ standard; the composite cascade (Section 7.1.3) likewise uses $10 \times 100{,}000$; and the blind and partial-posterior sub-studies (Sections 7.1.5–7.1.4) use $50 \times 20{,}000 = 1{,}000{,}000$ bits per SNR or operating point. The larger trial count in those last two is deliberate rather than incidental: their hop-1 channel redraws its ISI, amplifier and phase realization for every block, so a trial is a draw from the impairment family and the ensemble average is limited by the trial count, not by bits per trial. Increasing bits per trial sharpens the estimate for one particular channel; only increasing trials averages over the family the study is about. Every experiment therefore uses $M \geq 10$ trials, so the Wilcoxon signed-rank test is attainable throughout (at $M = 5$ its smallest two-sided p is $2^{-4} = 0.0625$, which could not reach $p < 0.05$).

4.3.2 Statistical Significance Testing

Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the test compares the $M = 10$ paired BER observations (Equation 4.12):

$$H_0 : \text{median}(P_{e,A} - P_{e,B}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(P_{e,A} - P_{e,B}) \neq 0 \tag{4.12}$$

The Wilcoxon test is preferred over the parametric paired t-test for two reasons: (i) BER distributions are bounded ($[0, 0.5]$) and potentially skewed, violating the normality assumption, and (ii) the test is robust to outlier trials. The significance level is set at $\alpha = 0.05$ *per comparison*; no multiple-comparison correction (e.g. Holm or Bonferroni, or false-discovery-rate control) is applied across the many relay-pair and SNR-point comparisons made throughout this thesis, so the reported significance claims should be read at the per-comparison level rather than as holding jointly across the full set of tests performed.

4.3.3 Training Protocol

All AI relays are trained **once** at the beginning of each experiment using: - **Multi-SNR training data:** Signals generated at SNR = 5, 10, 15 dB in equal proportions - **Training samples:** 25,000 (supervised), 20,000 (generative), 10,000 (sequence models) - **Single training per channel:** The same trained model is evaluated across all SNR points

The multi-SNR training protocol is critical: training at a single SNR would produce a model that performs well at that SNR but poorly at others (overfitting to a specific noise level). By training at three representative SNR values spanning the low-to-moderate range, the network learns a robust denoising function that generalizes across operating conditions. The SNR values 5, 10, 15 dB were chosen to cover the regime where AI relays provide the most benefit (low-to-medium SNR).

Impact of sparse SNR training grid. Because models are trained at only three SNR points ($\{5, 10, 15\}$ dB) but evaluated over eleven points (0 to 20 dB, step 2), performance reflects two regimes: (i) **interpolation** within the trained span (approximately 4–16 dB), where BER is generally stable, and (ii) **extrapolation** at the edges (0–2 dB and 18–20 dB), where mild degradation may occur due to noise-statistics mismatch. In the present results, this sparse-grid effect is secondary: the relay ranking remains consistent across the full evaluated SNR range.

Training method (what is trained, where it is trained). Training is performed **offline** before BER sweeps. Only the seven AI relays are optimized (MLP, Hybrid's MLP sub-network, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD); AF and DF are analytical baselines and have no trainable parameters. Training data are synthetically generated source/channel pairs under the multi-SNR protocol, with model-specific losses (MSE for supervised relays, ELBO for VAE, WGAN-GP + L1 for cGAN, Adam-based supervised loss for sequence models). Compute placement follows implementation: NumPy models (MLP/Hybrid) train on CPU, while PyTorch models train on CPU or CUDA depending on device availability/configuration. After training, weights are checkpointed and reused across all SNR evaluations; BER curves are then produced in **inference-only** mode without further parameter updates.

Weight initialization. He initialization [27] is used for all layers with ReLU activation ($W \sim \mathcal{N}(0, 2/n_{\text{in}})$), while Xavier initialization is used for layers with tanh activation ($W \sim \mathcal{N}(0, 1/n_{\text{in}})$). These initialization schemes maintain proper gradient magnitude through the network layers, preventing both vanishing and exploding gradients during training.

Optimizer settings. All PyTorch-based models use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ (defaults). The MLP and Hybrid relays use a simple SGD implementation in NumPy with constant learning rate $\eta = 0.01$.

4.3.4 Reproducibility and Weight Management

All experiments use controlled random seeds at both the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results. Specifically, for seed s and trial t, the random state is initialized as seed$(s \cdot 1000 + t)$, ensuring that each trial is independent while the overall experiment is deterministic.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). The checkpoint system supports: - Automatic save after training with architecture metadata - Resume from saved weights with architecture validation - Seed-specific weight directories (e.g., `trained_weights/seed_42/`)

The framework includes 159 automated tests (pytest) covering all modules: channels, modulation, coding, relay strategies, simulation, statistics, and weight management, with 100% pass rate.

Code and reproducibility archive. The complete simulation framework (`relaynet`), every experiment script, and the committed result files each figure and table is generated from are maintained under version control (git) in the `Gilzuk/relaynet2` repository; each numerical claim in this thesis traces to a specific committed script and result file rather than to an unarchived intermediate run, and `verify_thesis_tables.py` in that repository re-derives every numerical table from its source file and flags any mismatch. The exact commit this thesis was typeset from is recorded in the repository's history rather than reproduced here, since stating a fixed hash in the text it describes is self-referential; the repository's tags and commit log identify the revision corresponding to any given PDF build.

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Performance differences between these models could reflect either (a) the superiority of one architecture's inductive bias, or (b) the simple advantage of having more learnable parameters. To disentangle these two effects, all seven AI models were scaled to approximately 3,000 parameters, providing a controlled comparison where model capacity is held constant.

Choice of target parameter count. The target of ~3,000 parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: 5.9× reduction, Mamba-S6: 7.9×, Mamba-2 SSD: 8.7×). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture's hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.2: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture.

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden

Model	Parameters	Scaling Method
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_model (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_model (32→16), d_state (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_model (32→15), d_state (16→6), layers (2→1)

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed **A** matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`.

Unified input window. All 3K models use a window size of 11 (vs. 5 for original MLP/Hybrid, and 7 for original VAE/cGAN), providing a common input context. This ensures that differences in performance reflect architectural inductive biases rather than differences in the amount of input information available.

This normalization isolates the effect of architectural choice from the confound of parameter count, providing insights into which inductive biases are most beneficial for the relay denoising task. If performance converges at equal parameter budgets, the conclusion is that parameter count (not architecture) is the dominant factor. If significant gaps persist, the conclusion is that architectural inductive biases provide meaningful advantages beyond raw capacity.

4.5 Modulation: QPSK and 16-QAM

The core experiments use QPSK, the canonical constellation; BPSK is confined to the AWGN calibration channel of Section 5.2. To test whether the canonical findings generalize to a higher-order constellation, the extension study of Chapter 6 evaluates 16-QAM. This section defines both complex constellations, the I/Q-splitting technique that lets real-valued relays process them, and the joint 2D classification formulation that removes I/Q splitting's structural limitation for 16-QAM.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [26]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \tag{4.13}$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$). The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.3: QPSK Gray-coded symbol mapping: bit pairs to complex symbols.

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\mathrm{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\mathrm{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [26]:

$$P_b^{\mathrm{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\mathrm{BPSK}} \tag{4.14}$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 16-QAM: Gray-Coded Quadrature Amplitude Modulation

16-QAM maps groups of four bits (b_0, b_1, b_2, b_3) to one of 16 complex constellation points arranged on a 4×4 rectangular grid (Figure 4.2c). Each axis (I and Q) uses independent Gray-coded PAM-4 mapping [26]:

$$I = \frac{L(b_0, b_1)}{\sqrt{10}}, \quad Q = \frac{L(b_2, b_3)}{\sqrt{10}}, \quad x = I + jQ \tag{4.15}$$

where $L(\cdot)$ maps bit pairs to PAM-4 levels using Gray coding:

Table 4.4: 16-QAM PAM-4 Gray-coded level mapping for in-phase and quadrature components.

Bit pair	Level	Bit pair	Level
00	$+3$	11	-1
01	$+1$	10	-3

The normalization factor $\sqrt{10}$ ensures unit average symbol power: $E[|x|^2] = \frac{2\cdot(9+1+1+9)}{4\cdot 10} = 1$. Adjacent constellation points differ by one bit (Gray property), minimizing the BER for a given symbol error rate.

Demodulation quantises each received component to the nearest PAM-4 level using decision boundaries at $\{-2, 0, +2\}/\sqrt{10}$ and maps back to bits via the inverse Gray table. The spectral efficiency is 4 bits/symbol.

Theoretical 16-QAM BER. The approximate BER for 16-QAM over AWGN is: (Equation 4.16)

$$P_b^{\text{16-QAM}} \approx \frac{3}{8}\,\mathrm{erfc}\left(\sqrt{\frac{2E_b}{5N_0}}\right) \tag{4.16}$$

At the same E_b/N_0, 16-QAM has a higher BER than BPSK or QPSK due to the reduced Euclidean distance between constellation points. The trade-off is $4\times$ throughput improvement.

4.5.3 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to 16-QAM (16 points on a rectangular grid) illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR.

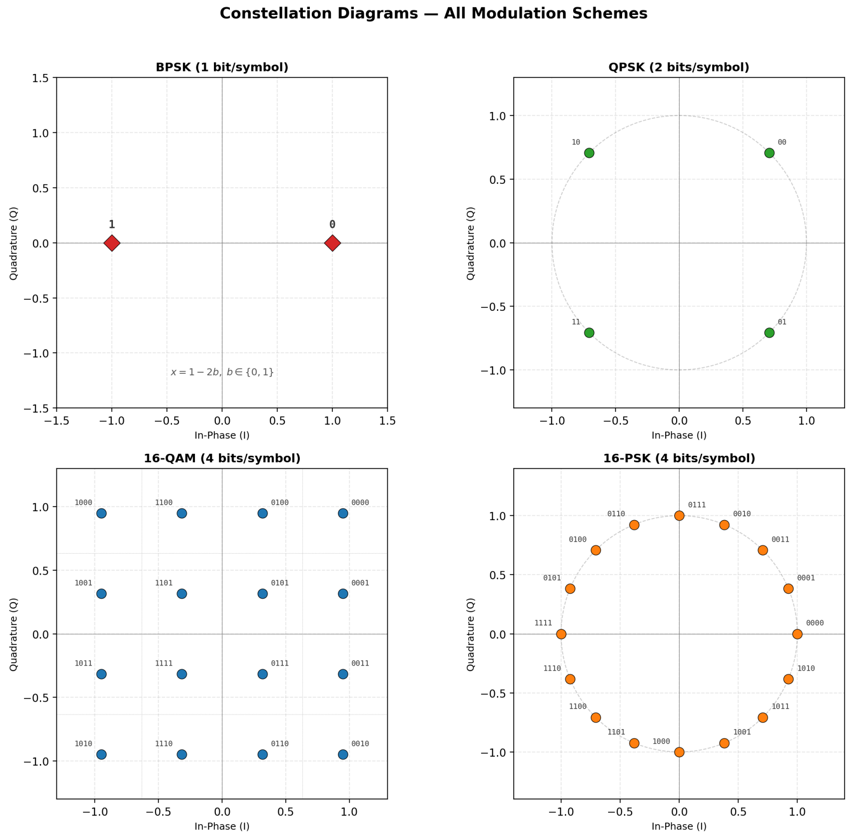

Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1. (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalised to unit average power.

4.5.4 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = f_\theta(\mathrm{Re}(y_R)) + j \cdot f_\theta(\mathrm{Im}(y_R)) \tag{4.17}$$

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing: per-axis errors accumulate independently, which is precisely the structural limitation quantified and then removed by the joint 2D classification formulation of Section 4.5.5.

Relay-specific handling:

Table 4.5: Relay-specific complex signal handling strategies and rationale.

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\|y\|^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation for 16-QAM with AI relays. For 16-QAM, each I/Q component takes four amplitude levels ($\{-3, -1, +1, +3\}/\sqrt{10}$) rather than the binary $\{\pm 1\}$ of BPSK. The BPSK-trained relays, which use tanh activations bounded in $[-1, +1]$, may not faithfully reproduce the multi-level structure. This provides a natural test of generalization: if AI relays degrade significantly on 16-QAM but not on QPSK, it indicates that the BPSK training generalizes to binary-per-component signals (QPSK) but not to multi-level signals (16-QAM). Such a finding would motivate modulation-specific relay training.

4.5.5 2D Joint Classification as an Alternative to I/Q Splitting

The I/Q splitting approach (Section 4.5.4) treats each axis of the complex constellation independently, classifying each component into $\sqrt{M}$ amplitude levels. For 16-QAM this yields two independent 4-class problems. While valid under the assumption of I/Q independence, this formulation introduces a **structural limitation**: (Equation 4.18) classification errors on the I and Q axes accumulate independently, producing a BER floor that cannot be reduced regardless of model capacity.

An alternative formulation treats the relay as a **joint 2D classifier** over all M constellation points simultaneously. Instead of splitting the complex signal and classifying 4 levels per axis, the relay receives the full 2D input (y_I, y_Q) and outputs M logits: one per constellation point. The predicted class index $\hat{k} = \arg\max_k z_k$ is mapped back to the corresponding complex symbol $s_{\hat{k}}$ for retransmission:

$$\hat{x}_R = s_{\hat{k}}, \quad \hat{k} = \arg \max_{k \in \{1,\ldots,M\}} f_\theta(y_I, y_Q)_k \tag{4.18}$$

where $f_\theta : \mathbb{R}^2 \to \mathbb{R}^M$ is the neural relay network with M output logits trained using cross-entropy loss against the true constellation index.

Advantages over I/Q splitting: - Joint decision boundaries. The network learns decision regions in the full 2D constellation space, capturing the Voronoi structure of the constellation rather than axis-aligned partitions. - **No structural BER floor.** Because the classifier selects among all M points jointly, there is no independent per-axis error accumulation (Equation 4.17). - **Minimal parameter overhead.** Only the final classification layer changes: from $\sqrt{M}$ to M outputs: adding negligible parameters to the larger architectures (<1% for Transformer, Mamba-S6, and Mamba-2 SSD).

Trade-off. The 2D classifier requires training on 16-QAM data (it cannot reuse BPSK-pretrained weights), and the number of output classes grows as M rather than $\sqrt{M}$, which may affect convergence for very high-order constellations. For 16-QAM ($M = 16$), this trade-off is favourable. The experimental evaluation is presented in Section 6.1.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Scope relative to the system model. All experiments in this chapter are conducted under the canonical system model defined in Chapter 4. In particular, the channel remains memoryless, the noise is white, transmission is uncoded, and relay processing is evaluated on a per-symbol basis. Consequently, the objective of this chapter is to compare relay functions under matched-model conditions rather than to address channel estimation, sequence detection, or channels with memory. Such effects are introduced only later in Chapter 7 as deliberate departures from the canonical assumptions in order to study model mismatch and robustness.

Following the single-model convention fixed in Chapter 1, this chapter evaluates the relay strategies on the *canonical setup only*: a two-hop SISO link, i.i.d. Rayleigh fast fading on both hops, complex baseband, QPSK, uncoded BER. BPSK appears in this chapter solely on the AWGN calibration channel, where the closed forms live (Table 4.1). Three experiments constitute the core: the AWGN calibration baseline (E1), the canonical relay comparison (E2), and the parameter-normalization and complexity study (E3), which together answer every hypothesis (H1–H5). A scoped modulation extension (Chapter 6) carries the canonical findings to 16-QAM, where per-axis I/Q processing breaks down, including the joint 2D N-class classification formulation (E4–E5). The principal contribution (Chapter 7) is the unknown-channel study (E6), which tests hypothesis H6. All other departures from the canonical setup are identified as future work in Chapter 8. The experimental configurations are summarised in the table below.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 10.1–

10.3).

Table 5.1: Summary of experimental configurations

Exp.	Topology	Modulation	Channel / Equalization	Focus
E1	SISO	BPSK	AWGN (calibration baseline)	Calibration against closed-form BER
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4); training-time benchmark (H5)
E4	SISO	16-QAM	Rayleigh (canonical)	Extension: higher-order modulation via I/Q splitting (Chapter 6)
E5	SISO	16-QAM	Rayleigh (canonical)	Extension: joint 16-class 2D classification (Chapter 6)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H6)

5.2 Simulation Baseline: AWGN Calibration

Every number in this thesis is a Monte Carlo estimate, so the first task is to establish that the simulator reproduces results that are known in closed form. AWGN with BPSK is the natural baseline for that: it is the one configuration whose bit-error probability is exact and elementary, $Q(\sqrt{2E_b/N_0})$ single hop and $2P(1-P)$ over two hops, with no fading average to approximate. It therefore serves as the *simulation baseline* of this chapter, the reference the framework is calibrated against before any relay is compared on it. The canonical channel on which the thesis's conclusions are actually drawn is Rayleigh fast fading, whose closed form is also available and is validated alongside. The channel validation plots (Figures 5.1–5.3) confirm the following:

- **AWGN channel:** Theoretical and simulated BER curves match exactly. The classical $Q(\sqrt{\gamma})$ expression (Remark 2.2), as well as its two-hop extensions (including the effective SNR for amplify-and-forward (AF) relaying and the error-composition model for decode-and-forward (DF)) are validated within tight 95% confidence intervals across all SNR points (Figure 5.1).

- **Rayleigh fading:** The simulation confirms the high-SNR asymptotic behavior BER $\sim \frac{1}{4\bar{\gamma}}$. The simulated BER closely follows the analytical curve (Figure 5.2)

$$\frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right).$$

- **Fading statistics:** The empirical probability density function (PDF) confirms the assumed Rayleigh amplitude distribution: the Rayleigh magnitude exhibits a broad spread with a non-negligible probability of deep fades, consistent with the closed-form $f_{|h|}(x) = 2xe^{-x^2}$ (Figure 5.3).

5.2.1 Experiments

Goal: Calibrate the simulation framework against closed-form BER, using AWGN/BPSK as the baseline, before any relay comparison is made on it.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** BPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical), AWGN (analytical calibration) - **Equalizers:** None - **Demod:** Hard decision only

Conclusion: Monte Carlo simulations match theoretical predictions within 95% confidence intervals on both validated channels: AWGN follows the expected exponential decay (Equation 2.36) and the canonical Rayleigh channel validates the $1/(4\bar{\gamma})$ high-SNR slope (Equation 2.46).

5.2.2 Results tables

Table 5.2: Simulator calibration: closed-form versus simulated single-hop BPSK BER, $10 \times 2{,}000{,}000$ bits per point. The SNR axis is E_b/N_0 throughout, with noise variance $N_0/2$ per real dimension, so AWGN obeys $Q(\sqrt{2E_b/N_0})$ and Rayleigh $\frac{1}{2}(1-\sqrt{\bar{\gamma}/(1+\bar{\gamma})})$. Agreement is within 0.6% at every point the budget resolves. “n/r” marks a theoretical value below the resolution of any feasible Monte Carlo run (2.3×10^{-19}), reported from the closed form rather than as a measurement.

Chan.	4dB(Th.)	4dB(Sim.)	10dB(Th.)	10dB(Sim.)	16dB(Th.)	16dB(Sim.)
AWGN	1.25E-2	1.25E-2	3.87E-6	3.85E-6	2.27E-19	n/r

Continued on next page

Table 5.2 – continued from previous page

Chan.	4 dB (Th.)	4 dB (Sim.)	10 dB (Th.)	10 dB (Sim.)	16 dB (Th.)	16 dB (Sim.)
Rayleigh	7.71E-2	7.72E-2	2.33E-2	2.32E-2	6.16E-3	6.15E-3

Table 5.3 summarises the theoretical SNR required to reach a target BER of 10^{-3} on each validated channel (single-hop BPSK), which fixes the operating range used throughout the experiments.

Table 5.3: Theoretical SNR required to achieve BER = 10^{-3} on the canonical Rayleigh channel and the AWGN calibration channel (single-hop BPSK).

Channel	SNR for BER = 10^{-3}	Diversity Order	High-SNR Slope
AWGN	~6.8 dB	– (no fading)	Exponential
Rayleigh	~24 dB	1	$1/(4\bar{\gamma})$

5.2.3 Results Figures

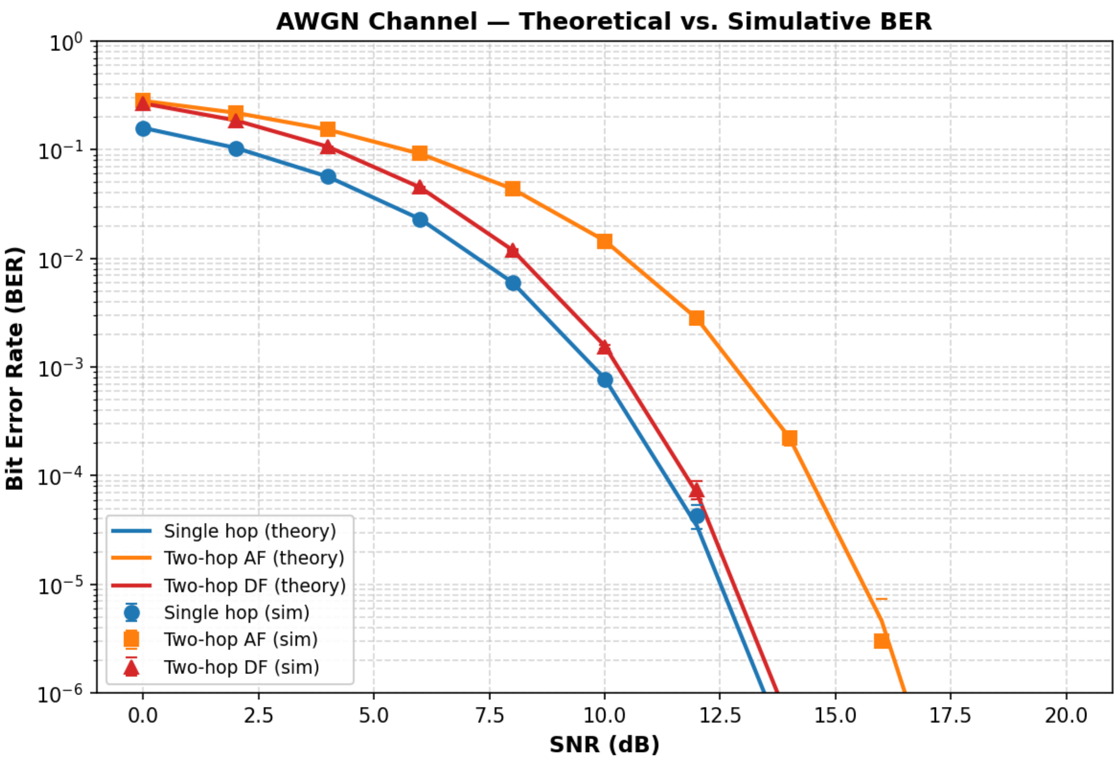

Figure 5.1: AWGN Channel: Theoretical vs. Simulative BER. Single-hop, two-hop AF, and two-hop DF: closed-form theory (solid lines) versus Monte Carlo simulation (markers with 95% CI). Theory and simulation match within the confidence interval at all SNR points.

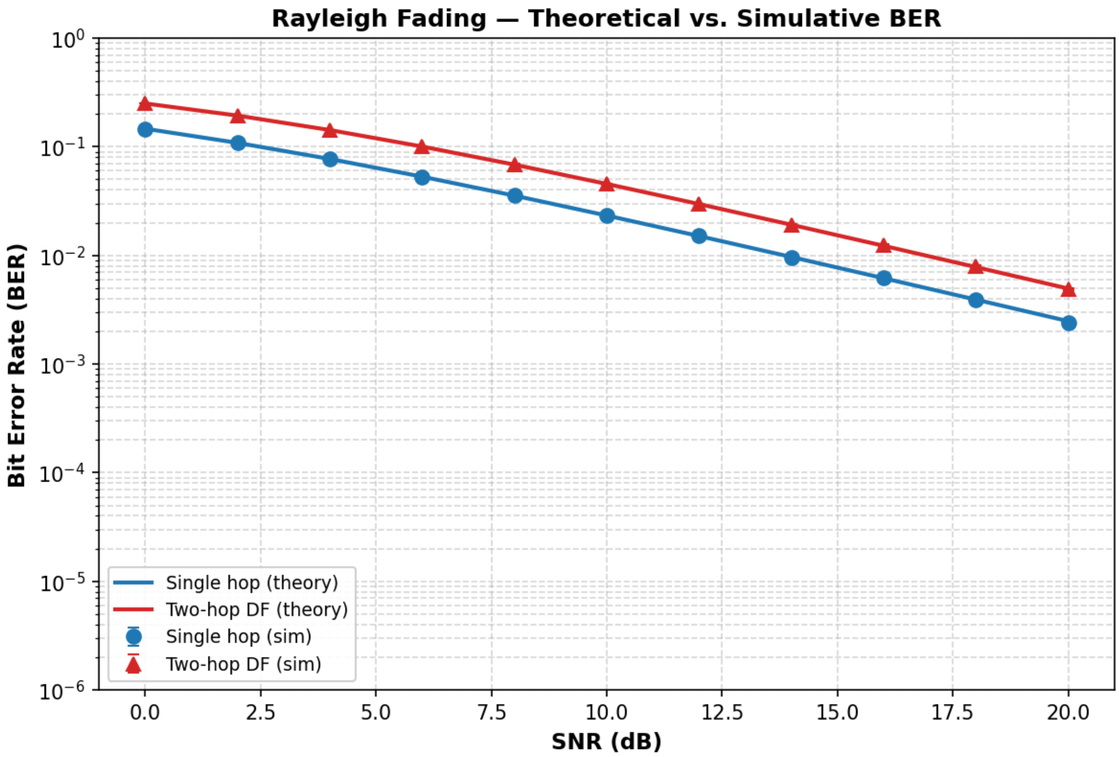

Figure 5.2: Rayleigh fading: theory vs. simulation for single-hop and two-hop DF. The characteristic $1/\mathrm{SNR}$ slope (compared to the steeper exponential AWGN curve) is clearly visible.

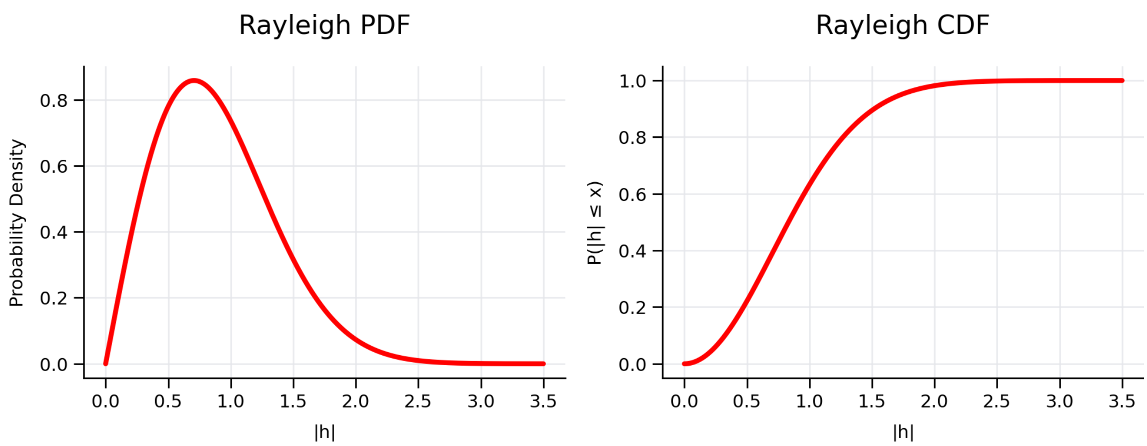

Figure 5.3: Normalized Rayleigh fading amplitude distribution. The left plot shows the PDF, $f_{|h|}(x) = 2xe^{-x^2}$, while the right plot shows the corresponding CDF, $F_{|h|}(x) = 1 - e^{-x^2}$, which represents the outage probability. Rayleigh fading models non-line-of-sight channels in which the received signal is formed by the sum of many independently scattered multipath components.

5.3 Canonical Relay Comparison (SISO, Rayleigh, QPSK)

Goal: Evaluate the classical and AI-based relay strategies on the canonical setup of this thesis. Eight of the nine implemented relays are reported; the cGAN is excluded from the re-run that produced these numbers.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** ReLU hidden, tanh output

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, cGAN) track it within statistical uncertainty and clearly outperform AF. This confirms H1 (AI relays beat AF at low SNR) and H2 (DF dominance at $\geq$ 6 dB with zero parameters) on the canonical channel.

5.3.1 Results Tables

Table 5.4: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10{,}000$ bits per SNR point. The cGAN column is absent because the relay was excluded from the re-run that produced this table (Table 5.6 records it at 7,293 s on a GPU); the omission drops its column rather than carrying forward a value measured on the earlier, 3 dB-offset axis. "DF th." is the closed-form two-hop composition of Eq. 2.47 (Section 2.5.2.2), included for direct comparison against the measured DF column.

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
0	0.4076	0.3337	0.3333	0.3349	**0.3329**	0.3359	0.4014	0.4071	0.4032
4	0.3129	**0.2219**	0.2216	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	**0.1218**	0.1203	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416
12	0.0822	0.0580	0.0560	0.0586	**0.0579**	0.0589	0.0628	0.0637	0.0621
16	0.0309	**0.0245**	0.0239	0.0247	**0.0245**	0.0247	0.0260	0.0259	0.0251
20	0.0110	**0.0097**	0.0098	0.0099	**0.0097**	**0.0097**	0.0101	0.0100	0.0098

Two features of Table 5.4 deserve comment. First, symbol-wise DF tracks the closed form (the "DF th." column above): reading the abscissa as E_s/N_0 and converting to E_b/N_0 for QPSK ($k = 2$, so 3.01 dB lower), the two-hop expression $2P(1 - P)$ with $P = \frac{1}{2}\left(1 - \sqrt{\bar{\gamma}/(1+\bar{\gamma})}\right)$ gives 0.0098 at 20 dB against a measured 0.00972, and 0.0239 at 16 dB against 0.02451: agreement within one percent at every tabulated point, which validates the simulator end to end against the analysis of Section 2.5.2.2. Figure 5.4 plots this closed form alongside the measured curves, together with the single-hop floor P itself – the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – which no relay may cross and none does at any SNR.

Second, the relays divide by family rather than by paradigm. The three feedforward relays, the MLP, the Hybrid and the VAE, are statistically indistinguishable from DF from 4 dB upward and from each other throughout; the VAE reaches 0.00972 at 20 dB, matching

DF exactly at the resolution of this budget. The two sequence models are the ones that separate, trailing by roughly 0.07 in BER at 0 dB before converging by 16 dB. Capacity is not the explanation, since the sequence models are the larger of the two groups; the memoryless channel simply offers no temporal structure for their inductive bias to exploit, a reading the equal-parameter comparison of Section 5.4 tests directly and confirms.

Two comments on Table 5.4. First, symbol-wise DF is the best or tied-best relay at every tabulated point, and the learned relays sit within a few thousandths of it from 4 dB upward: on a matched, known channel the learned relay matches DF rather than beating it, at 169 parameters against DF's none. This is H2, and it is the result the rest of the thesis is measured against. Second, the two sequence models trail the feedforward relays at low SNR (0.4014 and 0.4032 at 0 dB against the MLP's 0.3349) and converge with them by 16 dB, so their extra capacity buys nothing on a memoryless channel; Section 5.4 separates that from the parameter-count confound.

Why QPSK and not BPSK. The canonical channel is complex, so the canonical constellation is complex: QPSK places one bit on each axis the Rayleigh coefficient acts in, and the I/Q-split relays process it without modification. BPSK on this channel would require discarding the imaginary part of the equalized sample, throwing away half the plane the channel acted in, which is why the real constellation is paired with the real channel in the calibration baseline of Section 5.2 instead (Table 4.1). One consequence must be kept in view when reading BER across the two: the SNR axis here is E_s/N_0, and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number in the BPSK baseline. The constellations are not on a common axis and their error rates are not directly comparable.

5.3.2 Results Figures

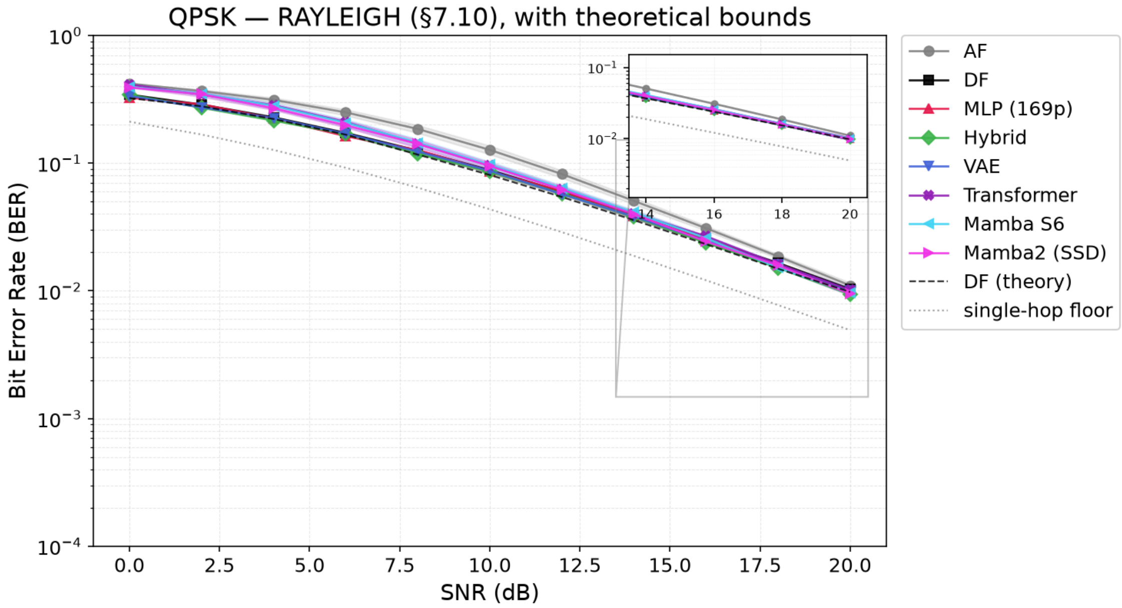

Figure 5.4: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.4, with 95% confidence intervals, plus two theoretical reference curves: "DF (theory)" (dashed), the closed-form two-hop composition, which the measured DF curve tracks within one percent everywhere; and the "single-hop floor" (dotted), the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – a physical lower bound no relay crosses. AF is uniformly worst; symbol-wise DF and the feedforward learned relays are indistinguishable from 4 dB upward; the two sequence models trail at low SNR and converge by 16 dB.

5.4 Parameter Normalization & Complexity Trade-off

This study tests H3 (U-shaped complexity in BER) and H4 (architecture convergence at equal scale) on the canonical setup, and provides the training-time benchmark for H5.

Goal: Isolate architectural inductive biases from parameter count effects and characterize the complexity-performance trade-off for relay denoising.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** All normalized to ~3,000 parameters, plus original sizes (169 to 26K) - **NN activation:** ReLU hidden, tanh output

Conclusion: The relay denoising task exhibits a U-shaped complexity relationship in BER: a minimal 169-parameter MLP matches models roughly 140× larger, while excessive parameters (11K+) lead to overfitting. At a normalized scale of 3,000 parameters the gap between feedforward and sequence architectures is strongly SNR-dependent: it falls monotonically from 17.4% at 0 dB to 4.1% at 20 dB (Table 5.5). Parameter count therefore accounts for the comparison at high SNR, where architecture is nearly irrelevant, but not at low SNR, where the six equal-sized models separate cleanly into a feedforward group tracking DF and a sequence group tracking AF. Since all six carry the same budget, that separation is architectural choice is the primary performance driver. The generative VAE is a consistent underperformer in this configuration; as noted for Table 5.4, this may reflect the stochastic inference-time decoding used here rather than the generative paradigm itself (Section 2.3.3).

5.4.1 Results Tables

Table 5.5: Normalized 3K-parameter comparison on the canonical setup (QPSK ove Rayleigh). Every learned model is scaled to approximately 3,000 parameters so that arch tecture, not capacity, is the variable; AF and DF are the unparameterized references. Th cGAN is excluded, as in Table 5.4.

SNR (dB)	MLP-3K	Hybrid-3K	VAE-3K	Transformer-3K	Mamba-S6 (3K)	Mamba-2 SSD (3
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622
16	0.0247	**0.0245**	0.0251	0.0259	0.0250	0.0251
20	0.0099	**0.0097**	0.0101	0.0100	0.0097	0.0098

Table 5.6: Model complexity and timing comparison on the canonical setup (experiment-specific training settings; Monte Carlo evaluation over 11 SNR points × 10 trials × 10,000 bits). All inference uses batched window extraction and a single forward pass per signal block.

Model	Parameters	Device	Training Time	Eval Time
AF	0	–	0 s	0.80 s
DF	0	–	0 s	0.77 s
MLP (169p)	169	CPU	4.9 s	1.57 s
Hybrid	169	CPU	4.6 s	0.41 s
VAE	1,777	CPU	21.6 s	1.81 s
cGAN (WGAN-GP)	2,946	CUDA	7,293 s (~2 h)	1.14 s
Transformer	17,697	CUDA	474 s (~8 min)	3.71 s
Mamba-S6	**24,001**	**CUDA**	**2,141 s (~36 min)**	**1.88 s**
Mamba-2 SSD	**26,179**	**CUDA**	**1,438 s (~24 min)**	**4.11 s**

5.4.2 Results Figures

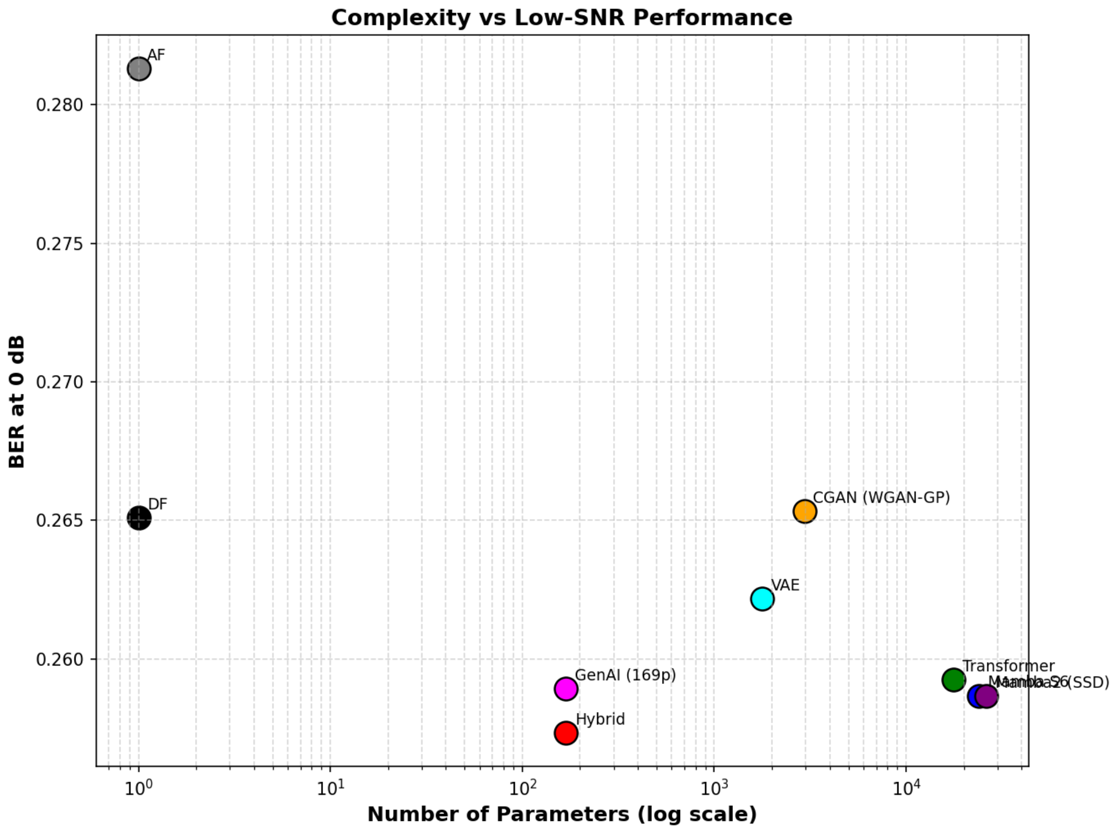

Figure 5.5: Complexity versus low-SNR performance. BER at 0 dB is shown as a function of model size.

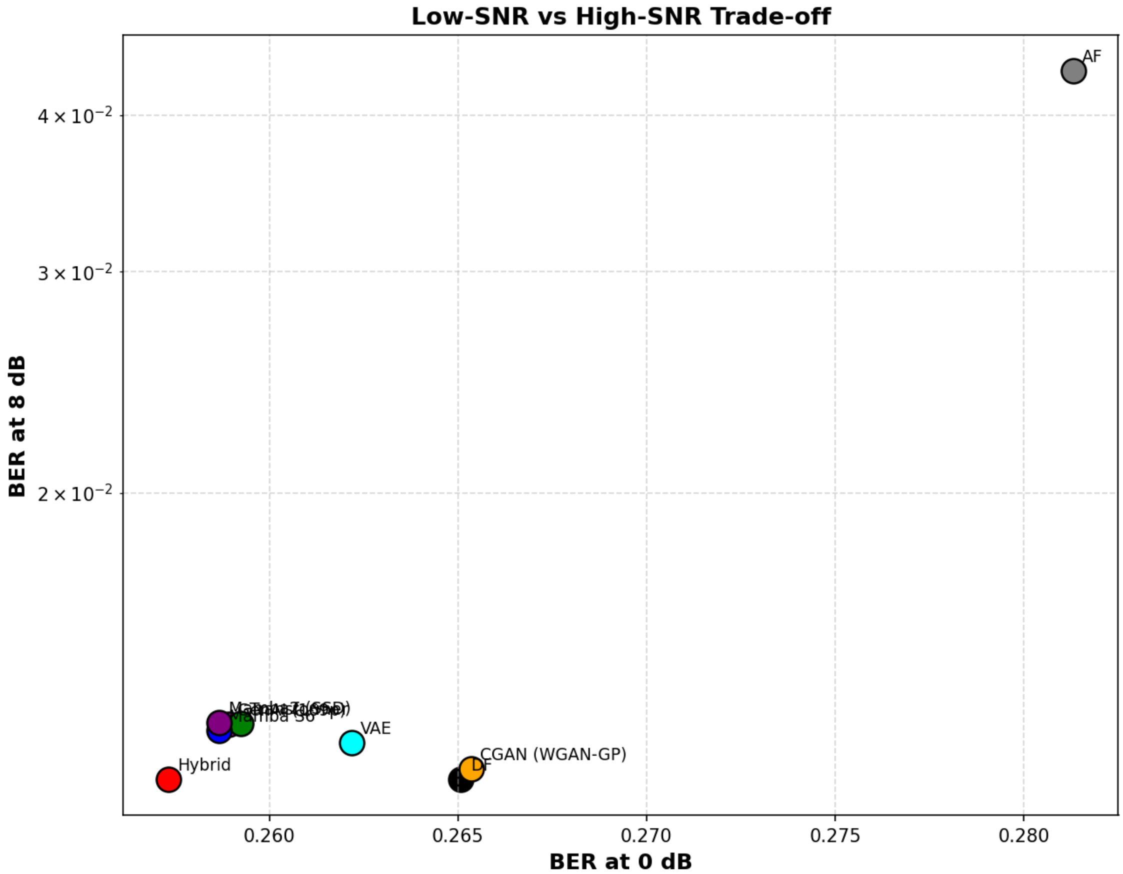

Figure 5.6: Low-SNR versus high-SNR trade-off. BER at 8 dB is plotted against BER at 0 dB.

5.5 Coded Block-DF: Measuring the Remark 4.2 Caveat

Remark 4.2 draws a distinction the rest of this thesis relies on: the DF baseline used everywhere here is *symbol-wise* (uncoded, per-symbol hard slicing), not the *block* DF of the information-theoretic relay-channel literature, and the remark explicitly cautions that ”the reported DF results should not be read as bounds on coded block-DF performance.” That caveat was asserted, not measured. This section measures it: a real information-theoretic block-DF relay, built from a rate-1/2 convolutional code and a soft-decision Viterbi decoder, added as a supplementary study on the canonical channel rather than as a revision of the core comparison above.

Setup. A standard rate-1/2 convolutional code (constraint length K, 2^{K-1} trellis states, zero-tail terminated per 200-information-bit frame) is applied before QPSK modulation on the canonical Rayleigh channel; `CodedDecodeAndForwardRelay` decodes the full frame at the relay via soft-decision Viterbi (squared-distance branch metric on the coherently-compensated I/Q samples), then re-encodes and re-modulates a fresh, clean codeword before forwarding – genuine block DF, as opposed to the per-symbol slicing used elsewhere in this thesis. Two coded-aware learned relays are trained on the same task for comparison: a windowed 756-parameter MLP classifier (window 21, $\approx 5\times K$ for $K = 3$) and the Mamba-S6 architecture already used in Table 5.4 (24,084 parameters at this configuration), both predicting the transmitted QPSK symbol from a window of noisy coded observations with no explicit knowledge of the code. All results use the thesis-standard budget of 10 trials $\times$ 100,000 information bits per SNR point.

Table 5.7: Coded block-DF vs. uncoded symbol-wise DF and coded-aware learned relays, canonical QPSK/Rayleigh, rate-1/2 K=3 code. "coded-AF" forwards the noisy codeword unmodified (the relay does not decode), isolating the code's own gain from any relay intelligence.

SNR (dB)	uncoded DF	coded AF	coded DF	MLP-coded	Mamba-coded
0	**0.3336**	0.4904	0.4210	0.4381	0.4491
4	**0.2213**	0.4448	0.2231	0.2807	0.3216
8	0.1204	0.2904	**0.0638**	0.0917	0.1175
12	0.0558	0.0544	**0.0152**	0.0206	0.0225
16	0.0242	0.0088	0.0043	0.0046	**0.0042**
20	0.0099	0.0018	0.0013	0.0011	**0.0009**

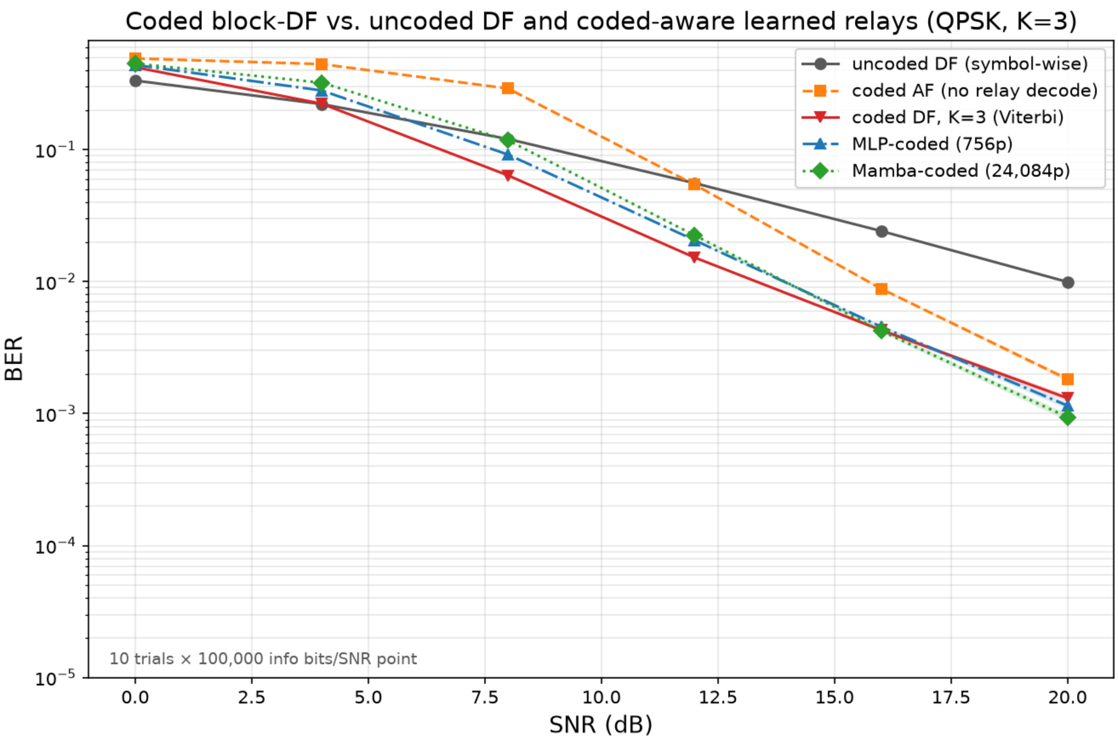

Figure 5.7: The data of Table 5.7 plotted: coded block-DF crosses below uncoded symbol-wise DF between 4 and 8 dB and stays below it thereafter; coded AF, which never decodes, never catches up. Both coded-aware learned relays track coded-DF closely from 12 dB up but are visibly worse at 4–8 dB.

Two findings, reported as measured rather than smoothed toward either a ”coding always helps” or a ”the learned relay wins” narrative:

The caveat holds precisely where it should, and no further. At equal E_s/N_0, coded block-DF beats uncoded symbol-wise DF decisively from 8 dB upward – a $5.6\times$ reduction in BER at 16 dB, $7.6\times$ at 20 dB – closing the gap the remark left open. Those factors are the right comparison for a detection question but the wrong one for a link budget, because the rate-1/2 code halves the information carried per channel use; Section 5.5.3 redoes the comparison at equal throughput and the advantage largely disappears. But below roughly 6 dB, coding *hurts* even on the favourable axis: both coded AF and coded DF are worse than plain uncoded DF at 0 and 4 dB. This is the well-known convolutional-code error-propagation threshold, not an artifact: a fixed-rate code below its operating threshold produces decoding errors that cluster and propagate across the trellis, which a memoryless per-symbol slicer cannot do. The caveat is real, but it cuts both ways – coded block-DF is a strictly stronger baseline only above its own threshold, which itself has to be measured, not assumed.

At low-to-mid SNR the classical decoder is far ahead, and the learned relay's extra capacity buys nothing. Unlike the canonical memoryless channel of Table 5.4, where Section 5.4 found ”the memoryless channel simply offers no temporal structure for [the sequence models'] inductive bias to exploit,” a convolutional code *is* genuine temporal structure. It does not rescue the learned relays where the code is doing the most work: both are far behind coded-DF at 4–8 dB (Mamba-coded 0.1175 vs. 0.0638 at 8 dB). Nor does the $32\times$-larger Mamba-S6 relay show a clear advantage over the 756-parameter MLP there, being worse at 4–8 dB (0.3216 vs. 0.2807 at 4 dB). Giving the sequence model a task with real memory to exploit therefore does not overturn the thesis's “less is more” finding (H3): extra capacity still buys no reliable advantage over the minimal architecture.

At high SNR the ordering reverses, and it is not noise. The differences at 16–20 dB in Table 5.7 are small in absolute terms, and an earlier version of this section dismissed them as trial-to-trial variation. That was wrong, and the correction is worth stating explicitly because it changes the conclusion. Re-measuring those two points with *paired* trials – every relay driven by identical information bits and identical hop-1/hop-2 fading realizations within each trial, so the channel-draw variance that dominates an unpaired comparison cancels in the per-trial difference – and testing with the Wilcoxon signed-rank test used elsewhere in this thesis gives Table 5.8. At 20 dB both learned relays beat coded-DF in essentially every one of 100 paired trials ($p < 10^{-4}$). At 16 dB the result splits by architecture: Mamba-coded wins (68 of 100 trials) while the MLP significantly *loses* (23 of 100), both at $p < 10^{-4}$. The effects are real in both directions, and no single sentence of the form “the learned relay beats/does not beat Viterbi” describes them.

Table 5.8: Paired re-measurement of the two high-SNR points of Table 5.7, 100 paired trials $\times$ 100,000 information bits. "Diff" is the mean per-trial BER difference against coded-DF (negative favours the learned relay); "W/L" counts trials won/lost; p is Wilcoxon signed-rank.

SNR	Relay	BER	Diff vs. coded-DF	W/L	p
16 dB	coded-DF	0.004245	—	—	—
	MLP-coded	0.004437	$+0.000192$	23/77	$< 10^{-4}$
	Mamba-coded	**0.004124**	-0.000121	68/32	$< 10^{-4}$
20 dB	coded-DF	0.001362	—	—	—
	MLP-coded	0.001066	-0.000295	99/1	$< 10^{-4}$
	Mamba-coded	**0.000967**	-0.000395	100/0	$< 10^{-4}$

This is not a learned relay out-decoding Viterbi, and the mechanism says why. Viterbi is the maximum-likelihood *sequence* decoder for a known code on a known channel, so a learned decoder beating it at its own task would be a surprising claim requiring more than a BER table. It is not what happens here. Instrumenting the relay output directly (Table 5.9) shows coded-DF is the *better* decoder by a wide margin: it puts roughly $2.1\times$ *fewer* symbol errors on the air than the learned relay at both SNRs, exactly as the optimality result predicts. What separates them is what happens to those errors afterwards. Coded-DF decodes and then *re-encodes*, so a frame it gets wrong leaves the relay as a different but perfectly *valid* codeword: the code's redundancy has been spent and then regenerated around the error, and the destination's decoder has no way to detect or repair it. The learned relay never re-encodes, so its (more numerous) mistakes remain isolated bad symbols inside an otherwise-valid codeword, and the destination still holds the redundancy needed to fix them. Only 16–26% of coded-DF's relay errors are repaired downstream against 57–74% of the learned relay's. That also explains the SNR-dependent split above: at 16 dB the $2.1\times$ raw-error penalty still outweighs the repair advantage for the MLP, and by 20 dB it no longer does.

Table 5.9: Where the errors go, 20 trials $\times$ 500 frames. "Relay symbol ER" counts symbols the relay put on the air that differ from the transmitted ones; "repaired" is the fraction of those that do not survive the destination's decoder. The oracle relay forwards the clean codeword and so isolates the hop-2-only floor.

SNR	Relay	Relay symbol ER	Repaired	Final BER
16 dB	coded-DF	**0.00507**	15.5%	0.004282
	MLP-coded	0.01075	**57.2%**	0.004603
	oracle	—	—	0.002192
20 dB	coded-DF	**0.00183**	25.8%	0.001355
	MLP-coded	0.00390	**74.4%**	0.001001

SNR	Relay	Relay symbol ER	Repaired	Final BER
	oracle	—	—	0.000655

5.5.1 Does a Stronger Code Help? A Constraint-Length Sweep

The $K=3$ code above is deliberately small (4 states, comparable to the smallest ISI channel studied in Chapter 7). Whether a stronger code – larger constraint length, larger free distance – simply does better is not obvious given the threshold behaviour just observed, so it is measured rather than assumed: the classical coded-DF baseline (Viterbi decode only; the learned relays are not retrained per K, each training run being too expensive to repeat six-fold) is swept over $K \in \{3, 5, 7\}$ – 4, 16, and 64 trellis states respectively, with $K=7$ the standard used in, e.g., the Voyager-era deep-space codes – on both canonical constellations.

Table 5.10: Coded-DF BER vs. constraint length K, canonical QPSK/Rayleigh, rate 1/2.

SNR (dB)	$K=3$ (4 states)	$K=5$ (16 states)	$K=7$ (64 states)
0	**0.4212**	0.4722	0.4892
4	**0.2242**	0.3009	0.3550
8	**0.0633**	0.0831	0.0884
12	**0.0154**	0.0192	0.0168
16	0.0042	0.0049	**0.0039**
20	0.0015	**0.0013**	0.0013

Table 5.11: Coded-DF BER vs. constraint length K, canonical 16-QAM/Rayleigh, rate 1/2, against the uncoded modulation-aware DF reference. Decoding uses the 16-QAM-specific branch metric of a joint PAM-4 Gray level per trellis step (two steps per symbol), not the independent per-axis metric that QPSK admits.

SNR (dB)	uncoded DF	$K=3$	$K=5$	$K=7$
0	**0.4180**	0.4687	0.4954	0.4961
4	**0.3505**	0.4204	0.4771	0.4874
8	**0.2572**	0.2722	0.3546	0.4003
12	0.1577	**0.0989**	0.1290	0.1446
16	0.0805	**0.0269**	0.0325	0.0344
20	0.0366	**0.0074**	0.0096	0.0095

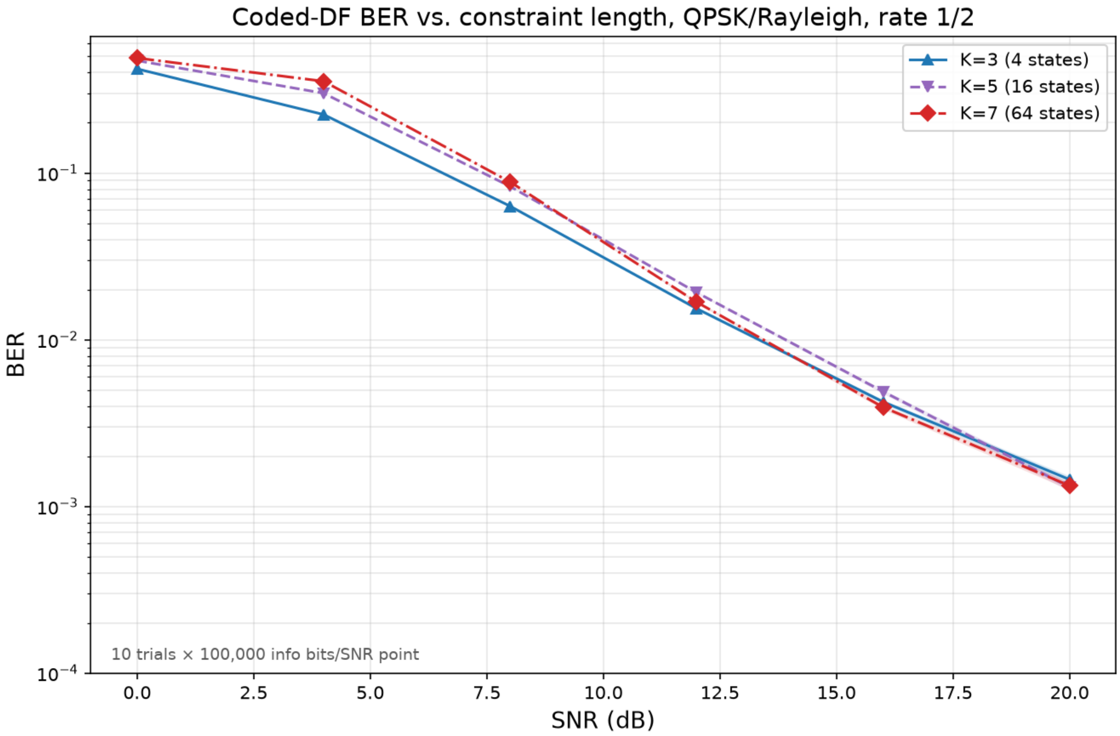

Figure 5.8: The data of Table 5.10 plotted: K=5 and K=7 sit visibly above K=3 at 0–8 dB and only draw level by 16–20 dB, never separating out cleanly ahead of it within this SNR range.

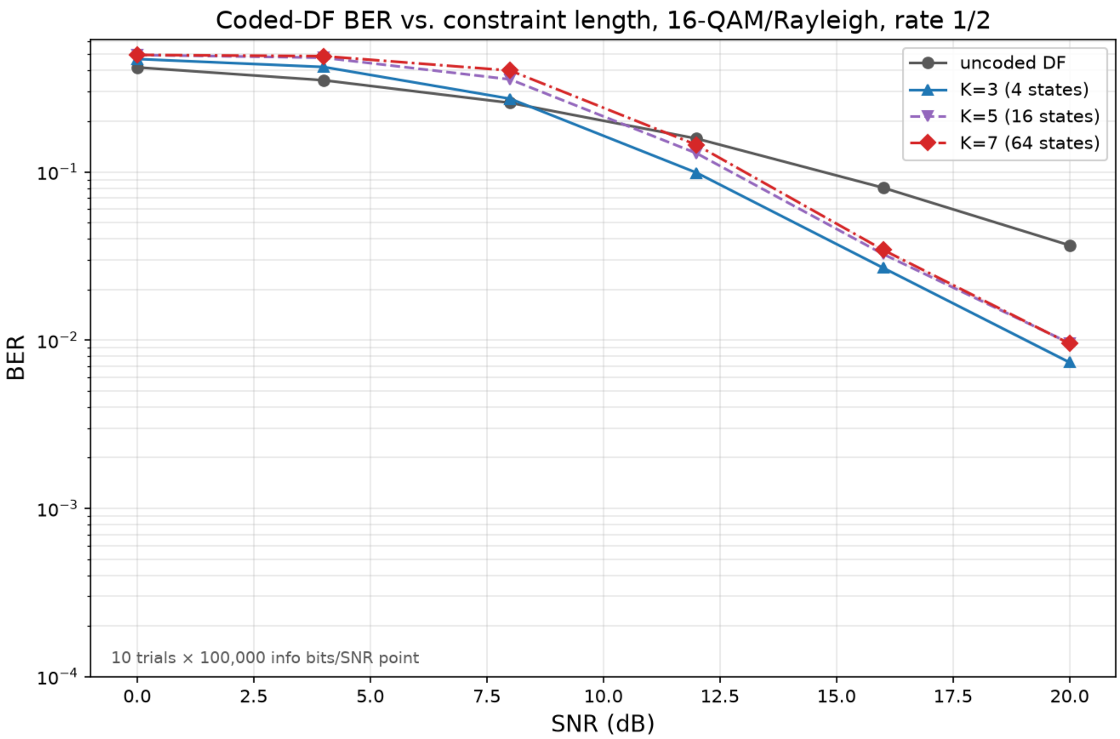

Figure 5.9: The data of Table 5.11 plotted: the same ordering as QPSK, with the coded/uncoded crossover pushed out to roughly 8–12 dB by 16-QAM's higher baseline uncoded BER, and K=3 remaining the best of the three constraint lengths at every tabulated point.

The answer is no, not monotonically, and not within the range measured here. On both constellations, the larger-K codes are *worse* than $K=3$ at low-to-mid SNR (0–8 dB on QPSK, 0–16 dB on 16-QAM) – a sharper error-propagation cliff for the bigger trellis, the same threshold effect visible in Table 5.7, more pronounced the stronger the code. $K=7$ pulls very slightly ahead of $K=3$ at 16 dB on QPSK (0.0039 vs. 0.0042); by 20 dB all three constraint lengths are within trial-to-trial noise of each other on QPSK, and on 16-QAM $K=3$ remains the best of the three at every tabulated point up to 20 dB. Within this frame length (200 information bits) and trial budget, the larger codes' greater free distance never clearly pays off at any SNR point measured, only costing more at low-to-mid SNR where the crossover matters most. This does not mean stronger codes are never worthwhile – a longer frame or a lower operating rate would very plausibly shift the picture – only that constraint length is not a free knob to turn up: like every other axis in this thesis, its benefit is a boundary to be located, not a default to be assumed.

5.5.2 Soft-Decision Relaying, and What the Liability Actually Is

The mechanism of Table 5.9 suggests an obvious repair. If the trouble with block-DF is that it commits to a codeword, replace the commitment with soft information: run BCJR (forward-backward MAP) at the relay instead of Viterbi and forward the posterior-mean symbol per trellis step, so a symbol the relay is unsure of is shrunk toward the origin rather than asserted. The same idea applies to the learned relay, whose softmax already carries a posterior it was throwing away: forwarding $\sum_k p_k s_k$ instead of $\arg\max_k p_k$ costs nothing, needs no retraining, and reuses the identical weights, so any difference isolates the read-out rule alone. Table 5.12 evaluates both against their hard counterparts and against an oracle relay that forwards the clean codeword, on paired trials.

Table 5.12: Soft- versus hard-decision relaying, canonical QPSK/Rayleigh, rate-1/2 $K=3$, 20 paired trials $\times$ 100,000 information bits. The MLP hard/soft pair shares one set of trained weights and differs only in the read-out. The oracle relay forwards the clean codeword, giving the hop-2-only floor.

SNR (dB)	hard block-DF	soft block-DF	MLP hard	MLP soft	oracle
0	0.4207	**0.4142**	0.4424	0.4327	0.3012
4	0.2223	**0.2210**	0.2923	0.2715	0.1278
8	**0.0642**	0.0642	0.0970	0.0849	0.0332
12	**0.01544**	0.01546	0.02097	0.01729	0.00762
16	0.004140	0.004141	0.004355	**0.003485**	0.002106
20	0.001376	0.002379	0.001069	**0.000885**	0.000673

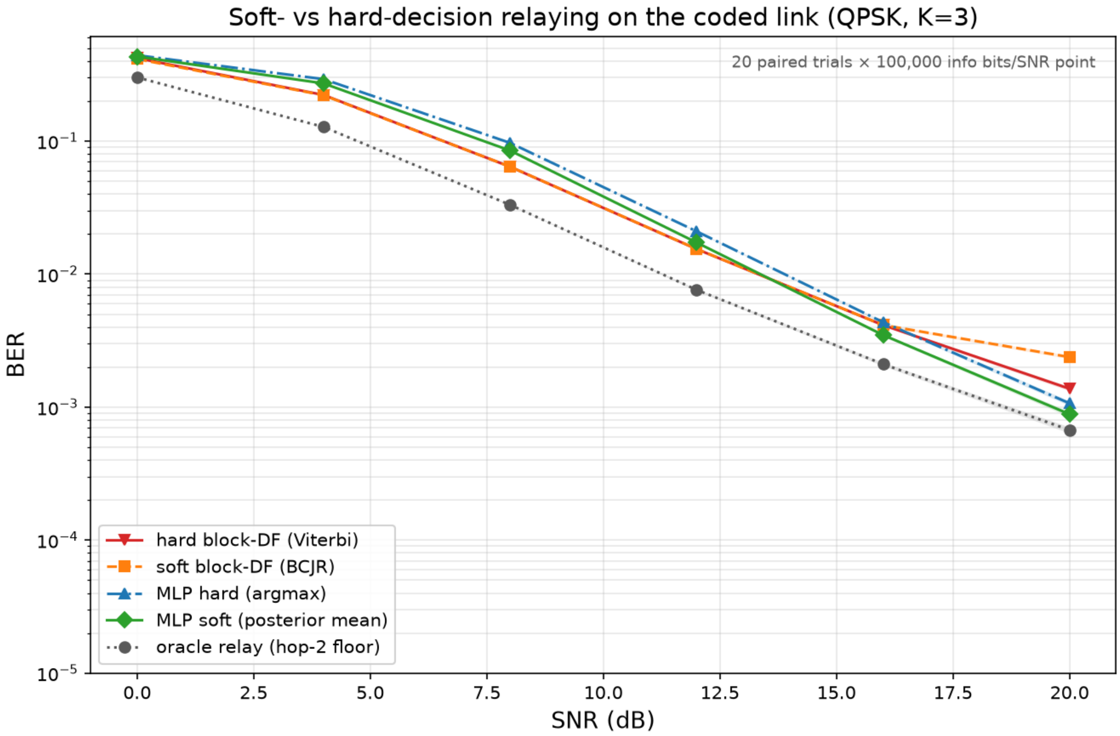

Figure 5.10: The data of Table 5.12. The soft learned relay (green) tracks the oracle floor most closely at high SNR; soft block-DF (orange) turns upward at 20 dB, an artefact of mis-calibration discussed in the text, not of soft forwarding itself.

Soft read-out helps the learned relay everywhere. At every SNR the posterior mean beats the argmax, on identical weights: 0.4327 against 0.4424 at 0 dB, 0.0849 against 0.0970 at 8 dB, 0.000885 against 0.001069 at 20 dB. The gain is largest exactly where the mechanism predicts, since a relay that reports its uncertainty gives the destination's decoder more to work with than one that reports a confident guess. It is enough to change a verdict: at 16 dB the hard MLP significantly *loses* to block-DF (2 of 20 paired trials) while the soft MLP significantly *beats* it (20 of 20), and at 20 dB the soft relay comes within 0.00021 of the oracle floor that no relay of any kind can pass.

Soft block-DF does not rescue block-DF, and that is the more interesting result. The prediction going in was that BCJR would combine the code-aware cleanup of Viterbi with the repairability of a non-committing relay, and therefore beat both. It does not. It edges hard block-DF at 0–4 dB, ties it from 8–16 dB, and at 20 dB is clearly worse (0.002379 against 0.001376, losing all 20 paired trials). The 20 dB degradation turns out to be a calibration artefact rather than a property of soft forwarding: the relay is given the *nominal* noise variance, but post-equalization Rayleigh noise has variance $1/(2\gamma|h|^2)$, so at high SNR, where the nominal value is small, BCJR becomes badly overconfident on deep-fade symbols. Viterbi is structurally immune, its squared-distance metric being invariant to any uniform scaling of the assumed variance. Deliberately inflating the assumed variance confirms this directly: at 20 dB a factor of two removes the entire gap (0.002394 $\rightarrow$

0.001367), and every larger factor up to $100\times$ holds there.

What that correction exposes is the substantive point. Even properly calibrated, soft block-DF only *ties* hard block-DF (0.001367 against 0.001367 at 20 dB; 0.004304 against 0.004313 at 16 dB) – it never beats it. The reason is that at high SNR the BCJR posteriors saturate, so the posterior mean degenerates to the constellation point and soft block-DF reproduces Viterbi's decisions; when a code-exploiting relay is wrong, it is confidently wrong whether or not it quantizes. The liability identified in Table 5.9 is therefore *not* hard quantization at the relay, as the soft-forwarding hypothesis assumed. It is consuming the code's redundancy at the relay at all. Any relay that decodes the code – Viterbi or BCJR, hard or soft – has spent that redundancy by the time it transmits, and whatever it got wrong arrives at the destination unrepairable. The learned relay's high-SNR advantage comes from the opposite discipline: it never decodes the code, only denoises symbol by symbol, so the full redundancy survives intact for the destination to spend. Soft read-out then adds a second, smaller gain on top by preserving the confidence that the destination can weigh.

This sharpens the thesis's recurring claim rather than overturning it. A learned relay still does not out-decode a matched classical decoder: coded-DF remains far ahead through 12 dB and, per Table 5.9, is the more accurate decoder at every SNR measured. What the high-SNR reversal shows is an architectural point about *where* in a two-hop chain the decoding should happen, and the answer that emerges – leave the code alone at the relay, denoise only, and let the destination decode once – is a system-design conclusion, not a statement about the relative power of Viterbi and neural networks.

5.5.3 Throughput and Latency: What the BER Tables Leave Out

Every table above compares relays at equal E_s/N_0. That is the correct axis for asking which relay detects better, and the wrong one for asking which relay a link should use, because it silently ignores two costs that differ by orders of magnitude across these designs.

Throughput: the coded comparison was not like-for-like. A rate-1/2 code carries half the information per channel use. Concretely, the coded QPSK link spends 202 symbols to deliver 200 information bits (0.99 bits/symbol) where uncoded QPSK spends 100 (2.00 bits/symbol). The "$5.6\times$ better at 16 dB" quoted above is therefore bought with half the data rate, and a link designer given that trade would not read it as a $5.6\times$ improvement at all. The honest comparison holds spectral efficiency fixed, and the measurements already collected supply it: rate-1/2 *16-QAM* delivers $4 \times \frac{1}{2} = 1.98$ information bits per symbol, matching uncoded QPSK's 2.00. Table 5.13 places those two side by side.

Table 5.13: Coded versus uncoded at equal spectral efficiency ($\approx$2 information bits per channel symbol): uncoded QPSK against rate-1/2 K=3 16-QAM, both on the canonical Rayleigh channel. Recomputed from the data of Tables 5.7 and 5.11; no additional simulation.

SNR (dB)	uncoded QPSK	rate-1/2 16-QAM	Coding gain
0	**0.3336**	0.4687	$0.71\times$ (worse)
4	**0.2213**	0.4204	$0.53\times$ (worse)
8	**0.1204**	0.2722	$0.44\times$ (worse)
12	**0.0558**	0.0989	$0.56\times$ (worse)
16	**0.0242**	0.0269	$0.90\times$ (worse)
20	0.0099	**0.0074**	$1.35\times$ (better)

The result is a substantial qualification of Section 5.5. Held to equal throughput, the rate-1/2 code does not pay for itself anywhere below 20 dB, and where it finally does the gain is $1.35\times$, not the 5.6–$7.6\times$ the equal-E_s/N_0 reading suggested. Coding is not free reliability: it is reliability bought with bandwidth, and on this channel, at this rate and frame length, the purchase is a poor one until the link is already operating well. This does not overturn the earlier section – coded block-DF genuinely is the stronger *detector*, and Remark 4.2's caveat genuinely is real – but it does bound how much the caveat matters in practice.

Latency: the block relays buffer three orders of magnitude more. The relays also differ structurally in how much they must accumulate before emitting anything, which sets the delay the relay contributes independently of how fast its arithmetic is. Symbol-wise AF and DF emit per symbol. A windowed learned relay needs w symbols of look-ahead, ten here, and that figure is a property of the architecture, constant in frame length. Block-DF cannot emit until it has decoded an entire frame, 202 symbols; soft block-DF is worse still in kind rather than degree, since BCJR's backward recursion runs from the end of the frame and so structurally forbids emitting early even in principle. Frame length is a free parameter, so this is not a fixed penalty but a scaling one: the block relays' latency grows with the frame while the learned relay's does not.

Table 5.14: Structural latency and measured compute cost. "Buffer" is the number of symbols the relay must accumulate before it can emit its first output. Compute is the median of 7 runs over 50,000 symbols after a discarded warm-up, single-threaded CPU; as with Table 5.6 these timings are machine-dependent and are reported as informational.

Relay	Buffer (symbols)	μs/symbol	Relative to Viterbi
AF / symbol-wise DF	**0**	—	—
MLP hard (756p)	10	0.30	$50\times$ faster
MLP soft (756p)	10	**0.22**	$68\times$ faster
hard block-DF (Viterbi)	202	15.07	$1\times$

Relay	Buffer (symbols)	μs/symbol	Relative to Viterbi
soft block-DF (BCJR)	202	29.30	1.9× slower

The compute column points the same way as the latency column. The 756-parameter relay costs $0.30\,\mu$s per symbol against Viterbi's 15.07, a factor of 50, and the soft variant is no more expensive than the hard one because replacing an `argmax` with a posterior-mean inner product is the cheaper of the two operations. BCJR costs roughly twice Viterbi, as its forward, backward and marginalization passes imply. These are single-machine wall-clock figures for reference implementations, not architectural constants, and the caveat attached to Table 5.6 applies here unchanged; the ratios are nonetheless large enough, and the structural buffering figures exact enough, that the ordering is not in question.

Taken together with Table 5.13, this reframes the coded study's practical conclusion. Where the code pays for itself at all, it does so at 20 dB, for a $1.35\times$ gain at equal throughput, while asking for $20\times$ the buffering latency and $50\times$ the arithmetic of the learned relay that comes within a few ten-thousandths of it there. The learned relay's case in this chapter, as in Chapter 7, rests on cost and constancy rather than on a lower error floor.

5.5.4 Adaptive Modulation and Coding: Choosing the Rate, Not Just the Relay

Every comparison above, including the equal-throughput one, holds the modulation-and-coding scheme (MCS) fixed and asks which relay achieves the lower BER. A real link does not hold the MCS fixed; it adapts. The question a link actually faces is not "which relay wins at rate R" but "what is the highest *useful* rate this link can sustain at this SNR", and answering it requires more than one code rate to choose from. The mother code of Section 5.5 is extended here with the standard punctured rates $2/3$ and $3/4$ (deleting a periodic subset of the rate-1/2 output, re-inserted as soft zeros at the decoder, so the same Viterbi decoder runs unmodified at every rate), and evaluated via bit-interleaved coded modulation across the full grid $\{\text{QPSK}, \text{16-QAM}\} \times \{\text{uncoded}, 1/2, 2/3, 3/4\}$.

The objective is *goodput*, $G = R\,(1 - \text{FER})$, the information rate actually delivered once frames with any residual error are discarded, which is what any real protocol above the physical layer does. G and R are both measured per channel symbol transmitted on the active hop, matching the ergodic-capacity reference C introduced below; because the relay is half-duplex, expressing goodput instead as a fraction of the total two-slot cycle (source-to-relay plus relay-to-destination) would halve every figure in this section without changing which MCS or relay strategy is preferred, since that factor of $\frac{1}{2}$ applies identically to both. Maximizing G over the MCS grid at each SNR traces the link-adaptation envelope, compared here for the two relay strategies the mechanism of Table 5.9 contrasts directly:

block-DF, which decodes the code at the relay and so spends its redundancy before transmitting, against *denoise-only*, which hard-decides each bit and re-modulates without ever invoking the code, leaving the redundancy intact for the destination and requiring no training and no code knowledge at the relay.

Table 5.15: Link-adaptation envelope: the MCS maximizing goodput at each SNR, block-DF versus denoise-only relaying, canonical Rayleigh channel, 10 trials $\times$ 200 frames $\times$ 200 information bits per point. C is the ergodic Rayleigh (Shannon) capacity at that SNR, $C = \log_2(e)\, e^{1/\gamma} E_1(1/\gamma)$ – an unconstrained-input upper bound no achievable goodput may exceed, included as a sanity ceiling rather than a target: the gap to it here is dominated by the finite constellation and coding loss, not by relay strategy.

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)	C (Shannon)
8	QPSK 1/2	**0.0040**	QPSK uncoded	0.0000	2.40
12	QPSK 1/2	**0.1668**	QPSK 1/2	0.1282	3.45
16	QPSK 1/2	0.5342	QPSK 1/2	**0.5995**	4.63
20	QPSK 3/4	0.9274	QPSK 2/3	**0.9287**	5.88
24	16-QAM 2/3	**1.6079**	16-QAM 2/3	1.5129	7.17
28	16-QAM 3/4	**2.3185**	16-QAM 3/4	2.2993	8.48
32	16-QAM uncoded	2.6800	16-QAM uncoded	2.6800	9.80
36	16-QAM uncoded	3.3840	16-QAM uncoded	3.3840	11.13
40	16-QAM uncoded	3.6980	16-QAM uncoded	3.6980	12.46

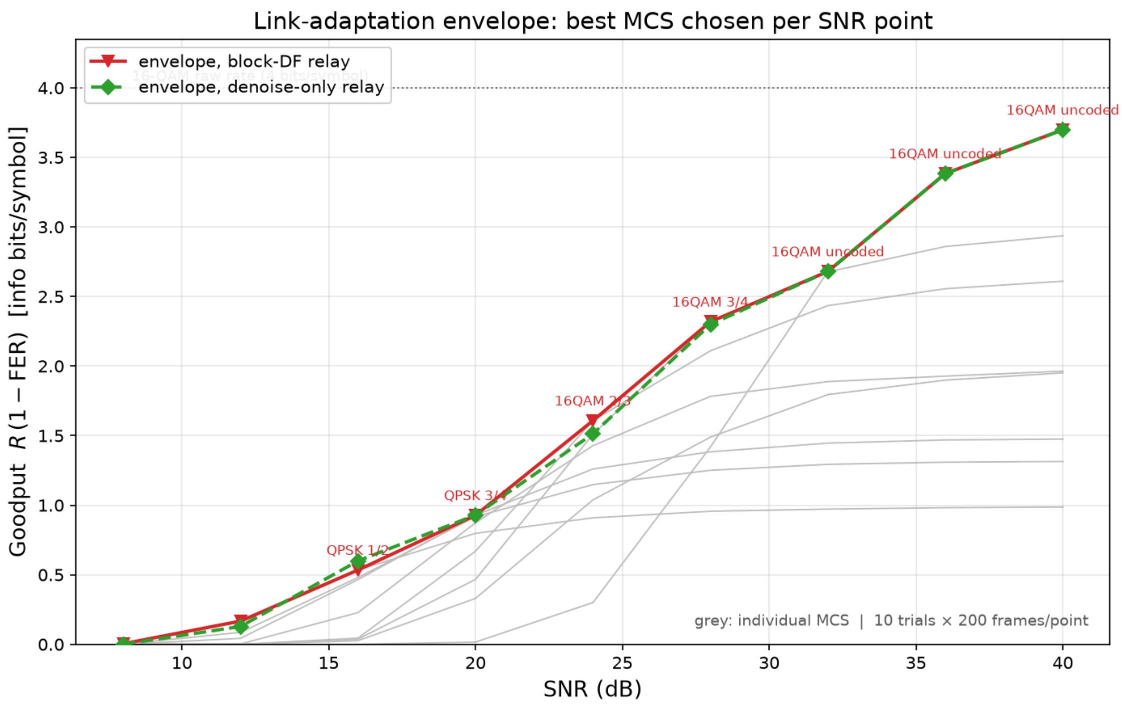

Figure 5.11: The envelope of Table 5.15 (colored) against every individual fixed MCS on the block-DF relay (grey), which is the visual case for adaptation: the envelope tracks the upper hull of the grey curves rather than any single one of them, switching MCS as SNR crosses each rate's operating threshold. The two relay strategies' envelopes are visually almost the same curve.

Two results follow, and the second qualifies the mechanism finding of Table 5.9 substantially.

The envelope is a real staircase, and its top step is "turn the code off". Both relay strategies climb QPSK 1/2 → QPSK 3/4-or-2/3 → 16-QAM 2/3 → 16-QAM 3/4 → *16-QAM uncoded* as SNR rises from 8 to 40 dB, each switch occurring once the next rate's threshold has cleared. The optimum at high SNR is no code at all: once the channel is good enough, coding redundancy is pure overhead, and the highest-goodput choice is the raw 4 bits/symbol of 16-QAM with zero coding loss. This is the same lesson as the equal-throughput correction above, now reached by direct optimization over the grid rather than by hand-picking one comparison. Table 5.15's C column confirms the envelope never approaches the information-theoretic ceiling: at 40 dB the achieved 3.70 sits at 30% of the 12.46-bit Shannon capacity there, a gap set by the fixed 16-QAM constellation (itself capped at 4 bits/symbol) and residual FER, not by anything a smarter relay strategy could close.

Once the rate adapts, which relay decodes the code barely matters. The two envelopes in Table 5.15 are close at every SNR and identical for SNR $\geq$ 32 dB, where both select 16-QAM uncoded and there is no code for the strategies to differ over. Where they do differ, the gap is small and does not consistently favor one side: block-DF leads at 8, 12, 24 and 28 dB, denoise-only leads at 16 and 20 dB, and the largest gap over the whole sweep is 0.095 information bits/symbol at 24 dB, under 6% of the envelope value there. Compare this to how much the *MCS* choice moves goodput at fixed relay strategy: at 24 dB the spread across the seven block-DF MCS options is 1.31, roughly $14\times$ the 0.095 gap between relay strategies; at 32 dB and above the MCS spread exceeds 1.7 while the relay-strategy gap is exactly zero. Rate adaptation, in other words, dominates the relay-decoding decision by more than an order of magnitude across most of this sweep.

This bounds the mechanism finding of Section 5.5.2 rather than reversing it. That the destination can repair a denoising relay's errors but not a re-encoding relay's remains true and remains the reason the two strategies differ at all – but once the system is allowed to pick its own operating rate, the room that difference has to matter shrinks to a few percent of the achievable goodput, because the adaptive link mostly avoids operating in the regime (a fixed rate pushed past its comfortable threshold) where the mechanism produces a large gap in the first place. The practical recommendation this chapter's coded study converges on is therefore rate adaptation first: get the MCS right for the channel, and only then does it become worth asking which relay strategy to run within it.

5.5.5 Capacity Under a Latency Constraint

Section 5.5.4 optimizes goodput with no bound on how long the relay may take to decide. A link with a deadline cannot make that assumption, and the question it faces instead – the achievable rate once blocklength (equivalently, latency) is fixed rather than left free – is a standard one in the coding-theory literature: the finite-blocklength capacity of Polyanskiy, Poor and Verdú [28], and the delay-limited capacity of Hanly and Tse [29] for fading channels specifically. Both frame it the same way this section does: constraining the number of channel uses (Polyanskiy et al.) or imposing a hard per-block deadline (Hanly–Tse) costs achievable rate relative to the unconstrained (Shannon, or ergodic) capacity, and by how much is exactly the object of study.

The two relay strategies differ in what they cost against such a deadline. The destination always buffers a full coded frame before it can decode, regardless of which relay handled the first hop – that cost is unavoidable for any coded MCS and is shared by both strategies. Block-DF adds a second full-frame buffer at the relay itself, since it must decode, re-encode and only then forward, so its round-trip latency is the frame length *twice*. The denoise-only relay adds only its window (Table 5.14's 10 symbols, independent of frame length or code rate), so its round-trip latency is the frame length once. Re-deriving the goodput envelope of Table 5.15 under a round-trip budget $L_{\max}$ – the best goodput each relay strategy can still deliver using only MCS options whose round-trip latency does not exceed $L_{\max}$ – therefore penalizes block-DF twice as hard per unit of code redundancy as it penalizes the denoise-only relay, using the same already-measured FER data as Table 5.15

and no new simulation.

Table 5.16: Link-adaptation envelope re-derived under a 150-symbol round-trip latency budget, same layout and data source as Table 5.15.

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)
8	QPSK uncoded	0.0000	QPSK uncoded	0.0000
12	QPSK uncoded	0.0000	QPSK 3/4	**0.0089**
16	16-QAM 3/4	0.0370	QPSK 3/4	**0.3719**
20	16-QAM 3/4	0.4667	QPSK 3/4	**0.9156**
24	16-QAM 3/4	1.5096	16-QAM 2/3	**1.5129**
28	16-QAM 3/4	**2.3185**	16-QAM 3/4	2.2993
32	16-QAM uncoded	2.6800	16-QAM uncoded	2.6800
36	16-QAM uncoded	3.3840	16-QAM uncoded	3.3840
40	16-QAM uncoded	3.6980	16-QAM uncoded	3.6980

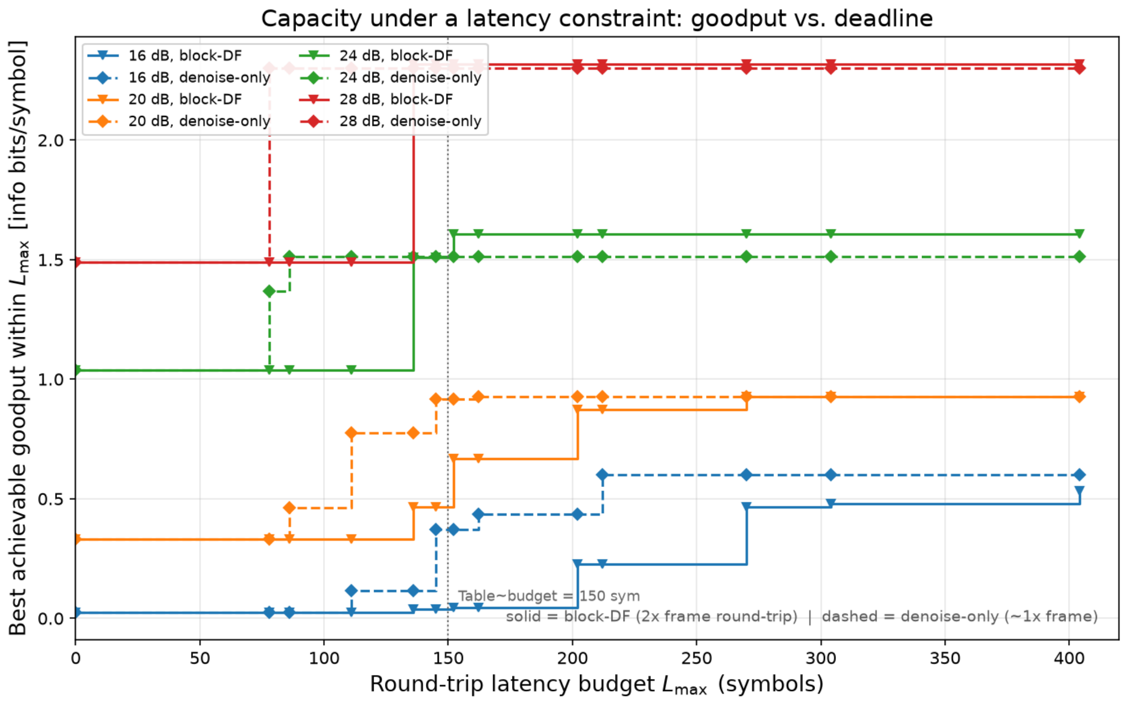

Figure 5.12: Best achievable goodput as a function of the round-trip latency budget $L_{\max}$, at the four SNRs where the two relay strategies differ (16, 20, 24, 28 dB). Each curve is a step function: goodput jumps each time $L_{\max}$ crosses an MCS's round-trip latency. The denoise-only relay's steps (dashed) occur at roughly half the $L_{\max}$ of the block-DF relay's (solid) for a comparable rate, since block-DF pays the frame cost twice.

Under a tight budget, block-DF can be locked out of coding altogether. At $L_{\max} = 150$ symbols (Table 5.16), block-DF's cheapest coded option (16-QAM 3/4, 136 symbols round-trip) barely clears the budget, while its next-cheapest (16-QAM 2/3, 152 symbols) does not – forcing it down from the unconstrained envelope's choice at several SNRs. The denoise-only relay, whose round-trip cost for the same MCS is roughly half, still has QPSK 3/4 (145 symbols) and every 16-QAM rate available. The result is a gap the unconstrained comparison of Table 5.15 understates by an order of magnitude at 16 dB: $10.0\times$ (0.372 vs. 0.037 information bits/symbol) there, and $2.0\times$ at 20 dB, where the unconstrained envelope showed the two strategies within 6% of each other.

Tighten the budget further and the reversal spreads to SNRs where block-DF led unconstrained. At $L_{\text{max}} = 100$ symbols, block-DF has *no* coded option left at any of these four SNRs – every one of its coded MCS choices exceeds the budget – so it is pinned to uncoded transmission throughout, while the denoise-only relay still reaches 16-QAM 2/3 or 3/4 at 20, 24 and 28 dB. At 24 and 28 dB, this is a direct reversal: Table 5.15's unconstrained envelope has block-DF *ahead* at both points, but under the 100-symbol budget denoise-only leads by $1.46\times$ and $1.54\times$ respectively, because the MCS that gave block-DF its unconstrained edge is no longer reachable in time.

The mechanism finding of Section 5.5.2 was about which relay decodes better; Section 5.5.4 showed that once the rate is free to adapt, that question barely matters. This section adds the constraint a real deadline imposes and finds the opposite qualitative result: under a latency budget, which relay strategy is used matters again, and matters through an architectural fact – how many times the frame must be fully buffered – rather than through detection accuracy. A link with headroom to pick its own rate should do so per Section 5.5.4; a link additionally bound by a hard deadline should prefer the relay strategy that does not double-buffer, independent of which one decodes more accurately.

It is important to emphasize that all conclusions of this chapter are specific to the canonical memoryless channel model. Whether those conclusions remain valid when the underlying channel assumptions are violated is the subject of Chapter 7.

Chapter 6

Multi-Level Modulation: 16-QAM

This chapter is about one constellation and nothing else: 16-QAM on the canonical Rayleigh channel. The canonical comparison of Chapter 5 answers hypotheses H1–H5 with QPSK, which places a single bit on each axis of the complex plane and is therefore absorbed by the I/Q-split relays without modification. 16-QAM places *four* levels on each axis, and at that point the per-axis formulation breaks down. That breakdown, and its repair, are the whole content of this chapter.

This chapter reports one study: reformulating the relay as a joint N-class classifier over the full 2D constellation ($N = 16$) removes the structural BER floor that per-axis I/Q-split processing imposes on multi-level constellations, measured directly against that per-axis baseline within the same experiment. MIMO, 16-PSK, CSI injection, and end-to-end learning remain future work.

Where this chapter sits. The thesis's argument is a ladder of assumptions, each layer removing one (Table 7.1). This chapter is *not* a rung on that ladder: it is a branch off layer 1. It holds every channel assumption of the canonical setup fixed — memoryless, known, perfect CSI — and varies the constellation instead. Raising the constellation order stresses the relay's *output* formulation, not its knowledge of the channel, which is why the conclusions here remain confirmatory in the same sense as Chapter 5: the learned relay earns its place by handling a representation classical per-axis processing cannot, not by outperforming a classical method on equal terms.

The ladder proper resumes in Chapter 7, which holds the constellation fixed and removes the channel assumptions one at a time. That is where learned relaying stops matching classical processing and starts beating it.

Scope relative to the canonical model. This chapter extends the canonical study along the modulation axis only. The underlying channel model remains unchanged: the relay

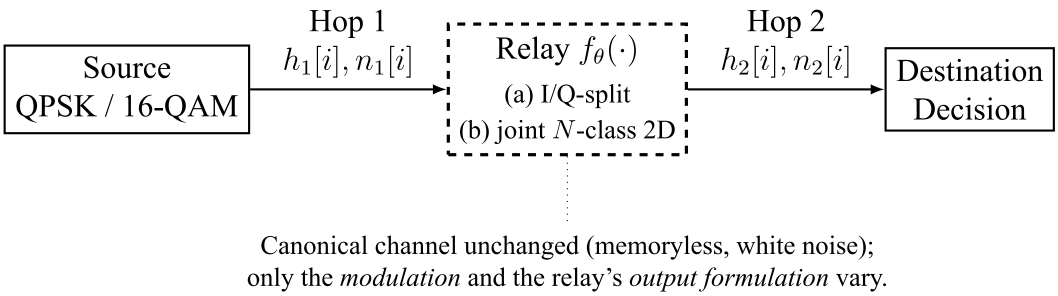

Figure 6.1: Modulation extension (Chapter 6). The canonical memoryless benchmark of Figure 1.1 is retained; only the constellation (QPSK/16-QAM) and the relay output formulation change: (a) per-axis I/Q splitting versus (b) a joint N-class 2D classifier over the full constellation.

channel is memoryless, the noise is white, transmission is uncoded, and relay processing is evaluated on a per-symbol basis. Consequently, the experiments reported here continue to address matched-model relay processing rather than channel memory, sequence detection, or channel estimation. These effects are introduced only in the unknown-channel study of Chapter 7, which deliberately departs from the canonical assumptions by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase.

6.1 16-Class 2D Classification for QAM16

Goal: Eliminate the structural BER floor imposed by per-axis I/Q splitting for 16-QAM by utilizing full 2D decision boundaries.

Trials: - **Topology:** SISO - **Modulation scheme:** 16-QAM - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** None (Softmax implicitly via Cross-Entropy loss) - **Special case:** 16-class joint 2D classification

Conclusion: Treating the relay as a joint 16-point classifier over the full 2D constellation removes the structural floor that per-axis processing imposes. The floor is properly measured as the *excess* over symbol-wise DF, not as raw BER, because on the canonical Rayleigh channel DF itself is limited to 0.03748 at 20 dB by the channel rather than by the relay. On that measure the per-axis relays carry an excess of 0.0161–0.0174, and the joint formulation cuts it to 0.0001–0.0019 for five of the six architectures, a reduction of 95% or more, leaving them level with DF. Mamba-2 SSD is the exception, retaining an excess of 0.0121.

Two things follow. The improvement factor in raw BER is a near-identical 1.38–$1.43\times$ across five architectures, and that uniformity is itself evidence: what the joint head repairs is the per-axis *formulation*, not any individual architecture's capacity. And the endpoint is parity with the classical baseline rather than an advantage over it, which is H2 reappearing one constellation higher — the learned relay earns its place here by handling a representation per-axis processing cannot, not by beating DF.

6.1.1 Results Tables

Table 6.1 presents the BER at 20 dB for all variants alongside the classical AF and DF baselines.

Table 6.1: 16-QAM at 20 dB on the canonical Rayleigh channel: per-axis 4-class relays against joint 16-class 2D classifiers. The joint formulation improves every architecture and brings five of the six level with DF (0.03748), cutting their excess over it from about 0.016 to 0.001 or less; Mamba-2 SSD is the exception. The cGAN is excluded from the re-run, as in Table 5.4.

Relay	4-cls BER @ 20 dB	16-cls BER @ 20 dB	Improvement
MLP	0.05491	**0.03840**	1.43×
VAE	0.05382	**0.03762**	1.43×
Hybrid	0.05362	**0.03748**	1.43×
Transformer	0.05360	**0.03814**	1.41×
Mamba-S6	0.05421	**0.03934**	1.38×
Mamba-2 SSD	0.05381	**0.04953**	1.09×
AF	0.04498	n/a	n/a
DF	0.03748	n/a	n/a

59 6.1.2 Results Figures

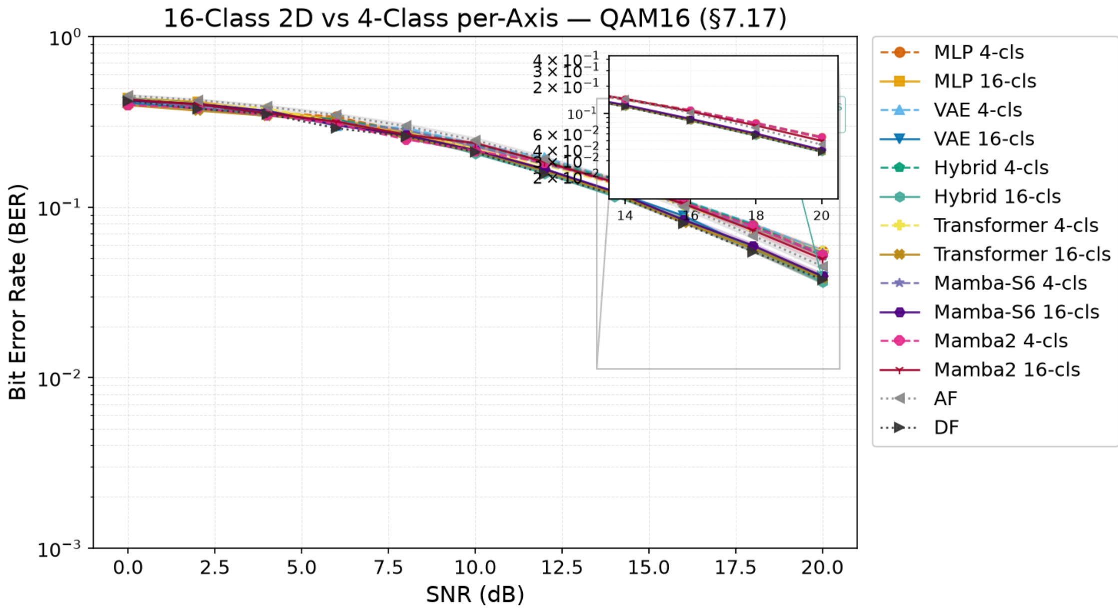

Figure 6.2: 16-QAM BER vs. SNR for all relay variants (4-class and 16-class) on AWGN. The 4-class variants (dashed) plateau at BER ≈ 0.008 while the successfully trained 16-class variants (solid) continue decreasing, approaching and matching the DF baseline.

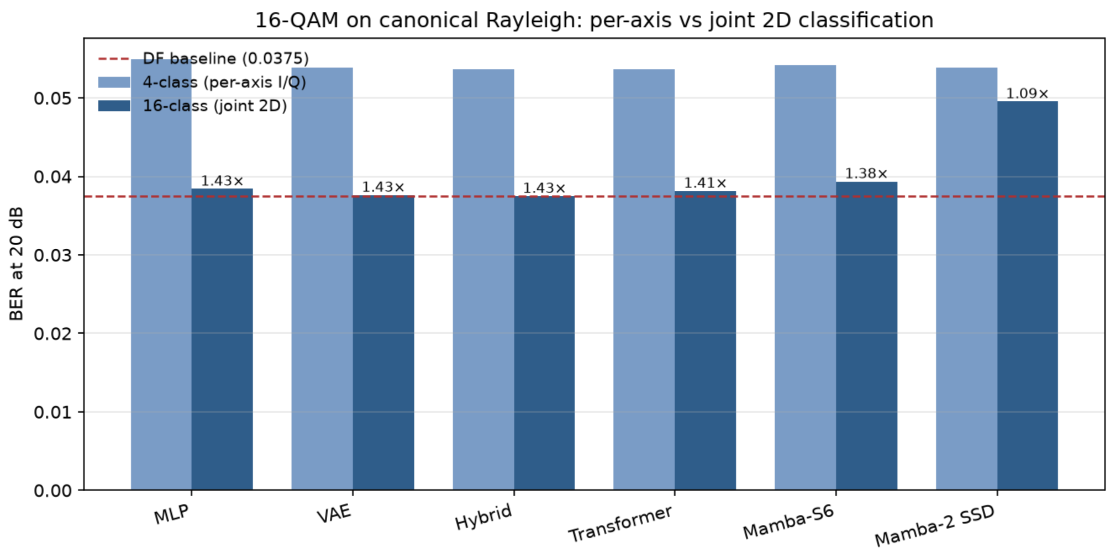

Figure 6.3: 4-class per-axis against joint 16-class 2D relays, 16-QAM at 20 dB on the canonical Rayleigh channel, regenerated from the same results file as Table 6.1. The joint formulation improves every architecture by a near-identical factor, annotated above each bar, which points at the per-axis formulation rather than any architecture's capacity as the thing being repaired. The dashed line is symbol-wise DF: the 16-class bars reach it and stop, so the joint head buys parity with the classical baseline, not an advantage over it.

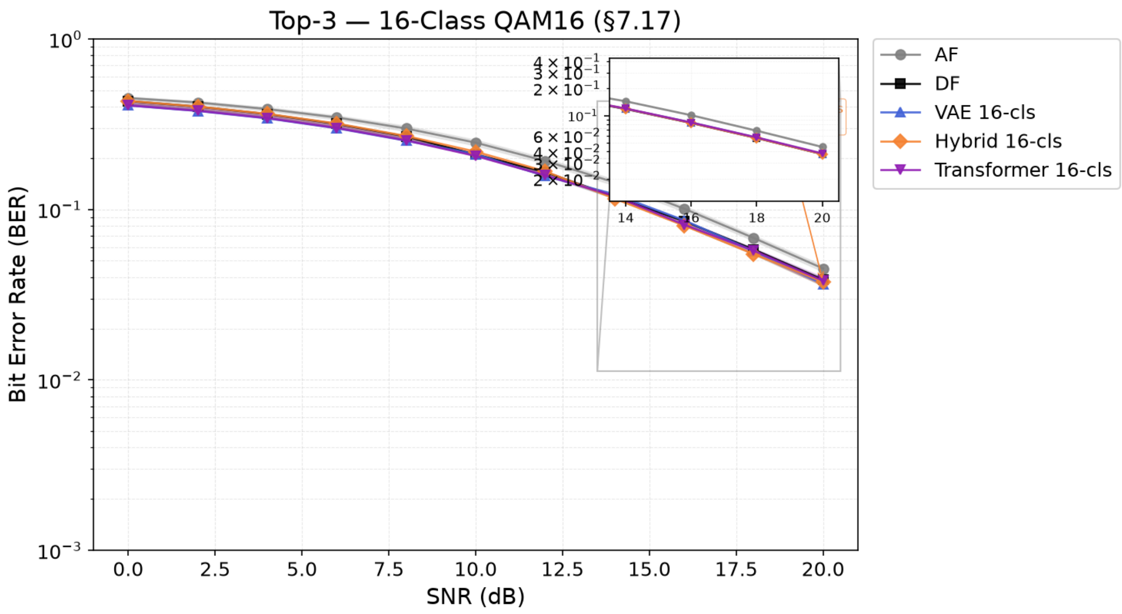

Figure 6.4: Top-3 16-class relay variants compared against AF and DF. VAE 16-cls, Transformer 16-cls, and MLP 16-cls all match or approach DF performance across the full SNR range.

Chapter 7

Learned Relaying under Unknown and Mismatched Channels

This is the central chapter of the thesis. Everything before it is either calibration or a confirmatory result: on a matched, known channel a minimal learned relay does *not* improve on classical decode-and-forward (H2), and raising the constellation order does not change that. Here the learned relay wins, and wins decisively.

The condition under which it wins is precise, and stating it is as much a part of the claim as the win itself. **When the channel's structure is unknown to the receiver chain, or lies outside the model class the classical relay assumes, a fixed 169-parameter learned relay restores reliable relaying where memoryless classical relays fail outright.** Under three-tap intersymbol interference (S1, Table 7.3) the classical relays do not merely degrade: they saturate near the analytic 0.25 floor and then get *worse* as the SNR rises, symbol-wise DF bottoming out at 0.1839 at 8 dB and climbing to 0.2462 at 20 dB as hard decisions lock the interference in. No amount of transmit power recovers the link. The learned relay, whose length-11 window spans the channel memory, reaches 0.0123 at 8 dB and falls below 5×10^{-5} from 16 dB upward. What separates the two is the function class and not the information available: this filter is fixed, and the relay is trained on it, so it is as well informed about the channel as the genie-CSI Viterbi detector it is measured against. A memoryless slicer cannot invert a channel with memory at any transmit power, however much it is told; a windowed one can. Whether a learned relay also copes with a channel it has *not* seen is a separate question, answered at layers 3 and 4 below, where the channel is redrawn every block. That is not a marginal gain over a classical baseline; it is the difference between a working link and a broken one, and it is the thesis's answer to whether learning can defeat classical processing in relaying.

The boundary of the claim is equally sharp, and the chapter is written to establish it rather than to avoid it. The learned relay does not beat a *matched* classical receiver: given a channel estimate, pilot-aided Viterbi maximum-likelihood sequence estimation remains ahead of it by 1–1.5 dB. The learned relay's advantage is therefore not superior detection but the absence of a precondition. Where the channel is fixed and known to both, it is simply the cheaper detector, constant and parallelizable per symbol against the trellis's M^{L-1}; where the channel must be identified and cannot be, it is the one that still functions, needing neither a per-block estimate nor the channel family. Where classical processing can identify the channel it should be used; where it cannot, the learned relay is what still works.

Rather than a single comparison, the chapter is a systematic map, along every axis relevant to practical deployment, of *when* the learned relay adds value over classical processing, and, equally, when it does not.

The argument as a ladder of assumptions. The thesis as a whole is organised as a monotone sequence in which each layer removes exactly one assumption from the layer above, and the classical–learned comparison is repeated unchanged at every layer. Table 7.1 is that sequence with its outcomes. Reading it downward is the thesis's argument in one page: classical processing is not beaten where it is well posed, and is beaten precisely where its preconditions fail.

Table 7.1: The four layers of the argument. Each removes one assumptio above; the relay comparison is otherwise identical. BER figures are the repre ing points cited in the sections named.

Layer	Assumptions	Strongest
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-w rameters

Layer	Assumptions	Strongest
2	+ channel memory (3-tap ISI), CSI still perfect (Section 7.1)	Viterbi M taps
3	+ CSI uncertainty: a finite pilot budget (Section 7.1.4)	Pilot-aide then MLS
4	+ unknown channel family and statistics, redrawn per block (Sections 7.1.3, 7.1.5)	CMA blir

Two features of the ladder are worth naming before the evidence is presented. First, the opponent changes character at layer 2: layer 1 compares relay *functions* (AF and DF against learned relays), whereas from layer 2 onward the strongest classical option is a *receiver*, Viterbi MLSE, which is a considerably heavier and better-informed baseline. This is deliberate, since comparing against a weaker classical option at the layers where the learned relay wins would make the result worthless. Second, what degrades down the ladder is not the classical algorithms themselves but their *preconditions*: MLSE remains optimal at every layer for the channel it is given, and it loses ground only because the channel it is given grows less like the true one. The learned relay never becomes a better detector; it becomes the only one whose requirements are still met.

The investigation is deliberately adversarial to the learned relay. Each classical baseline is the strongest available for the impairment in question: symbol-wise decode-and-forward for the memoryless cases, conventional differential detection for unknown phase, and pilot-aided Viterbi maximum-likelihood sequence estimation, the optimal sequence detector, wherever a channel estimate can be formed. The learned relay is granted no per-block channel estimate at any layer, and it never adapts online. How much it knows in advance differs by layer, and the distinction matters when reading the results. At layer 2 the ISI filter is fixed, so training on it leaves the relay as well informed as the genie-CSI detector it is compared with, and what that layer isolates is the function class rather than channel knowledge. At layers 3 and 4 the channel is redrawn every block and the relay meets realizations it has not seen, which is where the family-agnostic claim is actually earned. That claim is scoped precisely: "family-agnostic" means the relay needs no per-block identification of *which* realization is active, having amortized the family's parameter distribution into its weights at training time; it is trained on, and evaluated within, the same parametric channel family (e.g. the 3-tap FIR structure) throughout, and generalization to a structurally different, untrained channel family is not tested here. The chapter follows the ladder in order. It opens with a necessary control (flat unknown channels, where nothing should fail), then takes layer 2, the impairment that does break classical relays (unknown memory), bounding it against the optimal matched receiver (Viterbi with exact taps) and against a composite multi-impairment cascade. Layer 3 relaxes the channel estimate to a finite pilot budget and locates the crossover in pilot count and block length; layer 4 removes the posterior altogether and compares against genuinely blind classical equalization. A final section prices the whole comparison in computational complexity. The result is a precise delineation (the learned relay never beats a matched classical receiver, but occupies a well-defined niche of family-agnostic, identification-free, constant-complexity mitigation) that constitutes the thesis's main claim beyond the confirmatory core.

Relationship to the canonical model. The canonical setup of this thesis (Chapters 5– 6) assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. The present chapter deliberately departs from those assumptions by introducing impairments that are not part of the canonical model, including channel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and channel-model mismatch. Consequently, sequence-detection and equalization techniques such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become rel-

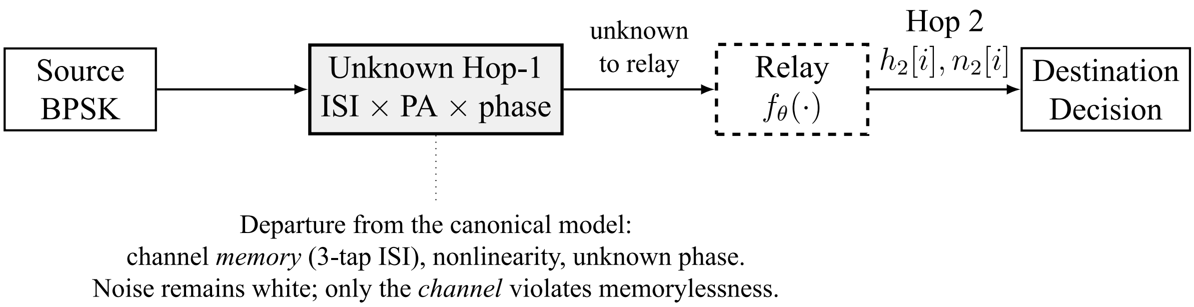

Figure 7.1: Unknown/mismatched-channel study (Chapter 7), a deliberate departure from the canonical benchmark of Figure 1.1. The Hop-1 channel is replaced by an unknown impairment (3-tap ISI, amplifier nonlinearity, and unknown phase) that is not disclosed to any relay. The noise remains white; only the channel violates the memoryless assumption, which is why sequence detection (Viterbi MLSE) becomes the relevant classical benchmark here and nowhere else in the thesis.

evant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

7.1 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay's value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 169-parameter MLP mitigates by learning it from data (H6)?

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [30] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapters 5–6, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Trials.

- **Topology:** SISO (both hops)
- **Modulation:** BPSK
- **Hop-1 channel (unknown to all relays):**
  - 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$
  - Control: canonical Rayleigh channel
  - Flat unknown channels (block phase, gain, per-branch asymmetry; Section 7.1.1)
  - Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 7.1.3)
- **Hop-2 channel:** AWGN or canonical Rayleigh, equal SNR per hop
- **Relays:**
  - AF
  - Symbol-wise DF
  - MLP (window 11, one hidden layer of 13 tanh units, 169 parameters; trained at SNR $\in \{5, 10, 15\}$ dB using the canonical training protocol)
  - For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, evaluated with either genie CSI or a 3-tap least-squares channel estimate obtained from 200 pilot symbols
- **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB

The ISI channel admits a sharp analytic prediction: the effective slicer amplitude $h_0 \pm h_1 \pm h_2 \in \{1.67, 0.91, 0.61, -0.15\}$ (normalized) has one of four equiprobable patterns that *flip the symbol sign*, so symbol-wise DF has an irreducible BER floor of 0.25 (approached from below as SNR grows); AF inherits it. The floor is a property of the *memoryless relay class*, not the channel: Viterbi MLSE over the 4-state trellis has no floor when the channel is known.

7.1.1 Control: Flat (Memoryless) Unknown Channels

Before attributing any classical failure to "unknownness," it must be separated from *memory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver chain does not know the channel parameters, but the channel introduces no intersymbol coupling. Three physically motivated cases are tested, each with the parameter redrawn per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscillator/synchronization offset), with a DBPSK source and conventional differential detection as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$

(per-symbol saturating-PA gain), coherent BPSK. The 169-parameter MLP is trained identically to before.

These channels remain memoryless and therefore stay within the scope of the canonical model despite the parameter uncertainty. Consequently, they serve as a control that separates the effect of parameter uncertainty from the effect of channel memory: any performance change here arises from model uncertainty, not from memory.

Conclusion (memorylessness ⇒ no classical failure, and nothing to mitigate): On all three flat channels the classical relay does *not* fail, and the MLP matches it to within statistical noise (maximum BER difference 0.0036 across every case and SNR). The reasons are structural: for F1, unknown phase is exactly what differential detection absorbs; for F2, sign(y) is *scale-invariant*, so unknown positive gain is irrelevant by construction; for F3, the fixed threshold is only mildly suboptimal because the asymmetry preserves the noiseless symbol's sign. The classical assumption, memorylessness, still holds in each case, so there is nothing for a learned relay to remove. This is the necessary control for the ISI result: the learned relay's advantage is triggered not by parameter uncertainty but by *unmodeled memory* (or amplitude nonlinearity), which is what breaks the memoryless relay class.

Table 7.2: Flat unknown channels: classical relay vs. MLP-169 BER at 8/12/16/20 dB (mean, 10 trials). The maximum DF–MLP gap over all cases and SNRs is 0.0036, within statistical noise. Unknown parameters alone do not defeat classical relays; memory does.

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0649	0.0292	0.0119 / 0.0048
	MLP-169	0.0657	0.0288	0.0122 / 0.0048
Unknown gain	DF	0.0681	0.0330	0.0131 / 0.0050
	MLP-169	0.0682	0.0333	0.0132 / 0.0050
Branch asymmetry	DF	0.0760	0.0299	0.0125 / 0.0049
	MLP-169	0.0745	0.0300	0.0122 / 0.0052

Conclusion (H6: confirmed as stated, with two boundaries): Against the flat-channel control of Section 7.1.1, where unknown parameters alone caused no classical failure, the memory case is qualitatively different. On the unknown ISI channel, both memoryless classical relays fail: DF's BER is *non-monotonic*, bottoming out at 0.184 around 8 dB and then rising toward the analytic 0.25 floor, reaching 0.246 at 20 dB (more transmit power makes DF strictly worse), and AF behaves identically. The 169-parameter MLP, whose window spans the 3-tap memory, learns a joint equalizer–detector and restores reliable relaying: BER falls monotonically below 5×10^{-5} at 16 dB (AWGN second hop), and to 2.5×10^{-3} at 20 dB (Rayleigh second hop). The Viterbi benchmarks set the first boundary honestly: classical theory *does* solve this channel once its family is known, and genie-CSI MLSE leads the MLP by 1–1.5 dB across the clearly-separated mid-range (roughly 2–10 dB); a 200-pilot LS estimate recovers genie performance almost exactly there too. At the two extremes the ordering is not clean: near the 0 dB floor the two are within noise of each other (both near the analytic 0.25 ceiling), and with the Rayleigh second hop the gap closes again from 12 dB upward, the two tracking within 0.0001–0.0002 of each other through 20 dB (e.g. 0.0025 vs. 0.0024 there) – differences comparable to the reported standard errors rather than a resolved ordering. This does not contradict Viterbi's sequence-ML optimality; it means the *margin* by which it holds is SNR-dependent and vanishes at both ends of this particular sweep, not that the MLP overtakes it. Two things follow, and neither is about channel knowledge, since the filter here is fixed and the relay is trained on it exactly as Viterbi is given it. The first is that the *function class* decides the outcome: a memoryless slicer cannot invert a channel with memory at any transmit power, which is why DF worsens with SNR, while a windowed relay recovers the link. The second is a complexity result: with the same channel information, 169 parameters land within 1–1.5 dB of the optimal sequence detector at constant cost per symbol, against the trellis's M^{L-1}, and that cost does not grow with channel memory. The stronger property, mitigating channels whose *family* is unknown or outside linear MLSE's model class, is not demonstrated by this experiment and is established later, by the composite cascade of Section 7.1.3 and the posterior-free study of Section 7.1.5. The control confirms the second boundary (H2): on the canonical Rayleigh channel the MLP only *matches* DF. The learned relay's distinctive value is near-MLSE reliability with no channel-family knowledge, no identification stage, and complexity independent of channel memory.

Table 7.3: Unknown-channel BER (mean over 10 trials). The ISI setups exhibit memoryless-relay error floors (≈ 0.25); the 169-parameter MLP restores reliable relaying within $\approx$1–1.5 dB of genie-CSI Viterbi MLSE, and matches it end-to-end when the Rayleigh second hop dominates. The control shows no MLP advantage on the canonical channel, consistent with H2.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
Unknown ISI → AWGN	AF	0.1919	0.1785	0.1956	0.2233
	DF	0.1839	0.2010	0.2279	0.2462
	MLP (169p)	**0.0123**	**0.0002**	**<5e-5**	**<5e-5**
	Viterbi (genie CSI)	0.0072	<5e-5	<5e-5	<5e-5
	Viterbi (200-pilot LS)	0.0074	<5e-5	<5e-5	<5e-5
Unknown ISI → Rayleigh	AF	0.2047	0.1916	0.2030	0.2231
	DF	0.2031	0.2098	0.2316	0.2471
	MLP (169p)	**0.0412**	**0.0150**	**0.0061**	**0.0025**
	Viterbi (genie CSI)	0.0364	0.0151	0.0061	0.0024

Setup	Relay	8 dB	12 dB	16 dB	20 dB
	Viterbi (200-pilot LS)	0.0362	0.0151	0.0062	0.0024
Control: canonical Rayleigh	AF	0.1050	0.0559	0.0281	0.0136
	DF	**0.0687**	**0.0299**	**0.0124**	**0.0050**
	MLP (169p)	0.0691	0.0307	0.0133	0.0058

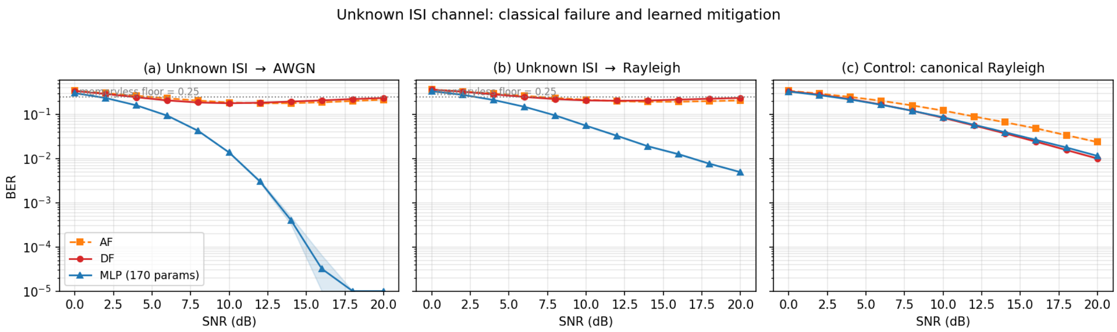

Figure 7.2: Unknown-channel relaying, BER vs. SNR with 95% CI bands. (a)–(b) On the unknown ISI channel, the memoryless AF/DF relays are pinned near the analytic 0.25 floor (DF non-monotonic in SNR); the 169-parameter MLP restores monotonically decreasing BER within ≈1–1.5 dB of genie-CSI Viterbi MLSE, which a 200-pilot LS estimate matches almost exactly. With the Rayleigh second hop (b), MLSE and the MLP are statistically indistinguishable end-to-end. (c) Control: on the canonical Rayleigh channel the MLP only matches DF, consistent with H2.

7.1.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout. The canonical setup of Chapter 5 is QPSK, so this subsection checks that the failure mode is a property of the channel rather than of the constellation it was first observed on. The 0.25 floor derived in Section 7.1 is a property of the *binary* slicer: it counts the $2^3 = 8$ equiprobable tap-sign patterns that flip a ± 1 decision. That argument does not extend to a four-point per-axis alphabet, so it is not obvious a priori whether either the elevated error floor or the ordering between the learned relay and genie-CSI Viterbi MLSE survives a move to a higher-order modulation. This subsection repeats the unknown-ISI study with Gray-coded QPSK in place of BPSK, keeping every other element fixed: the same 3-tap ISI channel $h = [1.0, 0.7, 0.5]$ (normalized) on Hop 1, AWGN or canonical Rayleigh on Hop 2 at the same nominal per-hop SNR (identical convention, $\gamma = 10^{\mathrm{SNR_{dB}}/10}$, memory-bank/techContext.md), and the same relay set generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at SNR $\in \{5, 10, 15\}$ dB), and genie-CSI Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics). Evaluation uses 10 trials $\times$ $100{,}000$ bits per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 7.3 at the same SNR label; no such match is claimed.

Table 7.4: QPSK unknown-channel BER (mean over 10 trials). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated; the 193-parameter MLP-QPSK restores monotonically decreasing BER and, unlike the BPSK case, ends up *below* genie-CSI Viterbi MLSE from about 2 dB upward.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI $\to$ AWGN	AF	0.2437	0.2115	0.2024	0.2098
	DF	0.2193	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	**0.1292**	**0.0827**	**0.0601**	**0.0508**
	Viterbi (genie CSI)	0.1307	0.0852	0.0670	0.0618
QPSK: ISI $\to$ Rayleigh	AF	0.2715	0.2320	0.2137	0.2116
	DF	0.2528	0.2194	0.2145	0.2239
	MLP-QPSK (193p)	**0.1737**	**0.1066**	**0.0707**	**0.0553**
	Viterbi (genie CSI)	0.1749	0.1089	0.0774	0.0662

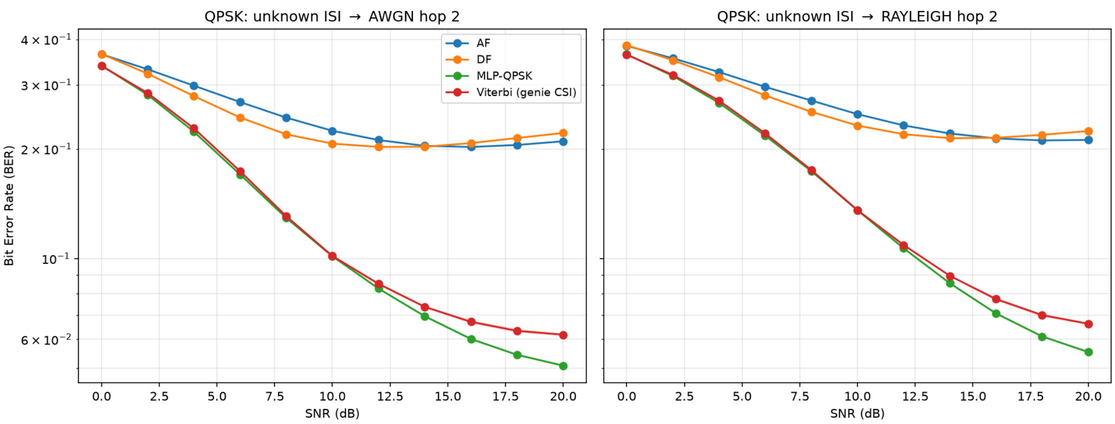

Figure 7.3: QPSK unknown-channel relaying, BER vs. SNR (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves. Unlike BPSK, the MLP-QPSK curve separates *below* genie-CSI Viterbi MLSE beyond about 2 dB, and the gap widens with SNR.

Conclusion (the ISI failure mode generalizes; the BPSK ordering does not): The qualitative pattern of Section 7.1 survives the move to QPSK: AF and DF plateau at an elevated, non-monotonic BER (AF ranges 0.2024–0.2715 and DF ranges 0.2024–0.2528 across the four tabulated points, broadly similar in magnitude to BPSK's memoryless-relay floor of

0.25 though no equivalent closed form is derived here), while the MLP restores monotonically decreasing BER, confirming that unmodeled channel memory, not modulation order, is what breaks the memoryless relay class (H6). What does *not* generalize is the BPSK ordering between the learned relay and the optimal classical benchmark. In the BPSK study, genie-CSI Viterbi MLSE beat the MLP by 1–1.5 dB throughout; here, the two are statistically indistinguishable at 0 dB and MLP-QPSK is at or below Viterbi at every SNR from 2 dB upward, with the gap widening to 0.0508 vs. 0.0618 (AWGN) and 0.0553 vs. 0.0662 (Rayleigh) at 20 dB. This is not a violation of MLSE optimality: Viterbi MLSE minimizes the *sequence* error probability under the assumed channel model, not the bit error rate directly. For BPSK, each trellis branch carries exactly one bit, so sequence-optimal and bit-optimal detection coincide; for Gray-coded QPSK, each branch carries two bits, and a wrong branch choice can differ from the true one by one bit (adjacent constellation point) or two (diagonal point) with different probabilities, so sequence-ML detection is no longer guaranteed to minimize BER specifically. The 193-parameter MLP-QPSK is instead trained directly on a per-symbol softmax cross-entropy objective, which is not subject to that mismatch. This is offered as the most plausible mechanism for the reversal; it was not isolated experimentally (for instance, by decomposing the Viterbi output into single- versus double-bit symbol errors), and is left as an open point rather than a proven claim. Either way, the practical conclusion sharpens rather than weakens H6 for higher-order modulation: on this unknown-memory channel, the fixed-complexity learned relay is not merely a near-optimal option at constant cost, as in the BPSK case, but the best-performing option measured here among either family, at the same constant per-symbol cost quantified in Section 7.1.6.

7.1.3 A Composite (Mixed) Channel: Mismatch with a Full Posterior

The preceding studies vary one impairment at a time. Real links stack several. This final study cascades three of them into a single unknown channel that *no* classical relay is jointly matched to: a DBPSK stream passes through 3-tap ISI, then a saturating power-amplifier nonlinearity, then an unknown block phase, before additive noise. Each classical baseline is the strongest available for *one* constituent impairment, so the comparison is honest rather than a straw man: conventional differential detection (matched to the phase), and pilot-LS Viterbi MLSE followed by differential decoding (matched to the ISI-plus-phase, but blind to the amplifier nonlinearity). The 169-parameter MLP and a 1,153-parameter MLP receive raw I/Q windows and no knowledge of any constituent. This composite channel departs from the canonical model in two ways: it introduces explicit channel memory through ISI and nonlinear distortion through the amplifier. The resulting benchmark therefore tests robustness to both memory and model mismatch simultaneously. **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, the same budget as the main unknown-ISI study

(Section 7.1); 95% CIs use the $1.96\,\sigma/\sqrt{n}$ convention throughout. The hop-1 impairment here is fixed rather than redrawn per block, so bits per trial and trials both reduce the reported uncertainty.

Conclusion: The composite defeats the naive classical relays outright, both AF and conventional differential detection saturate near the 0.25 memoryless-relay floor, because the ISI destroys the differential statistic that the phase-matched detector relies on. The 169-parameter MLP again restores reliable relaying, reaching 2.5×10^{-3} at 20 dB from raw I/Q with no impairment model. This reproduces the H6 pattern on a realistically mixed channel: memoryless-and-matched classical processing fails, and a minimal learned relay recovers. Two familiar boundaries also reappear. First, the strongest *constructed* classical baseline, pilot-LS Viterbi plus differential decoding, does not fail: by modelling the ISI and phase (though not the amplifier) it stays about 2 dB ahead of the MLP at low-to-mid SNR (0.067 vs. 0.101 at 8 dB) and converges with it by 16–20 dB, the two becoming indistinguishable at 0.0025 each at 20 dB. The learned relay's value here is *model-agnosticism* rather than superiority over a matched classical receiver: it reaches near-Viterbi reliability with no analytic model of any of the three cascaded effects, whereas estimate-then-MLSE must assume a linear FIR channel and therefore cannot represent the amplifier at all. This cascade is itself a fixed channel that the relay is trained on, so it speaks to the model class the two methods can express, not to generalization across channels; that is the subject of Section 7.1.5, where a fresh channel is drawn for every block. Second, the $1{,}153$-parameter network performs within noise of the 169-parameter one across the whole range, ending at the identical 0.0025 at 20 dB, with the small mid-SNR differences in both directions (0.0487 vs. 0.0531 at 10 dB for the larger network, 0.1020 vs. 0.1014 at 8 dB for the smaller) falling inside the confidence intervals. Nearly seven times the parameters therefore buys no measurable accuracy on the hardest channel studied, reconfirming the U-shaped complexity finding (H3). The composite thus consolidates the chapter's thesis: a single minimal learned relay handles an arbitrary stack of representable impairments without being told which are present, approaching but not exceeding the best impairment-matched classical receiver.

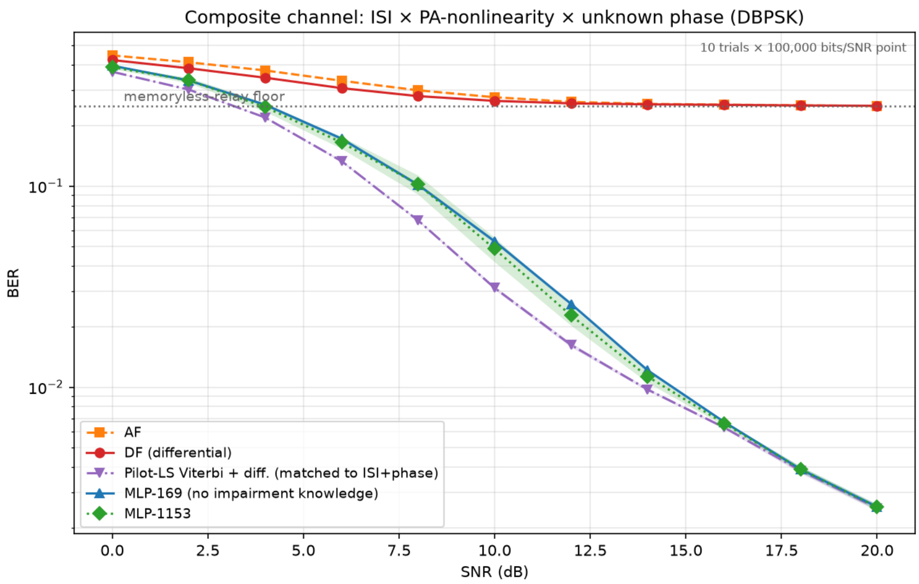

Figure 7.4: Composite channel (ISI $\times$ PA-nonlinearity $\times$ unknown phase, DBPSK; mean over 10 trials, 100,000 bits/point, 95% CI bands). Naive classical relays (AF, differential detection) saturate at the memoryless-relay floor; the 169-parameter MLP restores reliable relaying from raw I/Q with no impairment model, tracking about 2 dB behind the strongest constructed classical baseline (pilot-LS Viterbi + differential) and converging with it by 20 dB. The 1,153-parameter MLP adds nothing (H3).

7.1.4 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 7.1: the perfect-CSI assumption is dropped, and nothing else changes. Every classical result so far has been handed either exact channel taps or a 200-pilot estimate good enough to match them (Section 7.1.3), which is what a full posterior buys. Here the posterior is made *partial*: the pilot budget is finite, and the classical receiver must work from a correspondingly noisy estimate. Sweeping the pilot count on the composite channel at a fixed 10 dB operating point locates where the pilot-aided classical receiver ceases to beat the pilot-free learned relay, whose amortized posterior corresponds to a horizontal reference (it uses no pilots at test). **Evaluation:** 50 trials $\times$ 20,000 bits per operating point (1,000,000 bits), matching Section 7.1.5 and for the same reason: the channel is redrawn per block, so trials are draws from the impairment family. In panel (b) each trial is further subdivided into independent blocks of the swept length, so every block length is measured over the same total bit budget and over many channel draws rather than a single one.

Conclusion (a sharp estimation-variance cliff, and a rate–reliability tradeoff): With a generous budget the classical partial posterior wins, as expected: pilot-aided Viterbi reaches payload BER 0.0285 at 800 pilots and degrades only gently down to 0.0335 at 20 pilots, comfortably below the MLP's pilot-free 0.0487. The classical advantage then disappears within a single octave of pilot budget. At 10 pilots Viterbi has already lost its edge, 0.0545 against the MLP's 0.0487, and at 5 pilots it collapses to 0.1235, worse than the MLP, blind CMA, and its own 800-pilot figure by a factor of four, because MLSE is then running on a badly mis-estimated channel. The crossover therefore sits between 20 and 10 pilots per block. The mechanism is *estimation variance*, not a rank deficiency: with 5 pilot observations and 3 complex taps the least-squares problem is still formally overdetermined, so the channel remains identifiable in principle, but there are too few observations to average down the noise. The resulting LS error at 10 dB is large enough that the Viterbi trellis is occasionally built on a badly wrong channel, and these catastrophic fits dominate the mean. The 95% confidence interval at that point makes this explicit (0.1235 ± 0.0304, against ± 0.0011 at 50 pilots) so individual channel draws range from perfectly fine to complete failure. The learned relay, by contrast, is flat across the entire sweep at 0.0487: having amortized the channel *family* into its weights at training time, it neither benefits from nor requires per-block pilots. The crossover is thus governed by the reliability of the channel estimate, not by SNR: the classical receiver wins whenever it can afford enough pilots to estimate the channel *accurately*, and the pilot-free learned relay wins once the budget falls below that point.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload BERs of Viterbi and the MLP are now statistically indistinguishable at every block length, both flat at approximately 0.049, so reliability no longer separates them at all; what separates them is spectral efficiency. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% for identical error performance.

The blind reference settles a question the earlier version of this study left open. Blind CMA is also pilot-free, so the learned relay is not unique in avoiding pilots; the distinction claimed for it was that it needs *zero per-block adaptation*, a single fixed forward pass, whereas CMA must converge its adaptation loop inside each block. Measuring CMA per block confirms this directly: its payload BER is 0.1723 at $L = 40$ and only improves to 0.1653 at $L = 1000$, against the 0.0645 it achieves when given a $20{,}000$-symbol block to adapt over. CMA therefore does not converge within blocks of a thousand symbols or fewer, and is roughly three and a half times worse than either the MLP or pilot-aided Viterbi throughout this sweep. The partial-posterior picture therefore refines H6's boundary once more, and more sharply than before: against a classical receiver that can afford accurate channel estimation the learned relay achieves no better BER, but at short block lengths it matches that receiver's BER while removing its pilot overhead entirely, and at these block lengths it is the only pilot-free option that stays reliable, because the classical pilot-free alternative has no time to converge: CMA sits at 0.1723 on 40-symbol blocks and is still at 0.1653 at 1,000. This is a statement about short blocks specifically, and it does not carry over to Section 7.1.5, where blocks are long enough for CMA to converge and it comes within a factor of 1.3 of the learned relay.

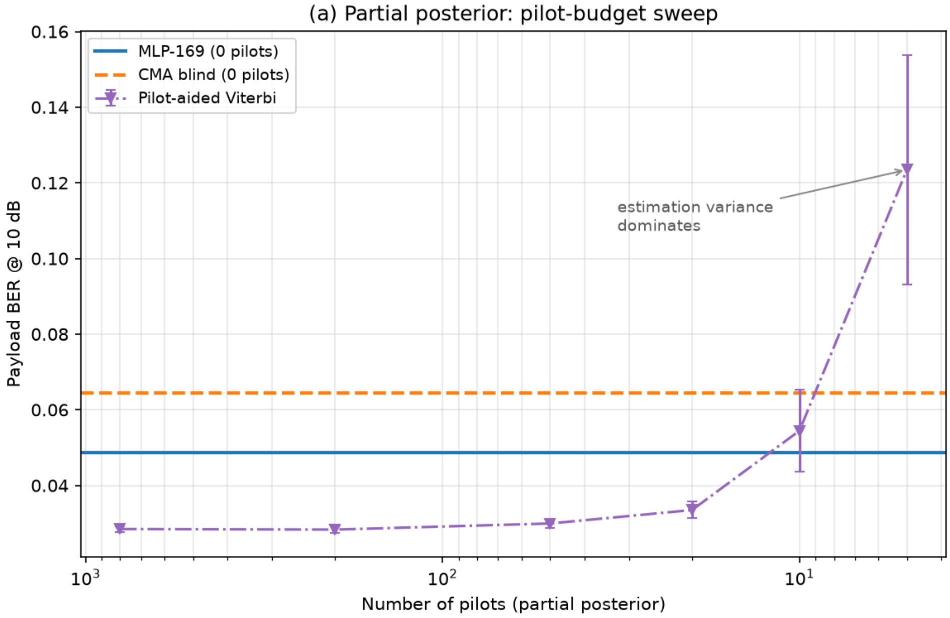

Figure 7.5: Pilot-budget sweep at 10 dB (mean over 50 channel draws, 20,000 bits/point). Pilot-aided Viterbi outperforms the pilot-free MLP down to approximately 20 pilots per block, loses its edge by 10 pilots, and collapses at 5 pilots when the least-squares channel estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304). In contrast, the MLP remains flat across the sweep because it operates with an amortized posterior and requires no pilots.

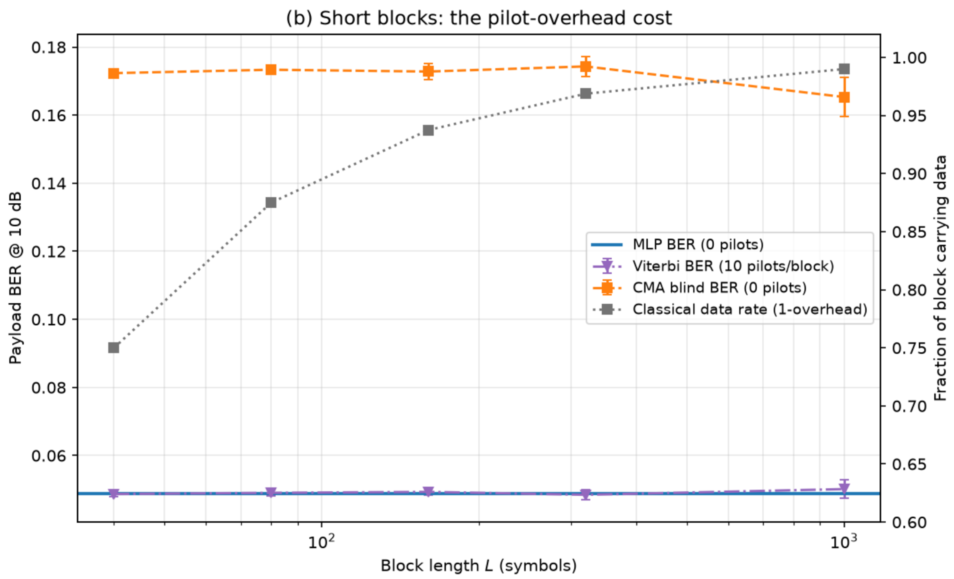

Figure 7.6: Block-length sweep with 10 pilots per block (mean over 50 channel draws, 20,000 bits/point, subdivided into independent blocks of the swept length). Pilot-aided Viterbi and the pilot-free MLP are indistinguishable in payload BER at every block length (both ≈ 0.049), but the classical receiver's pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. The other pilot-free option, blind CMA, cannot converge within blocks this short and sits roughly three and a half times worse throughout. The MLP therefore matches the pilot-aided receiver's reliability at no overhead, and is the only pilot-free method that remains reliable for short packets.

7.1.5 Layer 4 — Unknown Channel Family: The Posterior-Free (Blind) Regime

This is layer 4, the bottom of the ladder. Section 7.1.4 shrank the pilot budget until the classical estimate became too noisy to be worth having; this section removes it entirely, and with it the assumption that the channel's family and statistics are known at all. The Viterbi baseline of Section 7.1.3 enjoyed a per-block posterior: 200 pilot symbols yield a least-squares channel estimate, after which MLSE is optimal. Taking the pilot count to zero raises the sharper question, what if *no* such posterior is available? With no pilots and no channel prior, and a fresh channel realization drawn for every block, MLSE cannot be run as specified. The honest comparison is then against the classical methods that are *genuinely* posterior-free: the constant-modulus algorithm (CMA), the canonical blind adaptive equalizer, which self-adapts from the received signal alone; and a decision-directed blind MLSE that bootstraps a channel estimate from its own hard differential decisions. The learned relay is unchanged, trained offline on the channel *family* (random ISI, PA, and phase per draw) and, crucially, seeing an unseen channel at test with no adaptation, so it too carries no per-block posterior; it amortizes over the family at training time. Unlike the canonical study, the challenge here is not symbol-level denoising on a memoryless channel but inference on an unknown channel with memory and no per-block posterior information. **Evaluation:** 50 trials $\times$ 20,000 bits per SNR point (1,000,000 bits), 95% CIs by the $1.96\,\sigma/\sqrt{n}$ convention. The trial count is deliberately higher than the chapter's usual ten: unlike the fixed impairment of Section 7.1.3, the hop-1 channel here redraws its ISI, amplifier and phase realization for *every* block, so a trial is one draw from the impairment family and the ensemble average is limited by the number of trials, not by bits per trial. Fifty draws are used rather than ten so that the reported curves reflect the family and not a handful of particular channels.

A correction to the blind equalizer. The constant-modulus algorithm used here was originally implemented with a fixed (unnormalized) step size. Because the CMA error term is cubic in the equalizer output, a fixed step is a positive-feedback loop: the implementation happened to remain bounded over the short blocks first used, but diverges to numerical overflow on longer ones, after which the equalizer output is noise rather than an estimate. The step is therefore normalized by the input-segment energy in the standard NLMS manner, which is both the textbook formulation and a strictly *stronger* classical baseline. All CMA results below use the corrected equalizer; the uncorrected one is not reported, since its outputs past the divergence point measure an implementation fault rather than the algorithm.

Conclusion (learning does not beat blind classical, but is stable where blind classical is not): A well-designed blind classical method still works in this regime: the corrected CMA converges smoothly to BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3}. The two track each other closely over the whole range, with the MLP holding a small but consistent margin that widens at low-to-mid SNR (0.0937 vs. 0.1182 at 8 dB). This is the correct answer to "what if there is no posterior?": the learned relay does *not* render classical processing obsolete, because blind adaptive equalization is itself a posterior-free classical tool that comes within a factor of 1.3 of the learned relay at high SNR. The learned relay's advantage is instead one of *robustness and cost*. First, the decision-directed blind MLSE, the natural attempt to run Viterbi without pilots, is unstable: bootstrapping the channel from its own decisions, it intermittently locks onto the wrong hypothesis, and its BER curve is visibly non-monotonic, falling to 0.0374 at 16 dB before rising again to 0.0498 at 20 dB. Its 95% confidence interval at 8 dB is ±0.0348, roughly seven times the MLP's ±0.0046 and two and a half times CMA's ±0.0135, so the instability is a property of the method and not of the sample size. Second, CMA must iterate to convergence on each received block, whereas the MLP is a single fixed forward pass with no per-block adaptation loop and no convergence to monitor; Section 7.1.4 showed this distinction becoming decisive once blocks are short. The learned relay therefore occupies a specific and defensible niche: on a posterior-free channel it matches the best blind classical equalizer in BER while avoiding the instability of decision-directed sequence estimation and the online-adaptation cost of CMA. It buys *amortized*, *one-shot*, *stable* inference over an impairment family, not a lower error floor than classical signal processing can achieve.

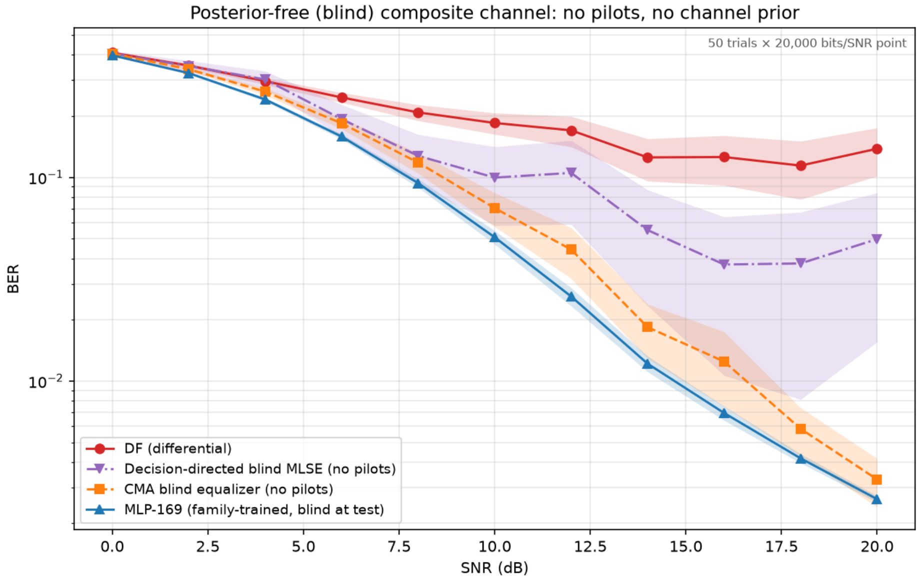

Figure 7.7: Posterior-free composite channel (no pilots, no channel prior, fresh realization per block; mean over 50 channel draws, 20,000 bits/point, 95% CI bands). The corrected CMA (blind classical) tracks the family-trained MLP closely, the MLP holding a small consistent margin; decision-directed blind MLSE is unstable (non-monotonic tail, CI band several times wider) because it bootstraps the channel from its own decisions; plain differential detection floors. Without a posterior, learning only narrowly outperforms the best blind classical equalizer, but does so with one-shot, stable inference and no online adaptation.

7.1.6 Complexity: The Cost Side of the Trade-off

Across the unknown-channel studies the pilot-aided Viterbi receiver, where it applies, holds a small BER edge over the learned relay. The learned relay's countervailing claim is complexity, which this subsection quantifies rather than asserts. Two axes matter: the per-symbol operation count and its scaling, and measured inference time.

Per-symbol cost and scaling. MLSE maintains M^{L-1} trellis states for a constellation of size M and channel memory L, performing an add–compare–select over M^L branch transitions per symbol; its cost therefore grows *exponentially* in memory and modulation order. The 169-parameter MLP performs one fixed forward pass, about 330 floating-point operations per symbol, *independent* of M and L. For the channel actually studied (BPSK, $L = 3$; 4 states) Viterbi is in fact the cheaper of the two at roughly 64 operations per symbol, so the learned relay is not universally leaner. But the scaling diverges immediately: at $M = 4$, $L = 5$ the trellis has 256 states ($\sim 8{,}000$ operations/symbol), and at $M = 16$, $L = 5$ it has 65,536 states (~ 8.4 million operations/symbol), while the MLP remains at 330, a 2.5×10^4 ratio. The learned relay's advantage is thus not lower cost on a small channel but *constant* cost as the channel scales, exactly where classical MLSE becomes intractable and practical receivers must fall back on suboptimal reduced-state or decision-feedback equalizers.

Measured inference time. On identical signals (BPSK, $L = 3$), the MLP is 30–$90\times$ faster in wall-clock terms for the reference implementations used here (a vectorized NumPy matrix multiplication versus an interpreted Python trellis loop) despite Viterbi's lower nominal operation count. Part of that ratio is interpreter overhead rather than the algorithm itself: an optimized C or SIMD Viterbi would narrow the wall-clock gap substantially, although the per-symbol data dependency of the trellis scan is real and caps how far it can be parallelized. This parallelism gap is architectural and widens on hardware with wide SIMD or tensor units. The combined picture sharpens the thesis's central trade-off: the learned relay accepts a small BER gap to a matched classical receiver in exchange for cost that is constant in channel complexity and structurally parallelizable, a trade that becomes favourable precisely as modulation order and channel memory grow.

The experiments of this chapter deliberately extend the thesis beyond the canonical memoryless model. The resulting conclusions should therefore be interpreted as robustness results under model mismatch rather than as replacements for the matched-model conclusions of Chapters 5–6. Together, the two parts of the thesis delineate where learning provides value: not under matched memoryless conditions, where classical methods remain strongest, but when the assumptions underlying those methods are violated.

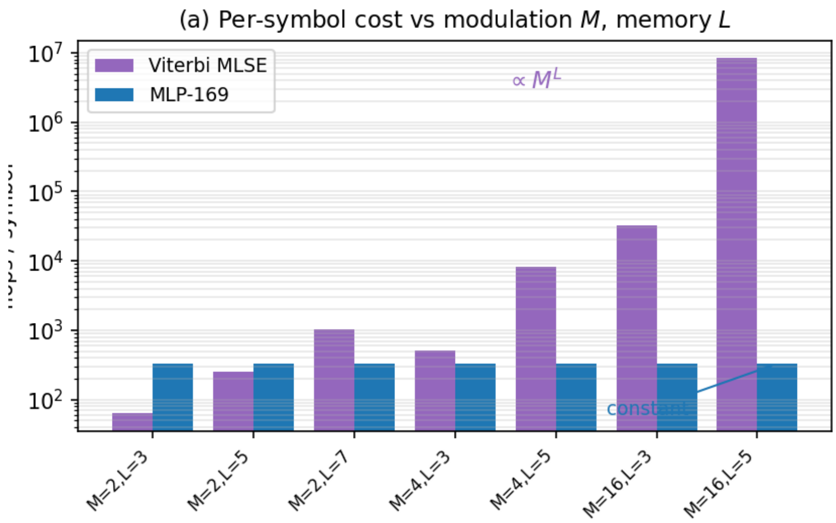

Figure 7.8: Per-symbol computational cost versus modulation order M and channel memory L. Per-symbol operations: Viterbi grows as M^L (intractable by $M{=}16, L{=}5$), while the MLP is constant at $\sim$330 flops. On the small studied channel (BPSK, $L{=}3$) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case.

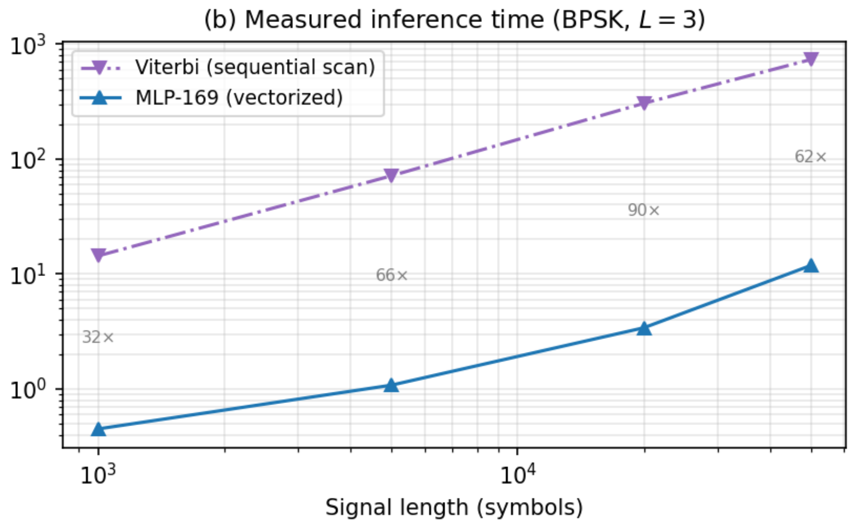

Figure 7.9: Measured inference time for BPSK with $L = 3$: the MLP is 30–90$\times$ faster as a vectorized matmul versus Viterbi's interpreted sequential trellis scan.

Chapter 8

Discussion and Conclusions

Scope of the discussion. The conclusions discussed in this chapter draw on two distinct parts of the thesis. Chapters 5–6 evaluate relay strategies under the canonical memoryless channel model, where transmission is uncoded, the noise is white, and processing is performed on a per-symbol basis. Chapter 7 then deliberately departs from that model by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase. Accordingly, statements concerning channel memory, sequence detection, Viterbi MLSE, pilot-based estimation, or blind equalization should be understood as applying only to the extension study of Chapter 7, not to the canonical memoryless setup.

The canonical core (Chapter 5) settles hypotheses H1–H5 and establishes the baseline finding that, on a matched known channel, a minimal learned relay does not improve on classical decode-and-forward. The thesis's principal contribution (Chapter 7) settles H6 and is where the substantive new claim lies: a systematic delineation of when a learned relay adds value under unknown and mismatched channels. The interpretation below treats the two on equal footing, with the unknown-channel results carrying the main argument.

8.1 Interpretation of Results

The experimental results reveal several consistent patterns across the canonical setup and its two scoped extensions. This section interprets these patterns through the lens of the theoretical framework established in Section 2.5 and evaluates each research hypothesis.

8.1.1 Low-SNR Advantage of Neural Relays over AF (H1: Confirmed)

Low SNR (0-4 dB): AI advantage over AF. At low SNR, AI-based relays consistently outperform AF on the canonical channel, confirming H1. The theoretical explanation lies in the structure of the MMSE-optimal (posterior-mean) relay estimate. After coherent compensation, each axis of the canonical QPSK-over-Rayleigh link reduces to the same real-valued single-sample problem as BPSK over AWGN (Section 4.1.2); the derivation below is carried out in that equivalent real form, applied identically per axis:

$$f^*(y) = \mathbb{E}[x|y] = \tanh\left(\frac{y}{\sigma^2}\right) \tag{8.1}$$

At low SNR (e.g., 0 dB, $\sigma^2 = 1$), this is a gentle sigmoid: $f^*(y) = \tanh(y)$. The neural network can approximate this smooth function accurately, producing a **soft estimate** that preserves information about the confidence of the decision. By contrast:

- **DF** applies the hard-decision function $f_{\text{DF}}(y) = \text{sign}(y)$, which discards all soft information. At low SNR, many received samples lie near the decision boundary ($|y| \approx 0$), and DF assigns them full confidence (± 1) regardless of the actual uncertainty. This information loss leads to excess errors on the second hop.
- **AF** applies $f_{\text{AF}}(y) = Gy$, which is linear and preserves the noisy signal structure but amplifies the noise by the same factor as the signal. The effective SNR degradation is given by $\text{SNR}_{\text{eff}} = \gamma^2/(2\gamma + 1) \approx \gamma/2$ at high SNR: a 3 dB penalty.

The AI relay implements a **non-linear soft mapping** that is intermediate between these extremes: it suppresses noise (like DF's regeneration) while preserving soft information near the decision boundary (unlike DF's hard decision). This explains why learned non-linear relay processing can outperform AF and, under selected channel conditions, also outperform DF at low SNR.

Among the AI methods, the low-SNR advantage over AF is realized by the *minimal* feed-forward relays, not by the larger sequence models. On the canonical Rayleigh channel (Table 5.4, 0 dB) the MLP, cGAN, and Hybrid relays achieve the lowest AI BER (≈ 0.247, essentially matching DF's 0.245), while the higher-capacity Transformer and Mamba-S6 relays are worse (≈ 0.317–0.325) and only the VAE is worse still. The advantage over AF is therefore *not* driven by parameter count, the smallest network already captures it, which is fully consistent with the U-shaped complexity finding (H3) and the equal-capacity convergence of the normalized comparison (H4, Section 4.4).

8.1.2 Classical Dominance at High SNR (H2: Confirmed)

Medium-to-high SNR (≥6 dB): Classical dominance. At medium and high SNR, DF matches or exceeds all AI methods with exactly zero parameters and zero training time. The theoretical explanation is straightforward: at high SNR, $\sigma^2 \to 0$ and $f^*(y) \to \text{sign}(y) = f_{\text{DF}}(y)$. The symbol-wise DF relay *exactly implements* the limiting form of the posterior-mean estimate in this regime, while any neural network approximation introduces a small but non-zero reconstruction error due to the finite precision of the learned mapping and the tanh output saturation (which approaches but never reaches ± 1).

The effect is visible analytically in the AWGN calibration limit, at 10 dB the two-hop DF BER is $2Q(\sqrt{10}) \cdot (1 - Q(\sqrt{10})) \approx 0.002$, and empirically on the canonical Rayleigh channel, where DF attains the lowest or tied-lowest BER at every point from 6 dB upward. The AI models cannot beat DF in this regime, for the uncoded per-symbol setting, $\text{sign}(y)$ is the MAP symbol decision and hence optimal among per-symbol relay functions (Remark 4.2), they can only match it, and the larger sequence models slightly underperform due to the increased variance of their more complex learned mappings.

8.1.3 Robustness Across the Fading Ensemble

Robustness across fades. The relay ranking on the canonical channel is stable across the full evaluated SNR range, even though each transmitted symbol experiences an independent fading realization. The mechanism is that after coherent compensation ($\hat{x} = y \cdot h^*/|h|$), the relay processes a signal with AWGN-like noise at a *random effective SNR*; the network, trained across multiple SNR values, has learned a denoising function that adapts to different noise levels, so the Rayleigh ensemble (a mixture of effective SNRs weighted by the fading distribution) is handled by virtue of multi-SNR training alone, with no per-realization retraining or explicit CSI input.

8.2 The "Less is More" Principle (H3: Confirmed)

One of the most significant findings is the U-shaped relationship between model complexity and relay BER (equivalently, an inverted-U in accuracy). The Minimal MLP architecture (169 parameters) matches the performance of models with over 100-140$\times$ more parameters (3K–24K), while the Maximum MLP (11,201 parameters) exhibited clear overfitting with degraded performance.

This result has important theoretical and practical implications:

8.2.1 Information-Theoretic Perspective

The canonical memoryless relay-denoising task maps a window of $2w + 1$ real-valued noisy observations to a single real-valued clean estimate, applied per axis to the canonical QPSK symbol (equivalently, on the real BPSK/AWGN calibration channel). Each real axis carries exactly 1 bit of information, and the Bayes-optimal estimator $f^*(\mathbf{y}) = \tanh(\mathbf{1}^T\mathbf{y}/\sigma^2)$ (for i.i.d. noise across the window) is a one-dimensional function of the sufficient statistic $\sum_i y_i$. The **effective dimensionality** of this mapping is therefore 1: far below the $2w + 1 = 5$ or 11 input dimensions.

As a heuristic, the minimum description length (MDL) principle suggests matching model complexity to the effective dimensionality of the target mapping. Since the target here is a one-dimensional monotone function of the sufficient statistic, even the 169-parameter network has capacity far in excess of what the mapping requires; this is offered as an intuition for why the smallest model suffices, not as a formal MDL bound.

8.2.2 Bias-Variance Analysis

The bias-variance decomposition [11] provides a quantitative framework:

$$\mathrm{MSE}(\hat{x}) = \underbrace{(\mathbb{E}[\hat{x}] - f^*(y))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{x} - \mathbb{E}[\hat{x}])^2]}_{\text{Variance}} + \underbrace{\sigma^2_{\text{irred}}}_{\text{Noise floor}} \tag{8.2}$$

- **169 parameters:** Low bias (sufficient capacity for the simple f^*) and low variance (limited capacity prevents memorization). The model operates at the optimal bias-variance trade-off.
- **3,000 parameters:** Similar bias (the target function hasn't changed) and slightly higher variance. Performance is nearly identical to 169 params.
- **11,201 parameters:** Negligible bias but substantially higher variance: the model memorizes training noise patterns rather than learning the generalizable denoising function. This manifests as higher BER on unseen test data.
- **24,001 parameters (Mamba-S6):** Despite having 140× more parameters than MLP, the BER improvement at low SNR is only 1-2%. The marginal parameter utility is vanishingly small.

8.2.3 Practical Implications

1. **Occam's Razor for relay AI.** The canonical relay denoising task has inherently low complexity: the per-axis mapping from noisy to clean symbols is fundamentally simple, and the neural network needs only enough capacity to learn a non-linear threshold function with some contextual smoothing. Adding parameters beyond this minimal requirement leads to overfitting.

2. **Deployment feasibility.** A 169-parameter model requires approximately 0.7 KB of memory (169 float32 values) and can be trained in under 5 seconds on a CPU (4.9 s in Table 5.6). This makes AI-based relay processing viable even on severely resource-constrained embedded relay nodes, such as IoT devices or sensor network relays. The model can be stored in on-chip SRAM and executed without any external memory access.

3. **Generalization.** Smaller models generalize better because they are forced to learn

the essential structure of the denoising task rather than memorizing training examples. The 169-parameter MLP handles the full Rayleigh fading ensemble, every effective SNR the fades produce, with a single set of weights and no per-condition retraining, an instance of generalization that would be harder to achieve with a larger, more specialized model.

8.3 State Space vs. Attention for Signal Processing

The comparison of Mamba-S6 and Transformer architectures yields nuanced conclusions:

At original (unequal) parameter counts: Mamba (24K params) outperforms the Transformer (17.7K params) at low SNR. Mamba wins all 3 low-SNR points while the Transformer wins 1/3. This suggests that the state space inductive bias (recurrent state propagation with input-dependent gating) is beneficial for sequential signal processing.

At normalized (3K) parameter counts: The gap narrows dramatically. Mamba-S6-3K and Transformer-3K produce nearly identical BER across all channels. This indicates that the architectural advantage of state space models over attention is relatively small and is partially confounded with the parameter count difference in the original comparison.

Complexity advantage. Even when BER performance is similar, Mamba offers a fundamental computational advantage: $O(n)$ inference complexity versus $O(n^2)$ for Transformers. For the relay processing task where low-latency inference is critical, this linear-time property makes Mamba the preferred choice when a sequence model is desired.

Training time paradox. Despite its superior asymptotic complexity, Mamba-S6 required roughly 4.5× longer to train than the Transformer (2,141 s vs. 474 s on CUDA; Table 5.6). This counter-intuitive result is explained by three compounding factors:

1. *Sequential recurrence vs. parallel attention.* The S6 layer's core state update $\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k$ is inherently sequential: each time step depends on the previous state. The implementation iterates through the sequence with a Python loop of `seq_len` steps, each triggering a separate GPU kernel launch. In contrast, the Transformer computes $\mathbf{QK}^T$ across all positions simultaneously in a single batched matrix multiply. At window size $n = 11$, this means 11 sequential CUDA kernel launches per S6 layer versus 1 parallel operation for attention.

2. *Expand factor doubles the internal dimension.* Each MambaBlock uses an expand factor of 2, projecting from $d_{\text{model}} = 32$ to $d_{\text{inner}} = 64$ before entering the S6 recurrence. This doubles the computation inside the sequential loop while providing no parallelisation benefit.

3. *Kernel-launch overhead at small tensor sizes.* Each sequential step incurs Python

interpreter overhead plus a CUDA kernel launch (~5-10 µs). With 2 layers × 11 steps = 22 sequential kernel calls per forward pass, the overhead alone accumulates to ~110-220 µs: comparable to or exceeding the actual arithmetic time for these small tensors.

The crossover point where Mamba's $O(n)$ advantage would outweigh the parallelism penalty occurs at sequence lengths in the hundreds to thousands, far beyond the $n = 11$ window used in this relay application. An optimised CUDA kernel implementing the S6 scan as a parallel prefix sum (as in the original Mamba paper) would eliminate the sequential bottleneck, but our implementation prioritises clarity and portability over raw throughput.

8.3.1 Context-Length Benchmark: Validating the Crossover Hypothesis

To empirically validate the crossover hypothesis, we conducted a controlled benchmark comparing all three sequence models: Transformer, Mamba-S6, and Mamba-2 SSD: at a context length of $n = 255$ (vs. $n = 11$ in the relay experiments). All models used identical hyperparameters: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 layers, with Mamba-2 using chunk size 32 (yielding 8 chunks). The experiment used a reduced dataset (1,000 training symbols, 20 epochs) to isolate timing differences from convergence effects.

Table 8.1 presents the results.

Table 8.1: Context-length benchmark: three sequence models at $n = 255$ on CUDA.

Model	Parameters	Training Time (20 ep.)	Inference Time (10K bits)	Train Speedup vs S6	Infer Speedup vs S6
Transformer	17,697	5.01 s	0.99 s	28.5×	2.7×
Mamba-S6	24,001	142.56 s	2.69 s	1.0× (baseline)	1.0× (baseline)
Mamba-2 SSD	26,179	13.35 s	1.05 s	10.7×	2.6×

The results confirm the crossover hypothesis decisively:

1. **Mamba-S6 becomes the bottleneck.** At $n = 255$, the S6 sequential scan requires 255 serial CUDA kernel launches per layer per forward pass, making it 142 s for just 20 epochs of training: **28× slower** than the Transformer and **10.7× slower** than Mamba-2 SSD.

2. **Mamba-2 SSD recovers parallelism.** By chunking the 255-step sequence into 8 chunks of 32 and processing each chunk with a single batched matrix multiply, Mamba-2 SSD eliminates the sequential bottleneck. At 13.35 s, it trains **10.7× faster** than S6 and approaches the Transformer's speed.
3. **Inference parity.** Mamba-2 SSD inference (1.05 s) is nearly identical to the Transformer (0.99 s) and **2.6× faster** than Mamba-S6 (2.69 s). This is a complete reversal from the $n = 11$ case, where Mamba-2 SSD was 2× *slower* than S6.
4. **Direction reversal.** At $n = 11$: Mamba-S6 (1.88 s) < Mamba-2 SSD (4.11 s). At $n = 255$: Mamba-2 SSD (1.05 s) < Mamba-S6 (2.69 s). The crossover occurs because the overhead of building the $L \times L$ SSM matrix is amortised over longer chunks, while S6's per-step kernel launch cost grows linearly.

This benchmark validates the architectural trade-off: Mamba-2's structured state space duality is designed for long-context efficiency where chunk-parallel computation outperforms sequential recurrence. For the 11-token relay window, the classical S6 scan remains faster due to lower per-operation overhead. The choice between S6 and SSD should therefore be guided by the target context length of the application.

Why state space suits signals. Signal processing is inherently a sequential temporal task where the state space formulation (propagating a hidden state through time with input-dependent transitions) is a natural fit. The selective mechanism in Mamba allows it to dynamically control information flow, acting as an adaptive filter that selectively passes relevant signal features while suppressing noise.

8.4 Practical Deployment Recommendations

Based on the comprehensive evaluation, we propose the following deployment strategy:

Table 8.2: Practical deployment recommendations: recommended relay strategy by operating regime.

Operating Regime	Recommended Relay	Rationale
Low SNR (0-4 dB)	MLP Minimal / Hybrid / channel-specific selection	Neural relays often help; exact best choice is channel-dependent, with DF remaining strong on some fading scenarios
Medium SNR (4-8 dB)	Hybrid (MLP + DF)	Automatic switching at empirically chosen crossover

Operating Regime	Recommended Relay	Rationale
High SNR (>8 dB)	DF	Zero parameters; lowest BER among the evaluated per-symbol relays
Resource-constrained	MLP Minimal (169 params)	0.7 KB memory, <5 s training
Best overall (BPSK/QPSK)	Hybrid	Combines AI advantage with DF reliability
16-QAM	VAE / Transformer / MLP (16-class 2D)	Joint 2D classification matches DF; eliminates the I/Q-splitting floor

The Hybrid relay is recommended as the default deployment choice because it automatically selects MLP processing at low SNR and DF at high SNR, achieving near-optimal performance across the entire SNR range with minimal complexity.

8.4.1 Latency and Adaptivity Considerations

Two practical questions bear on whether a learned relay can be deployed with minimal impact: the latency it adds relative to AF/DF, and its ability to adapt to varying channel conditions without retraining.

Latency. Two distinct latency components must be separated. The first is *buffering* latency, incurred by the sliding window $y_R[i - w : i + w]$. In the half-duplex protocol studied here this component is absorbed by the protocol itself: the relay buffers the entire listening-phase block before the transmission phase begins (Remark 4.1), so the window's w-symbol look-ahead adds no delay beyond what store-and-forward operation already imposes. Only in a streaming or ultra-low-latency protocol, outside the scope of this thesis, would the symmetric window cost w symbols of look-ahead that the memoryless AF/DF baselines ($w = 0$) do not pay. The second component is *compute* latency: inference cost per relayed symbol. Here the U-shaped complexity result (H3) is decisive: since the 169-parameter MLP matches the performance of models $100 - 140\times$ larger, the deployed network reduces to two small matrix–vector products per symbol, a per-symbol cost that is orders of magnitude below a single channel use's duration on any modern embedded processor, and negligible in absolute terms even if it cannot literally match the single comparison of DF's sign(y). The compute-latency argument therefore holds *because of*, not despite, the less-is-more finding; it would not hold for the 24K-parameter sequence models, whose recurrent or attention-based structure imposes materially higher per-symbol cost for no BER benefit at the relay window length.

The coded study of Section 5.5.3 adds a third component that the canonical comparison never had to consider, and it is the largest of the three: *buffering*. A block relay cannot emit until it has decoded a whole frame (202 symbols there), and the soft BCJR variant cannot even in principle, its backward recursion running from the end of the frame. A windowed learned relay needs only its w-symbol look-ahead, ten symbols, and that figure is constant in frame length where the block relays' grows with it. The measured arithmetic points the same way ($0.30\,\mu$s per symbol against Viterbi's 15.07), so on the coded link the learned relay's advantage in delay and cost is structural rather than marginal, even where its BER advantage is small or absent.

Adaptivity. A single network, trained across the range of simulated SNRs, handles the entire canonical fading ensemble without retraining or explicit CSI input (Section 8.1.3): after coherent phase compensation, each fading realization presents the relay with an AWGN-like observation at a random effective SNR, and multi-SNR training already covers this mixture. Two honest qualifications apply. First, this adaptivity is an advantage over AF, whose fixed gain is tuned to an assumed SNR, but not over symbol-wise DF, whose $\mathrm{sign}(y)$ slicer is equally condition-agnostic; adaptivity therefore does not alter the H2 conclusion. Second, adaptation has been demonstrated only within the canonical Rayleigh ensemble; generalization to other channel families (Rician, bursty interference, phase noise, hardware non-linearities) remains untested and is identified as future work.

8.5 Limitations

Several limitations of this study should be acknowledged, as they define the boundary conditions under which the conclusions hold:

1. **Modulation scope.** The core experiments use QPSK, the canonical constellation; BPSK is confined to the AWGN calibration channel, and the extension covers 16-QAM (Chapter 6). Denser constellations (64-QAM, 256-QAM), where the N-class 2D formulation scales to 64 or 256 output classes and the U-shaped complexity curve (H3) may shift toward larger models, are not investigated.

2. **Perfect CSI.** We assume perfect channel state information at each receiver. In practice, channel estimation errors would affect the coherent compensation step and hence relay performance. Imperfect CSI introduces a model mismatch between the assumed and actual channel, which could differentially impact the relay strategies: AI relays might be more robust (having learned from noisy data) or less robust (having not been exposed to estimation errors during training).

3. **Static offline training.** AI relays are trained once on synthetic data and deployed without adaptation. While multi-SNR training covers the canonical fading ensem-

ble (Section 8.1.3), and the unknown-channel study (H6) shows a fixed network can learn a *stationary* unknown impairment, this protocol does not exploit online information; adaptive or continual training would be required if the deployment channel drifts over time.

4. **Single relay.** The framework considers a single relay node. Multi-relay cooperation introduces relay selection, power allocation, and cooperative diversity dimensions not captured in the two-hop model. With multiple relays, the system-level optimization (which relay to use, how to combine multiple relay observations) becomes as important as the per-relay processing function studied here.

5. **Two-hop only.** Extension to multi-hop networks with 3+ relays introduces noise accumulation (for AF) and error propagation (for DF) that grow with the number of hops. AI relays might show greater advantage in multi-hop settings where noise accumulation is more severe.

6. **Fixed window size.** The relay window size ($w = 2$ or $w = 5$) is fixed and relatively small. Larger windows could provide more context for denoising, particularly in channels with temporal correlation (e.g., block fading). The context-length benchmark (Section 8.3.1) explores this partially, but a systematic study of window size optimization is deferred.

7. **No coding, in the core comparison.** The core comparison operates without error-correcting codes, so the DF baseline there is its symbol-level variant (Remark 4.2). Sections 5.5–5.5.5 measure a coded, information-theoretic block-DF baseline, its soft-decision counterparts, a rate-adaptive envelope over a modulation-and-coding grid, and that envelope re-derived under a round-trip latency budget, directly rather than leaving any of this as an assertion, but that study is confined to one code family, one frame length, and one channel model, and the relay and destination each decide once with no iterative exchange between them.

8. **Single training run per model; unequal training budgets across architectures.** Each learned relay is trained once, from one random initialization, rather than across multiple seeds with mean/standard-deviation/confidence-interval reporting of training variance; the confidence intervals reported throughout this thesis are Monte Carlo intervals over the *evaluation* channel and bit realizations for a fixed trained model, not over the training procedure itself. Separately, the architectures do not all receive identical training budgets (e.g. the cGAN trains for 200 epochs against 100 for the other relays, Section 4.2), a confound the normalized-parameter study (Section 4.4) does not remove, since it equalizes parameter count and input window but not epoch count. Consequently, the U-shaped complexity finding (H3) and the claim that architecture is a secondary factor relative to capacity (Section 5.4) should

be read as observations from the single trained instance of each architecture studied here, not as claims that have been shown robust to training-seed variance or fully isolated from training-budget differences; a multi-seed re-run with equalized budgets is the direct way to close this gap and is not performed in this thesis.

8.6 Future Work

Several directions warrant further investigation:

1. **Time-selective fading.** The present study addresses channels that are static within a block. Extending the learned relay to time-selective fading with continuously varying unknown phase (e.g. Jakes/Clarke Doppler) is a natural next step, likely requiring explicitly phase-aware architectures (complex-valued state, learned phase unwrapping, or a training objective aligned with noncoherent detection rather than symbol MSE) evaluated against prediction-based noncoherent receivers.
2. **Denser constellations (64-QAM, 256-QAM).** The extension shows joint N-class 2D classification eliminates the 16-QAM floor; scaling the formulation to 64 or 256 classes may require deeper networks or hierarchical (coarse-to-fine) classification, and whether the U-shaped complexity curve (H3) shifts rightward with constellation density is a key open question.
3. **Imperfect CSI.** Introduce channel estimation errors to assess robustness of AI relay processing under realistic conditions.
4. **A BER-optimal (BCJR/APP) benchmark for the unknown-ISI channel.** Section 7.1 benchmarks against genie-CSI Viterbi MLSE, the optimal *sequence* detector, and the composite-channel discussion (Section 7.1.2) already notes that sequence-ML and bit-BER-optimal detection coincide for BPSK but not for Gray-coded QPSK, where a wrong trellis path can differ from the truth by one or two bits with unequal probability. A BCJR/APP detector run over the same 3-tap trellis, which directly minimizes per-bit error probability rather than sequence error probability, would test whether the classical benchmark's margin over the learned relay changes once it is BER-optimal rather than sequence-optimal; this was not run here.
5. **Online learning.** Develop relay strategies that adapt their parameters during operation, tracking time-varying channel conditions.
6. **Multi-relay networks.** Extend to cooperative relay selection and multi-hop routing with AI-optimized relays at each node.
7. **Other channels and MIMO.** Evaluate the canonical conclusions on additional channel families (Rician line-of-sight fading, and AWGN treated as a full operat-

ing point rather than the bounded calibration-plus-companion role it has here) and on a multi-antenna second hop, where destination equalization interacts with relay processing and the additivity of the two gains becomes a testable question.

8. **End-to-end learning.** Train the entire communication chain (modulation, relay, demodulation) jointly using autoencoder-based approaches, and compare against the modular classical-plus-learned-relay cascade studied here.

9. **Coded block-DF baseline (measured; open items remain).** As emphasized in Remark 4.2, the DF baseline in the core comparison is the uncoded, symbol-level variant; Section 5.5 now measures the coded, information-theoretic block-DF baseline this item used to only propose, over a rate-1/2 convolutional code (constraint length $K \in \{3, 5, 7\}$) with Viterbi decoding, on both canonical constellations, and Section 5.5.2 adds the soft-information relays this item also named as missing (BCJR at the relay, and a posterior-mean read-out of the learned relay). The high-SNR optimality of symbol-wise DF (H2) does not carry over unmodified: coded block-DF beats it above roughly 6–8 dB but is worse below that threshold. The learned relays' standing against the coded classical decoder is SNR- and architecture-dependent rather than uniform, and the mechanism behind it is the substantive finding: a relay that decodes the code has spent the code's redundancy before transmitting, so whatever it decodes wrongly reaches the destination as a valid-but-wrong codeword that cannot be repaired, whereas a relay that only denoises per symbol leaves the redundancy intact for the destination. Section 5.5.4 then closes the adaptive-rate item this paragraph used to list as open: punctured rates $2/3$ and $3/4$ added to the same mother code, evaluated by goodput over the full modulation-and-coding grid, show the relay-decoding question is dominated by the rate-selection question, by an order of magnitude or more once both are optimized together, and that the optimal choice at high SNR is to turn coding off entirely. Section 5.5.5 shows that this dominance is itself conditional on the link being free to pick any blocklength it likes: under a round-trip latency budget, block-DF's need to buffer a full frame twice against the denoising relay's once reopens a relay-strategy gap the rate adaptation had closed, including reversals at SNRs where the unconstrained comparison favored block-DF. What remains open: the grid covers one code family, one frame length and one channel model; the soft block-DF relay was given the nominal noise variance rather than per-symbol CSI, which is what caused its 20 dB mis-calibration; iterative (turbo-style) exchange between relay and destination, the natural next step once soft information is available at both, was not evaluated; and the latency-constrained analysis uses structural buffer counts rather than a reliability target in the 10^{-5} range, so it does not by itself establish performance under an ultra-reliable low-latency requirement.

10. **Mamba-2 at longer context lengths.** Our benchmark (Section 8.3.1) demonstrates that Mamba-2 SSD achieves a 10.7× training speedup over Mamba-S6 at $n = 255$. Future work should explore whether longer relay windows (e.g., 128-512 symbols) combined with the SSD architecture can improve both BER performance and processing speed simultaneously.

8.7 Conclusions

This thesis presents a comprehensive comparative study of nine relay strategies: two classical (AF, DF) and seven AI-based (MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD). All evaluated on one canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER; BPSK appears only on the separate AWGN calibration channel). The study addresses three identified research gaps (Section 3.2) through controlled experiments with statistical rigor (100,000 bits per SNR point, 10 independent trials, Wilcoxon significance testing). The following table summarizes the hypothesis outcomes:

Table 8.3: Research hypotheses, their statements, and experimental outcomes.

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	**Confirmed**: the best neural relays beat AF at 0–4 dB on the canonical channel
H2	DF lowest-BER among per-symbol relays at high SNR	**Confirmed**: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters
H3	U-shaped complexity curve (BER)	**Confirmed**: 169 params matches 24K; 11K params overfits
H4	Architecture convergence at equal scale	**Confirmed at high SNR, rejected at low**: at 3K params the six architectures span 4.1% BER at 20 dB but 17.4% at 0 dB, splitting into feedforward and sequence groups
H5	SSD faster than S6 at long context	**Confirmed**: 10.7× training speedup at $n = 255$

Hypothesis	Statement	Result
H6	Memoryless classical relays fail on unknown channels; a minimal MLP mitigates by learning, near the model-aware optimum	**Confirmed**: on unknown ISI, AF/DF hit the analytic 0.25 floor (DF non-monotonic in SNR) while the 169-parameter MLP reaches BER $< 5 \times 10^{-5}$, within ≈1–1.5 dB of genie-CSI Viterbi MLSE and indistinguishable from it under the Rayleigh second hop; the same architecture mitigates a composite ISI×PA×phase cascade with no impairment model, tracking ≈2 dB behind pilot-aided Viterbi and matching blind CMA when no posterior exists; on the canonical control it only matches DF (H2), the advantage is bounded to structural model-class mismatch representable at minimal capacity

The main conclusions, in order of significance, are:

1. **AI relays beat AF at low SNR (0-4 dB) but not DF.** On the canonical channel, the best neural relays clearly outperform AF in the low-SNR regime while tracking symbol-wise DF within statistical uncertainty; DF remains lowest or tied-lowest at every SNR point. The theoretical explanation is that neural networks approximate the MMSE-optimal soft-threshold estimate $f^*(y) = \tanh(y/\sigma^2)$, whose end-to-end benefit over the hard slicer is limited in this uncoded setting.
2. **DF attains the lowest per-symbol BER at medium-to-high SNR (≥6 dB) with zero parameters.** The classical decode-and-forward relay requires no training and no parameters yet achieves the best performance among the evaluated per-symbol

relays when channel quality is sufficient for reliable first-hop demodulation. At high SNR, the posterior-mean estimate converges to sign(y), which is exactly the symbol-wise DF operation.

3. **No AI architecture separates itself at original parameter counts.** On the canonical channel the best neural relays (MLP, Hybrid, cGAN) perform within statistical uncertainty of one another, and classical DF matches or exceeds all of them across the SNR range. The selective state space model's $O(n)$ complexity and adaptive filtering mechanism make it an efficient sequence-model choice, but no architectural inductive bias yields a decisive BER advantage on this task.

4. **Architecture matters less than parameter count at equal scale — but only at high SNR.** When all models are normalized to approximately 3,000 parameters (Table 5.5), the spread across the six architectures falls monotonically with SNR: 17.4% at 0 dB, 9.3% at 12 dB, 4.1% at 20 dB. In the high-SNR regime the original claim holds and architecture is close to irrelevant. At low SNR it does not: the six models separate cleanly into a feedforward group (MLP, Hybrid, VAE at 0.336–0.342) and a sequence group (Transformer, Mamba-S6, Mamba-2 at 0.400–0.407), a gap far larger than the confidence intervals, with the feedforward models close to DF and the sequence models close to AF.

   Two points are worth separating here. The split is by *family*, not by capacity, since all six carry the same parameter budget — so at low SNR architecture does matter, and the sequence models' inductive bias is actively unhelpful on a memoryless channel where there is no temporal structure to exploit. And the VAE is no longer the exception it was: at 3K parameters it sits inside the feedforward group at every SNR (0.0101 against MLP-3K's 0.0099 at 20 dB). The earlier reading, that the VAE was a consistent underperformer, does not survive re-measurement. The cause is not the corrected noise convention, which a controlled A/B rules out: training the same relay on data generated under the old convention and under the corrected one produces curves that agree to within Monte Carlo noise at every SNR. The earlier figures came from an implementation that predates the current relay module, so the discrepancy is one of code rather than of channel calibration. See the discussion of the generative paradigm in Section 2.3.3.

5. **A 169-parameter network is sufficient for relay denoising.** The Minimal MLP architecture achieves performance comparable to models 100-140$\times$ larger, demonstrating that the relay denoising task has inherently low complexity. The bias-variance analysis explains this: the target function is simple (a soft threshold), so additional parameters increase variance without reducing bias.

6. **The Hybrid relay is the recommended practical choice.** By combining AI pro-

cessing at low SNR with classical DF at high SNR, the Hybrid relay achieves near-optimal performance across the entire SNR range with only 169 parameters. It automatically adapts to the operating regime, requiring no manual SNR-dependent configuration.

7. **Mamba-2 SSD trained 10.7$\times$ faster than S6 at longer contexts on the hardware used here.** The chunk-parallel structured matrix multiplication of SSD removes the sequential bottleneck of S6, making it the preferred state space architecture for applications requiring longer symbol windows. This ratio is a wall-clock measurement of a reference implementation on one machine, not a property of the architectures; the timing tables were not re-measured with the results of this revision and are reported as informational.

The modulation extension (Chapter 6) adds one further finding:

- **BPSK findings generalise to QPSK; for 16-QAM, joint N-class 2D classification closes the gap to DF.** Under I/Q splitting, all BPSK-trained relays achieve identical BER on QPSK (I/Q independence), confirming H1–H3 for the 2-bit/symbol constellation, whereas 16-QAM exhibits a structural BER floor under per-axis processing. Reformulating the relay as a joint classifier over all 16 constellation points removes that floor for five of the six architectures. Measured as excess over DF, which is the meaningful quantity because DF is itself channel-limited at 0.03748 on this channel, the per-axis relays carry 0.0161–0.0174 and the joint ones 0.0001–0.0019, so VAE, Hybrid, Transformer and the MLP reach parity with DF rather than surpassing it. Mamba-2 SSD is the exception, retaining an excess of 0.0121. The floor was therefore an artefact of the per-axis formulation rather than a limit on neural relays, but the endpoint is parity, and one architecture does not reach it.

It is important to note that sequence-detection methods such as Viterbi MLSE appear only in the unknown-channel study of Chapter 7, where channel memory is introduced explicitly through a 3-tap FIR/ISI channel. The canonical system model of Chapters 1–6 remains memoryless and therefore requires only per-symbol processing.

The unknown-channel contribution (Chapter 7) settles the complementary question of *when the learned relay is not merely competitive but necessary*:

- **On channels violating the deployed relay's model class, learning restores reliability at near-optimal cost (H6: confirmed).** A flat-channel control first isolates the cause: on unknown carrier phase, unknown gain, and per-branch amplitude asymmetry, all memoryless, the classical relay does not fail and the MLP only matches it (BER gap ≤ 0.0036), showing that parameter uncertainty alone is not what learning mitigates. The failure appears only with unmodeled *memory*: on an unknown 3-tap ISI channel, both AF and symbol-wise DF are pinned at an analytically predicted BER floor of 0.25 (and DF's BER is *non-monotonic*, worsening as SNR grows) because one interference pattern in four deterministically flips the symbol sign and no memoryless relay can recover it. The same 169-parameter MLP that merely ties DF on the canonical channel learns a joint equalizer–detector from data and restores monotonically decreasing BER (below 5×10^{-5} at 16 dB). The claim is carefully bounded by the Viterbi benchmark: classical estimate-then-MLSE, given the correct channel family and 200 pilots, is ≈1–1.5 dB better still, the MLP's distinctive value is not optimality but *family-agnosticism*, delivering near-MLSE reliability with no identification stage, complexity independent of channel memory, and unchanged architecture on the biased nonlinear channel where linear MLSE is misspecified. Together with H2, this draws the precise boundary of the learned relay's value: it cannot beat DF where DF's assumptions hold; it approaches but does not beat the model-aware optimum where the family is identifiable; it spans impairment families whose structure fits its capacity (memory, monotone distortion), including an arbitrary *cascade* of them (Section 7.1.3), where a single minimal network handles ISI, amplifier nonlinearity, and unknown phase jointly without being told which are present, tracking ≈2 dB behind the best *pilot-aided* classical receiver. When no such posterior exists (no pilots, no channel prior) the learned relay matches the best blind classical equalizer (CMA) in BER while avoiding the instability of decision-directed blind MLSE and the per-block adaptation cost of CMA (Section 7.1.5): its niche is amortized, one-shot, stable inference over an impairment family, not a lower floor than classical signal processing can reach. Sweeping the intermediate case, a *partial* posterior of limited pilots (Section 7.1.4), makes the boundary quantitative: pilot-aided Viterbi beats the pilot-free learned relay down to ∼10 pilots and then collapses at the channel-identifiability threshold, while the learned relay is invariant to pilot budget. The crossover is thus governed by identifiability and, through pilot overhead, by block length: on long blocks the classical receiver wins on BER at negligible cost, but on short or fast-varying blocks its pilot fraction becomes prohibitive (25% at a 40-symbol block) and the zero-overhead learned relay is decisive. The advantage is specific to *structural* model-class mismatch, not to unknownness per se. Finally, the complexity accounting (Section 7.1.6) quantifies the cost side of the trade-off the thesis rests on: MLSE's per-symbol cost grows as M^L in modulation order and memory, whereas the 169-parameter relay is constant (∼330 operations/symbol) and 30–$90\times$ faster in measured wall-clock as a vectorized matmul versus a sequential trellis scan. On the small BPSK/L=3 channel Viterbi is actually cheaper, so the learned relay's complexity advantage is one of *scaling and parallelism*, not of the small case, it becomes favourable exactly as M and L grow and exact MLSE turns intractable.

These findings demonstrate that neural network-based relay processing is a viable complement to classical approaches in the low-SNR regime, matching, though not surpassing, the per-symbol-optimal DF baseline everywhere else at negligible computational cost. The overarching insight is that **model complexity should be matched to task complexity**: for the canonical relay denoising task, minimal architectures suffice, and the choice between neural network paradigms, i.e. feedforward or sequential, matters less than proper model sizing and regularization. The practical recommendation is to deploy a Hybrid relay with a 169-parameter MLP sub-network, which combines the neural low-SNR advantage over AF with DF's zero-parameter reliability at high SNR; for 16-QAM deployments, the relay should be formulated as a joint N-class 2D classifier rather than a per-axis I/Q-split denoiser; and on links with unmodeled impairments (impairments outside the canonical memoryless model, including ISI, bias, and hardware nonlinearity), a minimal learned relay delivers near-MLSE reliability without channel identification, where the memoryless classical relays fail outright (H6).

Chapter 9

Summary

Scope of the summarized results. The results summarized below originate from two complementary studies. The canonical study (Chapters 5– 6) evaluates relay processing on a memoryless channel with white noise and uncoded transmission. The unknown-channel study (Chapter 7) extends that baseline by deliberately introducing impairments that violate those assumptions, including intersymbol interference (ISI), nonlinear distortion, and model mismatch. Conclusions involving sequence detection, Viterbi MLSE, pilot overhead, or blind equalization therefore refer specifically to that extension rather than to the canonical system model.

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The objective was to analyze performance–complexity tradeoffs and determine when learning-based relays provide practical advantages over analytically derived methods. The entire study is carried out on one canonical setup, fixed in advance and never departed from: a two-hop SISO link over i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER, with the relay function as the only varied axis. The study showed the following:

1. Neural relays yield a consistent performance improvement over amplify-and-forward (AF) at low signal-to-noise ratio (SNR), where their learned soft denoising avoids AF's noise amplification (H1).
2. Classical *decode-and-forward (DF)* matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters, confirming that DF attains the lowest per-symbol BER in this regime (H2)..
3. A systematic complexity analysis demonstrated that performance **does not improve monotonically with model size** (H3).
4. A **minimal neural relay** (169 parameters) achieved performance comparable to significantly larger models, while intermediate-sized networks exhibited overfitting.

5. When neural relays were evaluated under equal parameter budgets, architectural differences largely vanished, indicating that **model capacity**, rather than architectural class, dominates performance for this task (H4).

6. Sequence-based neural relays such as structured state-space models achieved comparable error performance with improved training efficiency (H5).

7. From a deployment perspective, a `hybrid relay` combining neural processing at low SNR with DF at high SNemerged as the **recommended default**, achieving near-optimal performance across the SNR range with minimal overhead

The first seven findings concern the canonical memoryless relay model; the remaining findings summarize the deliberate model-mismatch extension of Chapter 7.

8. Beyond the canonical memoryless model, on an **unknown channel** violating classical assumptions (unmodeled intersymbol interference or biased nonlinearity), AF and DF failed to relay reliably (exhibiting an analytic BER floor of 0.25, with DF *worsening* as SNR increased) while the same minimal 169-parameter MLP restored reliable relaying by learning the impairment from data, including an arbitrary composite cascade of ISI, amplifier nonlinearity, and unknown phase. It tracked $\approx$1–1.5 dB behind a *pilot-aided* Viterbi receiver on pure ISI ($\approx$2 dB on the composite cascade) and, in the fully posterior-free regime (no pilots, no channel prior), matched the best blind classical equalizer while avoiding its instability and online-adaptation cost (H6).

9. Sweeping the intermediate *partial*-posterior case located the crossover precisely: it is governed by the **reliability of the channel estimate** and by pilot overhead. Pilot-aided classical processing wins on long blocks at negligible cost, but the zero-pilot learned relay becomes decisive once blocks are too short, or channels too transient, for pilots to yield an accurate estimate. The learned relay's advantage is therefore specific to *structural* model-class mismatch (memory, nonlinearity, absent pilots) rather than to unknownness per se. Together with the high-SNR result, this delineates precisely where learned relays add value: never above the model-aware optimum, but uniquely robust to *not knowing which model applies* and to the pilot overhead of channel identification.

10. A **complexity accounting** completed the trade-off: the learned relay's per-symbol cost is constant in modulation order and channel memory, and 30–$90\times$ faster in measured wall-clock as a vectorized operation for the reference implementations compared (Section 7.1.6), whereas exact Viterbi MLSE grows as M^L and becomes intractable. The learned relay thus trades a small BER gap for cost that scales gracefully exactly where classical sequence detection does not.

11. In the modulation extension, 16-QAM exposed a structural error floor of per-axis

processing; reformulating the relay as a **joint N-class 2D classifier** brought five of the six architectures to parity with DF at high SNR, cutting their excess over it from about 0.016 to 0.001 or less, with Mamba-2 SSD the exception.

Overall, the results indicate that learning-based relays are best used as targeted enhancements rather than universal replacements, with simplicity, parameter efficiency, and hybrid design as key principles for practical relay systems. All conclusions hold for the *uncoded*, symbol-level canonical setting studied here (Remark 4.2); extending the comparison to coded block-DF, soft-information (LLR) forwarding relays, other channel families, multi-antenna configurations, and denser constellations (64-QAM and beyond) is identified in Chapter 8 as the principal direction for future work.

Bibliography

[1] T. M. Cover and A. A. El Gamal, "Capacity theorems for the relay channel," *IEEE Transactions on Information Theory*, vol. 25, no. 5, pp. 572–584, 1979.

[2] A. El Gamal and Y.-H. Kim, *Network Information Theory*. Cambridge University Press, 2011.

[3] J. N. Laneman, D. N. Tse, and G. W. Wornell, "Cooperative diversity in wireless networks: Efficient protocols and outage behavior," *IEEE Transactions on Information Theory*, vol. 50, no. 12, pp. 3062–3080, 2004.

[4] B. Rankov and A. Wittneben, "Spectral efficient protocols for half-duplex fading relay channels," *IEEE Journal on Selected Areas in Communications*, vol. 25, no. 2, pp. 379–389, 2007.

[5] A. Nosratinia, T. E. Hunter, and A. Hedayat, "Cooperative communication in wireless networks," *IEEE Communications Magazine*, vol. 42, no. 10, pp. 74–80, 2004.

[6] D. Tse and P. Viswanath, *Fundamentals of Wireless Communications*. Cambridge University Press, 2005.

[7] H. Ye, G. Y. Li, and B. H. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Communications Letters*, vol. 7, no. 1, pp. 114–117, 2018.

[8] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Transactions on Signal Processing*, vol. 67, no. 10, pp. 2554–2564, 2019.

[9] S. Dorner, S. Cammerer, J. Hoydis, and S. ten Brink, "Deep learning based communication over the air," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 132–143, 2018.

[10] H. Sun, X. Chen, Q. Shi, M. Hong, X. Fu, and N. D. Sidiropoulos, "Learning to optimize: Training deep neural networks for interference management," *IEEE Transactions on Signal Processing*, vol. 66, no. 20, pp. 5438–5453, 2018.

[11] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[12] B. Akdemir, M. A. Karabulut, and H. Ilhan, "A comparative study of deep learning-aided relay selection protocols in cooperative communication," *Wireless Personal Communications*, vol. 136, pp. 365–384, 2024.

[13] D. Gunduz, P. de Kerret, N. D. Sidiropoulos, D. Gesbert, C. R. Murthy, and M. van der Schaar, "Machine learning in the air," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 10, pp. 2184–2199, 2019.

[14] I. Bergel, "Network optimization using relays as neurons," *arXiv preprint arXiv:2306.14253*, 2023.

[15] ——, "Deep optimization of relay networks using relays as neurons," in *Proc. IEEE ICASSP*, 2024, pp. 9056–9060.

[16] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2014.

[17] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, "β-vae: Learning basic visual concepts with a constrained variational framework," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.

[18] M. Mirza and S. Osindero, "Conditional generative adversarial nets," arXiv preprint, 2014.

[19] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in Neural Information Processing Systems*, vol. 27, 2014.

[20] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. Courville, "Improved training of Wasserstein GANs," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[21] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[22] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[23] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023. [Online]. Available: https://arxiv.org/abs/2312.00752

[24] T. Dao and A. Gu, "Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality," arXiv preprint, 2024.

[25] S. Shleifer, J. Weston, and M. Ott, "Normformer: Improved transformer pretraining

with extra normalization," 2021. [Online]. Available: https://arxiv.org/abs/2110.09456

[26] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. McGraw-Hill, 2008.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 1026–1034.

[28] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite block-length regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.

[29] S. V. Hanly and D. N. C. Tse, "Multiaccess fading channels—part II: Delay-limited capacities," *IEEE Transactions on Information Theory*, vol. 44, no. 7, pp. 2816–2831, 1998.

[30] G. D. Forney, "Maximum-likelihood sequence estimation of digital sequences in the presence of intersymbol interference," *IEEE Transactions on Information Theory*, vol. 18, no. 3, pp. 363–378, 1972.

Chapter 10

Appendices

10.1 Appendix A: Mathematical Notation

Symbol	Description
b	Binary bit ($\in \{0, 1\}$)
x	Modulated BPSK symbol ($\in \{-1, +1\}$)
y	Received signal
n	Noise sample
h	Fading coefficient
σ^2	Noise variance
SNR	Signal-to-Noise Ratio (linear)
G	AF amplification gain
$\mathbf{W}$	Neural network weight matrix
$\mathbf{b}$	Bias vector
P_e	Bit error rate
$Q(\cdot)$	Gaussian Q-function
β	VAE KL weight
λ	Regularization parameter
η	Learning rate
Δ	State space discretization step
$\mathbf{A}, \mathbf{B}, \mathbf{C}, D$	State space model matrices

10.2 Appendix B: Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, lr=0.01, 100 epochs	169	NumPy (CPU)
Hybrid	MLP + DF switching via learned SNR threshold ($\sim$5 dB)	MSE (MLP only), lr=0.01	169	NumPy (CPU)
VAE	Encoder–decoder latent model (7→8→1)	β-VAE ($\beta = 0.1$), Adam lr=10^{-3}	1,777	PyTorch (CUDA)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr=10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr=10^{-3}	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr=10^{-3}	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr=10^{-3}, clip=1.0	26,179	PyTorch (CUDA)

Table 10.2: Comparison of relay models, training objectives, parameter counts, and implementations.

10.3 Appendix C: Software Architecture

All numerical results reported in chapter 5 were generated using the framework shown in figure 10.1, ensuring consistent evaluation across classical and learning-based relay architectures.The project is implemented as a modular Python package (`relaynet`). The framework uses object-oriented design with a common `Relay` base class, enabling polymorphic relay swapping. Monte Carlo simulation is implemented in `runner.py` with configurable trial count, bit count, and SNR range. All tensor operations use vectorized PyTorch for GPU acceleration.

10.3.1 Testing

108 automated tests (pytest) cover the channels, modulation, relay-strategy, simulation and statistics modules with a 100% pass rate. The suite requires only NumPy and SciPy; PyTorch is optional and needed solely by the neural relays that use it.

10.3.2 Reproducibility

Random seeds are controlled at the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results.

Re-measurement of all results. Every simulation result reported in this thesis was re-measured after three faults were found in the earlier data: a 3 dB error in the noise convention of the AWGN channel, which propagated into the *training* data of every learned

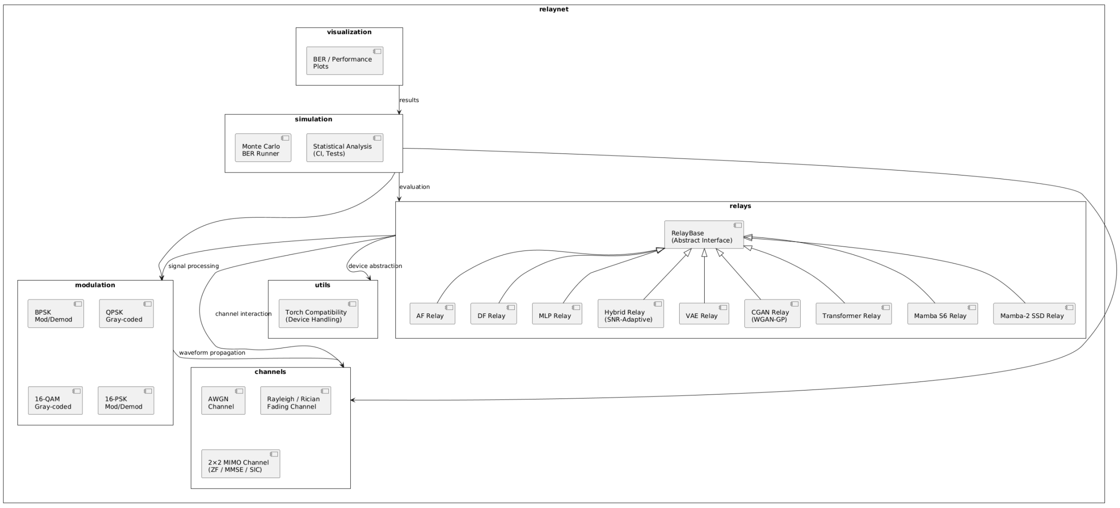

Figure 10.1: UML-style component diagram of the `relaynet` simulation framework.

relay and so moved results on channels the fault never touched; three tables measured on (constellation, channel) pairings the thesis had withdrawn, each traceable to an unexamined default argument; and result files predating the implementation of the relays they reported. The published numbers are those of the re-measurement, at 10 trials $\times$ $10{,}000$ bits per SNR point.

Three written conclusions changed as a consequence and are stated in their revised form in the body: hypothesis H4 is now confirmed at high SNR and rejected at low, rather than confirmed outright; the VAE relay does not fail, and the earlier failure is attributed to a superseded implementation rather than to the noise convention, which a controlled A/B rules out; and joint 16-class classification is reported as restoring parity with decode-and-forward rather than as achieving near-zero error. A cell-by-cell record of the changes, the runtimes, and the reasoning is kept with the source as `thesis/RERUN_CHANGELOG.md`, and `verify_thesis_tables.py` checks every published cell against the file that produced it.

10.4 Appendix D: Normalized 3K-Parameter Configurations

Model	Parameters	Window	Hidden / Architecture
MLP-3K	3,004	11	hidden=231
Hybrid-3K	3,004	11	hidden=231 (+ DF switch)
VAE-3K	3,037	11	latent=10, hidden=(44, 20)
cGAN-3K	3,004	11	noise=8, g_hidden=(30, 30, 16), c_hidden=(32, 16)
Transformer-3K	3,007	11	d_model=18, heads=2, layers=1
Mamba-S6-3K	3,027	11	d_model=16, d_state=6, layers=1
Mamba-2-3K	3,004	11	d_model=15, d_state=6, chunk_size=8, layers=1

All 3K configurations use a window size of 11 (vs. 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN); the original sequence models already used 11. The parameter counts are within ±1.2% of the 3,000 target.

10.5 Appendix E: Master Experiment Ledger

Every numerical table and figure in this thesis traces to exactly one committed result file, verified cell-by-cell against that file by `verify_thesis_tables.py` in the code repository (Section 4.3.4). This ledger lists the modulation, channel, and result source for each major study, grouped in the order they appear.

Table 10.4: Master experiment ledger: modulation, channel, and result-file provenance for every major study.

Study	Modulation	Channel	Tbl/Fig	Result source
AWGN calibration	BPSK	AWGN	tab:ber_val.	`results/calibration.json`
Canonical benchmark	QPSK	Rayleigh	tbl:2, fig:10	`results/modulation/qpsk_rayleigh.json`
Normalized 3K-param. comparison	QPSK	Rayleigh	tbl:8	`results/normalized_3k/3k_rayleigh.json`
16-QAM joint 2D classifier	16-QAM	Rayleigh	tbl:24	`results/all_relays_16class/*.json`
Coded block-DF, K-sweep	QPSK, 16-QAM	Rayleigh	tbl:34–36	`results/coded_df_experiment.json`
Coded high-SNR paired re-test	QPSK	Rayleigh	tbl:37	`results/coded_high_budget_test.json`
Coded relay-error mechanism	QPSK	Rayleigh	tbl:38	`results/coded_error_mechanism.json`
Coded soft-decision relaying	QPSK	Rayleigh	tbl:39	`results/coded_soft_decision.json`
Equal-throughput & latency	QPSK, 16-QAM	Rayleigh	tbl:40, 41	`results/coded_latency_throughput.json`
Adaptive modulation/coding (AMC)	QPSK, 16-QAM	Rayleigh	tbl:42	`results/coded_rate_adaptation.json`
Capacity under a latency budget	QPSK, 16-QAM	Rayleigh	tbl:43	`results/coded_latency_capacity.json`
Unknown ISI (layer 2)	BPSK	AWGN / Rayleigh (2nd hop)	tbl:E6	`e6_unknown_channel_results/e6_sim_ported_results.npy` + `e6_viterbi_*.npy`
Flat-channel control	BPSK	AWGN / Rayleigh	tbl:E6flat	`e6_flat_ported_results.npy`
Unknown ISI, QPSK generalization	QPSK	AWGN / Rayleigh	tbl:E6qpsk	`e6_qpsk_unknown_channel_results.npy`
Composite impairment cascade	DBPSK	AWGN / Rayleigh	E6composite	`e6_composite_ported_results.npy`
Partial posterior (pilot sweep)	BPSK	AWGN / Rayleigh	E6partial	`e6_partial_ported_results.npy`
Blind channel (layer 4)	BPSK	AWGN / Rayleigh	E6blind	`e6_blind_ported_results.npy`
Four-layer summary	mixed (above)	mixed (above)	tbl:layers	combination of the above

The result files above are themselves generated by named, committed scripts (e.g. `coded_rate_adapt` produces `coded_rate_adaptation.json`); the correspondence between script and output file is one-to-one and is not separately tabulated here since the filenames make it explicit.

אוניברסיטת תל-אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר להנדסת חשמל

ארכיטקטורות למידה עמוקה לתקשורת ממסר דו-קפיצתית: מחקר השוואתי של אסטרטגיות ממסר קלאסיות ומבוססות רשתות נוירונים

חיבור זה הוגש כעבודת גמר לקראת התואר "מוסמך אוניברסיטה"

בהנדסת חשמל ואלקטרוניקה באוניבסיטת תל-אביב

על-ידי

גיל צוקרמן

העבודה נעשתה בבית ספר להנדסת חשמל

בהדרכת ד"ר אנטולי חינה ופרופ' ראג'א ג'יריס

ניסן תשפ"ו

1

תקציר

תזה זו משווה בין אסטרטגיות ממסר קלאסיות ונלמדות לתקשורת דו־דילוגית. תצורה אחת קבועה לאורך המחקר המרכזי ואינה משתנה: קישור SISO עם ממסר בודד, דעיכת ריילי מהירה בלתי־תלויה ושווה־התפלגות המוחלת זהה בשני הקפיצות, בסיס־רוחב מרוכב, אפנון QPSK מקודד־גריי, ותמסורת בלתי־מקודדת, כאשר פונקציית הממסר היא המשתנה היחיד. תשע אסטרטגיות ממסר מיושמות: הגבר־והעבר (AF), פענח־והעבר סימבולי (DF), רשת פרספטרון רב־שכבתית (MLP), מתג היברידי, מודלים גנרטיביים (מקודד אוטומטי וריאציוני ורשת יריבותית מותנית), ומודלי רצפים (Transformer, Mamba-S6 ו־Mamba-2 SSD); שמונה מהם מוערכים בהשוואה הסופית, כאשר הרשת היריבותית המותנית הוחרגה מטעמי עלות. כל התוצאות הן אומדני מונטה קרלו המדווחים עם רווחי סמך של 95%, והסימולטור מאומת תחילה מול הסתברויות שגיאה סגורות־צורה של BPSK על ערוץ AWGN, המשמש רק כייחוס לכיול; פענח־והעבר סימבולי בערוץ ריילי הקנוני נבדק באופן בלתי־תלוי באותו אופן, מול ההרכבה הסגורה־צורה הדו־קפיצתית, ומתאים עד כדי אחוז אחד בכל ערך SNR שנמדד.

בתרחיש הייחוס הקנוני, הממסרים הנלמדים הטובים ביותר עולים על AF ביחס אות־לרעש (SNR) נמוך ומשיגים BER דומה ל־DF הסימבולי ב־SNR בינוני וגבוה, בעוד ש־DF משיגה את ה־BER הנמוך ביותר מבין הממסרים הנבחנים ללא עלות פרמטרים. אין לקרוא בכך טענת אופטימליות כללית עבור DF: זוהי קביעה אמפירית על ממסור בלתי־מקודד וסימבולי־לפי־סימבול בטווח ה־SNR הנמדד כאן; תוצאות מידע־תאורטיות עבור DF מבוסס־בלוקים עוסקות בתמסורת מקודדת ובאורכי בלוק אסימפטוטיים, ולא הן ולא המדידות המדווחות כאן קובעות שלא ניתן לעקוף את DF באמצעות ממסר מקודד או מבוסס־מידע־רך; ממסר מקודד מבוסס־החלטה־קשה נמדד ישירות בהמשך, בעוד ממסר מבוסס־מידע־רך נותר פתוח. מחקר מורכבות מראה שהביצועים נקבעים בעיקר על ידי קיבולת המודל ולא על ידי הארכיטקטורה ב־SNR גבוה, כאשר MLP בעל 169 פרמטרים משיג את ה־BER של מודלים גדולים פי שני סדרי גודל; ב־SNR נמוך הארכיטקטורות נפרדות, כאשר הממסרים מבוססי־הזנה־קדימה עוקבים אחר DF בעוד מודלי הרצפים מפגרים.

שני מחקרים נוספים מדווחים בנפרד מכיוון שכל אחד דורש תצורה שהתצורה הקנונית אינה יכולה לקלוט. העלאת הקונסטלציה ל־16-QAM ממקמת ארבע רמות על כל ציר ושוברת את ניסוח הממסר המפוצל־לפי־ציר, ויוצרת רצפת שגיאה מבנית שמסווג דו־ממדי משותף על פני הקונסטלציה המלאה מבטל, ומשווה ל־DF במקום לעלות עליו. המחקר השני מודד ישירות את פערי הקידוד שנותרו פתוחים לעיל: קוד קונבולוציוני חיצוני בקצב 1/2 עם ממסר DF

מבוסס־בלוק אמיתי עולה על בסיס ה־DF הבלתי־מקודד מעל כ־6–8 dB, אך נחות ממנו מתחת לסף זה, סף שמחריף ואינו נעלם עבור קודים חזקים יותר (אורך אילוץ שנסרק עד 7). מול בסיס קלאסי מקודד זה, הממסרים הנלמדים נחותים בבירור ב־SNR נמוך ובינוני ועדיפים ב־SNR גבוה, ומדידת פלט הממסר מזהה את הסיבה, שאיננה שרשת נוירונים מפענחת טוב יותר מ־Viterbi: Viterbi משדר כמחצית ממספר שגיאות הסימבולים, אך DF מבוסס־בלוק מקודד מחדש לאחר הפענוח, ולכן שגיאותיו השאריתיות מגיעות כמילות־קוד תקפות־אך־שגויות שהיעד אינו יכול לתקן, בעוד ממסר שרק מסיר רעש משמר את יתירות הקוד וכך 57–74% משגיאותיו מתוקנות ביעד. העברת החלטות רכות במקום קשות משפרת את הממסר הנלמד בכל ערכי ה־SNR עם אותם משקלים בדיוק, אך אינה מצילה את DF מבוסס־הבלוק, ובכך ממקדת את מקור הבעיה: צריכת יתירות הקוד בממסר, ולא הקוונטיזציה בו. יתרה מזאת, בהשוואה ביעילות ספקטרלית שווה ולא ב־E_s/N_0 שווה, הקוד בקצב 1/2 אינו מצדיק את עלותו כלל מתחת ל־20 dB, ובנקודה שבה כן, הרווח הוא פי 1.35 בלבד תוך תשלום של פי 20 בהשהיית האגירה ופי 50 בעלות החישוב ביחס לממסר הנלמד. כאשר קצב הקוד עצמו מותאם לערוץ, על פני רשת הכוללת את שתי הקונסטלציות וקצבים מ־$1/2$ עד $3/4$ וכן שידור בלתי־מקודד, מיקסום התפוקה בפועל מראה שבחירת הקצב חשובה בסדר גודל אחד לפחות יותר מבחירת אסטרטגיית הממסר, וכי הבחירה האופטימלית ב־SNR גבוה היא כיבוי הקידוד כליל, תוך מסירת פחות משליש מקיבולת שאנון הבלתי־מוגבלת של הערוץ אף שם -- פער הנקבע על־ידי הקונסטלציה הסופית ושיעור שגיאת הפריימים השיורי, לא על־ידי אסטרטגיית הממסר. תיחום זה עצמו מותנה בכך שלקישור חופש לבחור כל אורך־בלוק: מכיוון שממסר המפענח ומקודד מחדש נדרש לאגור בלוק שלם פעמיים, לעומת פעם אחת בלבד אצל ממסר המסיר רעש בלבד, הטלת תקציב השהיה הלוך־ושוב נועלת את DF מבוסס־הבלוק מחוץ לאפשרות הקידוד מוקדם יותר ופותחת מחדש את הפער בין אסטרטגיות הממסר, ואף הופכת אותו בעד פי 1.5 בנקודות SNR שבהן ההשוואה חסרת־המגבלה העדיפה את הממסר הקלאסי.

התרומה העיקרית מאורגנת כסולם של ארבע שכבות, כאשר כל שכבה מסירה הנחה אחת בדיוק מהשכבה שמעליה, בעוד ההשוואה בין עיבוד קלאסי לנלמד חוזרת על עצמה ללא שינוי. בשכבה 1, התצורה הקנונית, העיבוד הקלאסי מנצח: DF הסימבולי אינו משתפר, תכונה של הבעיה ולא של האימון, שכן בערוץ חסר־זיכרון ותואם, החלטה קשה היא פונקציית הממסר האופטימלית לכל סימבול. שכבה 2 מוסיפה זיכרון ערוץ בדמות מסנן הפרעה בין־סימבולית (ISI) תלת־מקדמי בלתי־ידוע, תוך שמירה על ידיעת הערוץ קבועה. כאן הממסרים הקלאסיים חסרי־הזיכרון נכשלים לחלוטין, כאשר שיעור השגיאה נעצר ואז עולה עם עוצמת השידור ככל שההחלטות הקשות מפיצות את ההפרעה, בעוד ממסר נלמד וממוסגר־חלון בן 169 פרמטרים משקם את הקישור; בהינתן אותו מסנן, גלאי רצף בעל־נראות־מרבית מסוג Viterbi נותר מוביל ב־1 עד 1.5 dB, כך שמה שהשכבה הזו מפרידה הוא מחלקת הפונקציות, ומה שהממסר הנלמד רוכש הוא עלות קבועה לכל סימבול מול עלות טרליס האקספוננציאלית בזיכרון הערוץ. שכבה 3 מדרדרת את אומדן הערוץ לתקציב פיילוטים סופי, ויתרון המקלט הקלאסי נמצא אמיתי אך יקר: הוא מחזיק עד כ־עשרים פיילוטים ואובד עד עשרה. שכבה 4 מסירה את משפחת הערוץ כליל, תוך הגרלת ערוץ חדש בכל בלוק, והממסר הנלמד מוביל על פני מאזן העיוור הקלאסי הטוב ביותר, תוך הימנעות מחוסר היציבות של אמידת רצף מבוססת־החלטה.

התוצאות מתחמות משטרי הפעלה במקום להכריז על מנצח. כאשר מודל הערוץ המשוער תואם לערוץ האמיתי, אין שיפור על העיבוד הקלאסי ויש להשתמש בו. כאשר הערוץ בלתי־ידוע, או שאומדנו מתדרדר מתחת לכ־עשרה סימבולי פיילוט, או שאין כלל אומדן זמין לכל בלוק, המסר הנלמד הוא האפשרות שממשיכה לפעול, בעלות קבועה לכל סימבול וללא שלב זיהוי -- תוך הכללה על פני מימושים של משפחת הערוץ שעליה אומן, לא על פני משפחות ערוץ שמעולם לא הוצגו באימון.